

## North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2016

### FINAL REPORT



*Prepared for:*

North Coast - Skeena First Nations Stewardship Society  
Prince Rupert, BC

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## EXECUTIVE SUMMARY

The North Coast (Areas 3 and 4) recreational fishery harvests multiple species of salmon and groundfish, which can originate from both BC and US stocks, including individuals with Coded Wire Tags. Effort and harvest data, are required for management of fisheries and monitoring of indicator stocks. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) collected these data via a creel survey program that consisted of access point interviews and aerial overflights, for the months of June, July and August 2016. Lodge-based recreational harvest and effort were monitored by DFO via the collection of electronic daily fishing data submitted at the conclusion of the fishing season, and combined with the non-lodge data from the creel survey. Survey design and implementation, and analytical methods were similar to and consistent with those from the past several fishing seasons, though not identical. From June 1<sup>st</sup> to August 31<sup>st</sup>, 2,027 vessel interviews (involving 6,192 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 159 survey shifts. 24 aerial effort surveys were conducted with 60% of flights scheduled on weekdays and 40% on weekend days or holidays. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, and length measurements from Pacific Halibut (*Hippoglossus stenolepis*). Survey effort was adequate, as evidenced by the relatively good precision of the harvest estimates for most of the main taxa.

Total fishing effort was estimated to be 12,581 vessel trips. In total, an estimated 12,233 Chinook Salmon were caught, with 10,043 fish kept and 2,190 released. It was estimated that 42,066 Coho Salmon were caught, with 37,350 fish kept and 4,716 released. Total Pacific Halibut catch was estimated to be 21,245 individuals, with 13,771 fish kept and 7,475 released. Biomass of harvested Pacific Halibut was estimated at 98.8 net tonnes (217,918 lb), including 48.7 t (107,464 lb) in Area 3 and 50.1 t (110,454 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 130.9 t (288,604 lb) including 64.6 t (142,322 lb) in Area 3 and 66.4 t (146,283 lb) in Area 4. Total harvest estimates included 1,803 Lingcod (*Ophiodon elongates*), 4,291 rockfish (*Sebastes* spp.), and 568 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 630 Chinook Salmon, including length measurements (median 76 cm, n = 548), adipose fin-clip incidence rates (11.4% were clipped; n = 615), and scale samples (n=491). For Coho Salmon, 1,718 fish were biosampled, including the measurement of fork lengths (median 66 cm, n = 1450), and assessment of adipose fin-clip incidence rates (1.9% were clipped; n = 1,715). For 57 Chinook and 26 Coho salmon with fin clips, heads were collected for submission to the Salmon Head Recovery Program. Biosampling of Pacific Halibut included length measurements for 848 individuals (median = 83 cm). Average net and round weights were 7.8 kg (17.2 lb) and 10.3 kg (22.7 lb) in Area 3, and 6.7 kg (14.7 lb) and 8.8 kg (19.5 lb) in Area 4, respectively.

Recreational fishing effort, salmon CPE, and salmon catch in Areas 3 & 4 have been increasing over the last 20 years, in some cases dramatically. Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. The increasing trends highlight the need for continued monitoring of this fishery into the future. Cost savings could be gained by replacing overflights with trailer counts, assuming that data from previous studies can be used to generate a regression (boat counts vs. trailer counts) with a strong coefficient of determination ( $R^2$ ).

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## INTRODUCTION

The North Coast recreational fishery occurs in DFO's Pacific Region, in Management Areas 3 and 4. The fishery harvests multiple species, with Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon being the main targeted species, but also Pacific Halibut (*Hippoglossus stenolepis*), Lingcod (*Ophiodon elongates*), and other groundfish, rockfish (*Sebastes* spp.) and invertebrate species. Indeed, sport fishing on the North Coast is a popular activity for both residents and visitors to the area, and effort has increased substantially over the last 20 years (Robichaud et al. 2016). In light of increasing pressure, there is a need to understand harvest from the Area 3 and 4 recreational fishery, with respect to catch rates and stock origins (harvested fish can originate from both BC and US stocks). Moreover, Coded Wire Tag data collected from this recreational fishery via a creel survey program, are important components of the annual stock assessment data, which are required to monitor trends and manage the fisheries for Nass and Skeena river Chinook and Coho salmon stocks. As such, the creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition and improved scientific understanding of factors that might limit salmon production.

The creel survey program has run during each recreational fishing season from 2008 to 2015, estimating fishing effort and species-specific catch rates (e.g., Van Tongeren and Winther 2010, Robichaud et al. 2016). Starting last year (in 2015), the responsibility for carrying out the creel survey shifted from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey followed the same design used by Fisheries and Oceans Canada in the past, which was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002). After a successful first year, NCSFNSS was contracted to run the creel survey again in 2016. This report documents the 2016 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

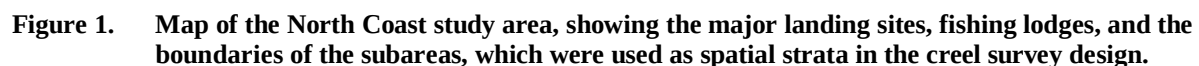
## SCOPE

The geographic scope of the creel survey was the same as in the previous seven years (Van Tongeren and Winther 2010, Robichaud et al. 2016), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island; Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).

The temporal scope of the full study was from June 1<sup>st</sup> through August 31<sup>st</sup>, 2016. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea.

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut and Lingcod.





The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used recently by DFO (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from June through August 2016. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month and day type (weekdays vs. weekend days). Spatial stratifications



were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The eight objectives for this study were:

- 1) Continue development of software for data management and analysis comparable to DFO's CREST program;
- 2) Update analytical scripts which will allow incorporation of logbook data and trailer counts into the analysis
- 3) Provide estimates of the catch and effort of all species targeted by the recreational fishery and associated estimates of precision;
- 4) Provide DFO and PSC with monthly in-season catch estimates for all salmon species and Pacific Halibut harvested in the North Coast recreational fishery;
- 5) Provide communication materials to recreational harvesters to convey the use and importance of their catch data in fisheries management;
- 6) Collect biological data , such as fin clip incidence for Coho and Chinook salmon, scale samples from Chinook Salmon, and Pacific Halibut lengths from as many fish as possible during angler interviews;
- 7) Produce a final written report to be available to a broad audience; and
- 8) Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests and First Nations.

Objectives 1 and 2 were required in order to meet any subsequent objective. Objectives 3 and 7 were met by completing this report. Objective 4 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 5 will be met subsequent to the publication of this report, as its contents will be based on the results presented herein. Objective 6 was met by biosampling fish at access points throughout the study period: fish were measured, fin clip incidence data was collected for Coho and Chinook salmon, scales were collected from Chinook Salmon, heads were collected from adipose-clipped Chinook Salmon, and all biological data were submitted to DFO periodically throughout the study period. Objective 8 was met in late 2016.

## METHODS

A complete census of catch and effort was available for lodge-based anglers. However, the remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each month and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1)

conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, kept fish were measured and sexed. Chinook and Coho salmon were checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook Salmon. Chinook Salmon were also examined for flesh colour;
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The methods used were adapted from those developed and documented for the Strait of Georgia Creel Survey (English et al. 2002). Statistical precision associated with creel survey catch and effort estimates were based on those documented in English et al. (2002) and Blakley et al. (2003).

This procedure provides a statistically unbiased estimate of catch per effort, provided the anglers interviewed are representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that was conducted from that site in recent years (see Van Tongeren and Winther 2010). Wherever possible, the busiest access points were selected preferentially in order to obtain the maximum number of interviews. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Ideally, interviews would capture data about angler activity in all creel subareas.

Interviewing sessions were to be separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00 and PM shifts from 13:00 to 21:00. In total, 159 shifts were scheduled.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 2-3 weekend days and usually on 5-6 weekdays each month. Timing of the overflights was targeted for peak fishing periods, but may have been off-peak on some occasions due to weather or scheduling conflicts.

### **Angler Activity Patterns**

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002).

The first weighting factor,  $W1$ , expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site  $a$ , were consistent for landing site  $a$ , even during the days when no interviews occurred. A specific  $W1$  was calculated for each landing site during each month and day type:

$$W1_{mda} = \frac{N_{md}}{K_{mda}} \quad (\text{Equation 1})$$

where  $N_{md}$  was the total number of type  $d$  days in month  $m$ ; and  $K_{mda}$  was the number of days during which interviews occurred at landing site  $a$ , on type  $d$  days during month  $m$ .

The second weighting factor,  $W2$ , expanded the numbers of interviews conducted, to account for the boats that were *not* interviewed (or that refused interviews). That is, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific  $W2$  was calculated for each surveyor ( $u$ ) operating on each surveying date ( $k$ ) at each landing site ( $a$ ):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku} \left( \frac{LF_{aku}}{LF_{aku} + LN_{aku}} \right)}{LF_{aku} - ri_{aku}}, \quad (\text{Equation 2})$$

where  $LF_{aku}$  was the number of interviewed boats that were fishing (a subset of these,  $ri_{aku}$ , refused an interview, or had an incomplete interview that could not be used),  $LN_{aku}$  was the number of interviewed boats that were not fishing, and  $LM_{aku}$  was the number of boats that were not interviewed by surveyor  $u$ , during surveying date  $k$ , at landing site  $a$ . This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term  $A_{mdsakut}$  to denote the number of non-lodge boats reporting activity during time-block  $t$ , as interviewed by surveyor  $u$  on survey date  $k$ , at landing site  $a$ , in subarea  $s$ , during month  $m$ , and on day type  $d$  ( $n_{mdsaku}$  was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{mdst} = \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot A_{mdsakut}). \quad (\text{Equation 3})$$

**Table 1. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns and CPE, for all levels of each factor, 2016.**

| Month  | Day Type | Subarea |    |    |    |    |     |    |    |    |    |    |    |
|--------|----------|---------|----|----|----|----|-----|----|----|----|----|----|----|
|        |          | 3I      | 3J | 3K | 4A | 4B | 4C  | 4D | 4E | 4F | 4G | 4H | 4I |
| June   | WD       | 20      | 37 | 1  | 31 | 43 | 47  | 30 | 5  | 87 | 13 | 34 | 6  |
|        | WE       | 9       | 33 | 3  | 25 | 69 | 70  | 35 | 6  | 67 | 14 | 37 | 4  |
| July   | WD       | 16      | 30 | 9  | 8  | 56 | 93  | 35 | 5  | 97 | 27 | 51 | 10 |
|        | WE       | 12      | 31 | 9  | 22 | 25 | 106 | 27 | 2  | 68 | 36 | 57 | 12 |
| August | WD       | 8       | 11 | 6  | 22 | 53 | 74  | 21 | 4  | 59 | 9  | 32 | 9  |
|        | WE       | 7       | 14 | 6  | 21 | 38 | 69  | 15 | 1  | 47 | 10 | 31 | 6  |

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{mds} = \sum_t A'_{mdst} \quad (\text{Equation 4})$$

The proportion of non-lodge boats ( $P_{mdst}$ ) that were actively fishing during each of the hourly time-blocks ( $t$ ) was calculated for each month, day type, and subarea:

$$P_{mdst} = \frac{A'_{mdst}}{\sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad (\text{Equation 5})$$

Using this method, 72 unique angler activity patterns were to be estimated (i.e., 3 months  $\times$  2 day types  $\times$  12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 72 blocks. In the end, some blocks contained too few interviews (Table 1), so it was decided to pool activity data over subarea. The equation for non-lodge angler activity was thus

$$P_{mdt} = \frac{\sum_s A'_{mdst}}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})}, \quad (\text{Equation 6})$$

and the associated variance was:

$$S^2_{P_{mdt}} = \frac{(P_{mdt})(1-P_{mdt})}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})}. \quad (\text{Equation 7})$$

### **Catch Per Effort Estimation**

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview ( $i$ ), the guided status ( $g$ ), month ( $m$ ), and the catch ( $C$ ) of each species ( $r$ ) in each subarea ( $s$ ) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{msrgi} = \frac{C_{msrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\}. \quad (\text{Equation 8})$$

Mean  $CPE$  was calculated as the average catch, over all  $n_{msg}$  boat trips:

$$\hat{CPE}_{msrg} = \frac{\sum_{i=1}^{n_{msg}} C_{msrgi}}{n_{msg}}. \quad (\text{Equation 9})$$

The variance for the estimate of mean catch per effort was calculated as:

$$S^2_{\hat{CPE}_{msrg}} = \frac{\sum (CPE_{msrgi} - \hat{CPE}_{msrg})^2}{(n_{msg} - 1)}. \quad (\text{Equation 10})$$

In the event that a month/subarea-specific sample size was  $< 5$ , adjacent subareas would be pooled together as follows:

- if 3I, 3J or 3K are too low, then it will be combined with the other two to give a 3IJK regional average CPE;
- if 4A, 4B or 4C are too low, then it will be combined with the other two to give a 4ABC regional average CPE;
- if 4D, 4E or 4F are too low, then it will be combined with the other two to give a 4DEF regional average CPE; and
- if 4G, 4H or 4L are too low, then it will be combined with the other two to give a 4GHL regional average CPE.

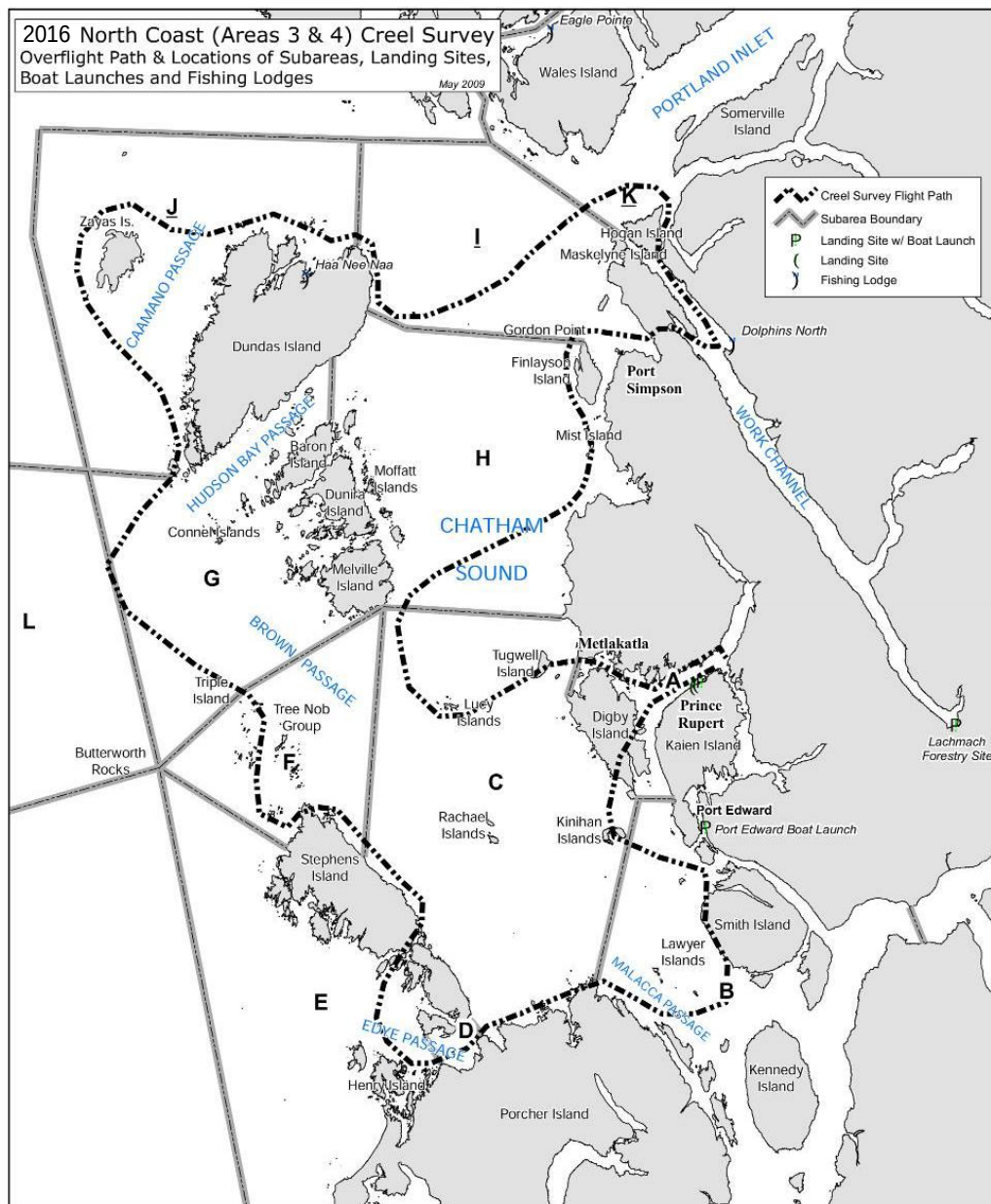
Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip.

### **Angler Effort Estimation**

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

#### *Lodge Effort Census*

A complete census of lodge-based angling effort was collected from lodge log books by DFO personnel (I. Winther, DFO Prince Rupert, pers. comm.). DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodge based vessels were considered unavailable to the flights on change-over days (days when the lodges received new clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight). Change-over days occurred twice a week but timing was often different for each lodge, making 29% (2/7) of the lodge effort not visible to flights due to change-over days. 2) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water



**Figure 2.** Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert, and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

**Table 2.** Summary of overflight sampling effort (number of flights shifts by month and day type), 2016.

| Month        | Weekday   | Weekend/ |           |
|--------------|-----------|----------|-----------|
|              |           | Holiday  | Total     |
| June         | 5         | 3        | 8         |
| July         | 6         | 2        | 8         |
| August       | 5         | 3        | 8         |
| <b>Total</b> | <b>16</b> | <b>8</b> | <b>24</b> |



during the day. Thus, approximately 60% of the remaining vessel activity was available to the flight counts on ‘regular’ days (i.e., not change-over). 3) Vessels on mid-day break or on change-over days were excluded from the numbers of vessels visible to the flights. All of the lodge catch was reported from Area 3, without subarea-specific effort information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

### Non-lodge Effort Surveys

Table 2 shows the number of overflight surveys scheduled for each month and day type. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of small (< 30 ft) and large ( $\geq 30$  ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey  $o$  (conducted during month  $m$  and on day type  $d$ ), observers tallied the total number of recreational boats that were actively fishing at time  $t$  on their  $i^{\text{th}}$  pass over subarea  $s$ ,  $V_{mdsoit}$ . For each visit to a subarea within Management Area 3, the relative proportion of boats was calculated,  $\rho_{mdsoit}$ , and the total number of lodge boats ( $\lambda_{mdo}$ ) apportioned accordingly:

$$\lambda_{mdsoit} = \rho_{mdsoit} \cdot \lambda_{mdo} \quad . \quad (\text{Equation 11})$$

The number of non-lodge boats was hence:

$$v_{mdsoit} = V_{mdsoit} - \lambda_{mdsoit} \quad . \quad (\text{Equation 12})$$

Since angling occurs over the course of the entire day, the number of non-lodge boats that were observed was divided by the proportion of average daily number of boats active ( $P_{mdst}$ ) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, hence the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea,  $B_{mdso}$ :

$$B_{mdso} = \sum_i \frac{v_{mdsoit}}{P_{mdst}} \quad . \quad (\text{Equation 13})$$

The variance of  $B_{mdso}$  was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} \text{if } z = x / y, \quad \text{Var}(z) &= (y^{-2})\text{Var}(x) + (x^2 y^{-4})\text{Var}(y) \quad ; \\ \text{if } z = xy, \quad \text{Var}(z) &= (y^2)\text{Var}(x) + (x^2)\text{Var}(y) \quad ; \text{ and} \\ \text{if } z = cx, \quad \text{Var}(z) &= (c^2)\text{Var}(x) \quad \text{where } c \text{ is a constant.} \end{aligned} \quad (\text{Equation 14})$$

Since there was no variance associated with the boat counts,

$$S_{B_{mdso}}^2 = (v_{mdsoit})^2 (P_{mdst})^{-4} S_{P_{mdst}}^2 \quad . \quad (\text{Equation 15})$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys conducted,  $n_{mds}$ , as:



$$\hat{B}_{m_{ds}} = \frac{\sum_{o=1}^{n_{m_{ds}}} B_{m_{ds}o}}{n_{m_{ds}}}, \quad (\text{Equation 16})$$

with a variance of:

$$S_{\hat{B}_{m_{ds}}}^2 = (n_{m_{ds}})^{-2} S_{B_{m_{ds}o}}^2. \quad (\text{Equation 17})$$

Total monthly fishing effort, was calculated for each day type and subarea by multiplying the average daily effort by the number days of day type  $d$  that occurred in month  $m$ :

$$E_{m_{ds}} = \hat{B}_{m_{ds}} \cdot N_{m_{ds}}. \quad (\text{Equation 18})$$

with a variance of:

$$S_{E_{m_{ds}}}^2 = S_{\hat{B}_{m_{ds}}}^2 \cdot N_{m_{ds}}^2 \quad (\text{Equation 19})$$

The standard error of the estimate of the total monthly fishing effort, after pooling over day types, was:

$$S_{E_{ms}} = \sqrt{\frac{\sum_d S_{E_{m_{ds}}}^2}{\sum_d n_{m_{ds}}}}. \quad (\text{Equation 20})$$

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of month and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of month and day type during which the data gap occurred.

### **Catch and Harvest Estimation**

Total catch was calculated for each month, subarea and species by multiplying total angling effort by catch per effort, and then summing over day type:

$$C_{msr} = \sum_d (E_{m_{ds}} \cdot \hat{CPE}_{msrg}) \quad (\text{Equation 21})$$

where  $g = \text{'all'}$  (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{msr}} = \sqrt{\sum_d \left( E_{m_{ds}}^2 \frac{S_{CPE_{msrg}}^2}{n_{msg}} + CPE_{msrg}^2 \frac{S_{E_{m_{ds}}}^2}{n_{m_{ds}}} + \frac{S_{CPE_{msrg}}^2}{n_{msg}} \frac{S_{E_{m_{ds}}}^2}{n_{m_{ds}}} \right)} \quad (\text{Equation 22})$$

where  $n_{m_{ds}}$  and  $n_{msg}$  are the sample sizes for E and CPE estimates, respectively.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released.

### **Biosampling**

When given permission by the anglers, observers collected biological data from Chinook and Coho salmon and Pacific Halibut. All three species were sexed. Chinook and Coho salmon

were examined for clipped adipose fins (mark rate data used by DFO for coded wire tag analyses) and head tags. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook Salmon (used by DFO to determine age, and stock composition).

Ideally, all anglers would allow their catch to be biosampled, and that the examined fish were an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had observers ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol so that, in 2016, anglers were asked if their catch could be biosampled before any other interactions took place. Another new detail, in 2016 no fin clip data were accepted on the word of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

### **Biomass of Harvested Pacific Halibut**

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned}\text{Weight(lb)}_{\text{net}} &= 6.921 \times 10^{-6} \times \text{FL(cm)}^{3.24} \\ \text{Weight(lb)}_{\text{round}} &= 9.166 \times 10^{-6} \times \text{FL(cm)}^{3.24}\end{aligned}\quad (\text{Equation 23})$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

### **Statistical Analyses**

Comparisons of CPE, effort, catch or harvest by month or day type were performed within each fish taxa using Kruskal-Wallis tests (Zar 1984). When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

### **Trailer Census**

During each interview stint at Rushbrooke or Port Edwards, between noon and 2 PM (i.e., similar to the timing of aerial overflights), the number of parked trailers at the landing location was recorded. Theoretically, trailer counts could be used as an index of fishing effort, and are more economical than overflights and are not weather dependant. DFO has been collecting trailer count data for years and we are maintaining the time series.

Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. The location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively.

## RESULTS

### Angler Interviews

Over the three month study period, there were 159 observer shifts, during which 2,027 interviews were conducted, involving 6,192 non-lodge anglers (Table 3; Figure 3). July was the busiest month, with 800 interviews conducted (average 15.1 interviews per shift), which represented 39% of the total number of study period interviews (Figure 1). In all there were 47 refusals, or 2.3% refusal rate. The highest refusal rate (3.1%) was at Rushbrooke Boat Launch, where 57% of the refusals occurred. The refusal rates for Prince Rupert Yacht Club and Port Edward Boat Launch were 2.1% and 1.4%, respectively.

On average, 25% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 17 of 592 trips interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 23% and 53% of the time, respectively (Table 4).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F being the overwhelming favourite (Table 5). Subareas 4C, and 4B were the next most likely areas for anglers to have reported activity. The subarea with the lowest number of trips reporting activity was 4E. Only 47 interviews were of parties that only targeted shellfish, and those were largely in subarea 4A (Table 6).

### Angler Activity Patterns

Monthly non-lodge angler activity patterns estimates are shown in Figure 4. There was virtually no difference between the weekday and weekend/holiday activity patterns for June, July or August. Angler activity patterns showed some changes across the three-month survey period. As expected, there was a shift over time away from late-afternoon angling with the reduction in daylight. In June and July, peak activity was about an hour after the beginning of the flight.

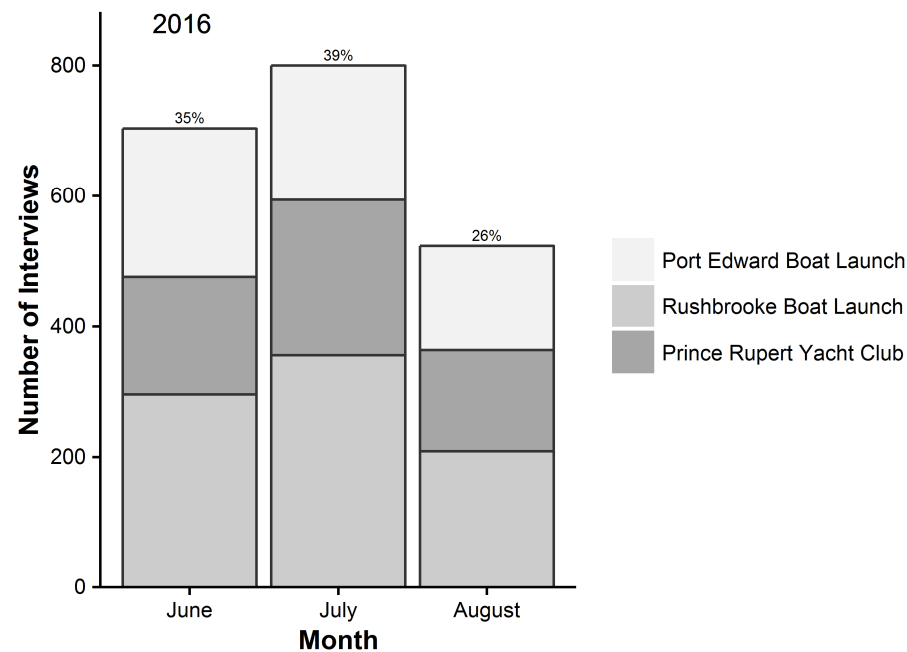
### Catch Per Effort Estimates

Sample sizes were adequate ( $n \geq 5$ ) to produce reliable estimates of non-lodge CPE in most combinations of months and management areas, except in subarea 3K in June (Table 7). However, when the interview data were split between guided and non-guided groups, monthly sample sizes did not suffice in some of the management areas (e.g., 3K, 4B, 4D-E, and 4L) where guiding was less common (Table 7). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPE calculations (Table 8). Average number of fish harvested per non-lodge boat trip (HPE) was calculated for each species by month, subarea, and by trip type (Appendix 2).

HPE varied significantly among months (Table 9) for Chinook ( $\chi^2 = 11.0$ ,  $df = 2$ ,  $P = 0.004$ ), Coho ( $\chi^2 = 17.5$ ,  $df = 2$ ,  $P < 0.001$ ), and Pink salmon (*O. gorbuscha*,  $\chi^2 = 16.1$ ,  $df = 2$ ,  $P < 0.001$ ). For Chinook Salmon, median HPE in June and July was significantly higher than in August (Table 9; Figure 5). For Coho Salmon, HPE in all months differed significantly, with highest values observed in July (Table 9; Figure 5). For Pink salmon, median HPE in June was

**Table 3. Monthly number of shifts numbers of boats interviewed per landing site, 2016. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.**

| Landing Site               | June       |              |           |             | July       |              |           |             | August     |              |           |             | TOTAL        |              |            |             |
|----------------------------|------------|--------------|-----------|-------------|------------|--------------|-----------|-------------|------------|--------------|-----------|-------------|--------------|--------------|------------|-------------|
|                            | # ints     | # anglers    | # stints  | ints/stint  | # ints     | # anglers    | # stints  | ints/stint  | # ints     | # anglers    | # stints  | ints/stint  | # ints       | # anglers    | # stints   | ints/stint  |
| Port Edward Boat Launch    | 226        | 607          | 20        | 11.3        | 206        | 610          | 17        | 12.1        | 160        | 488          | 15        | 10.7        | 592          | 1705         | 52         | 11.4        |
| Prince Rupert Yacht Club   | 182        | 601          | 19        | 9.6         | 238        | 811          | 20        | 11.9        | 155        | 507          | 18        | 8.6         | 575          | 1919         | 57         | 10.1        |
| Rushbrooke Boat Launch     | 295        | 830          | 20        | 14.8        | 356        | 1111         | 16        | 22.3        | 209        | 627          | 14        | 14.9        | 860          | 2568         | 50         | 17.2        |
| <b>TOTAL</b>               | <b>703</b> | <b>2,038</b> | <b>59</b> | <b>11.9</b> | <b>800</b> | <b>2,532</b> | <b>53</b> | <b>15.1</b> | <b>524</b> | <b>1,622</b> | <b>47</b> | <b>11.1</b> | <b>2,027</b> | <b>6,192</b> | <b>159</b> | <b>12.7</b> |
| % of Total Interviews      |            | 35%          |           |             |            | 39%          |           |             |            | 26%          |           |             |              |              |            |             |
| Estimated Non-Lodge Effort |            | 3,368        |           |             |            | 4,631        |           |             |            | 2,733        |           |             |              | 10,732       |            |             |
| % of Total Effort          |            | 31%          |           |             |            | 43%          |           |             |            | 25%          |           |             |              |              |            |             |
| % of Effort Sampled        |            | 21%          |           |             |            | 17%          |           |             |            | 19%          |           |             |              | 19%          |            |             |



**Figure 3. Monthly number of interviews conducted at each of three landing sites, 2016.**

**Table 4. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2016.**

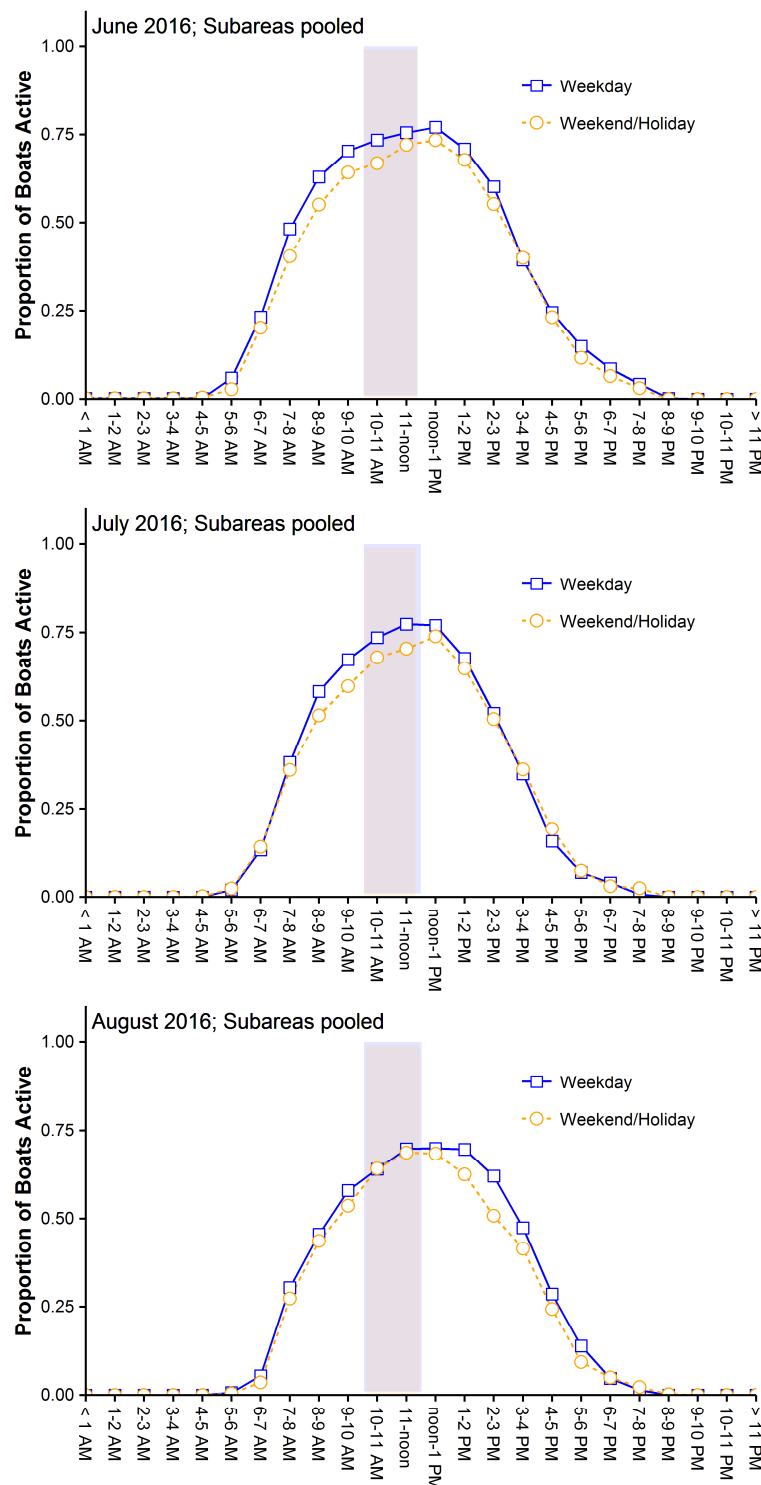
| <i>Landing Site</i>             | <i>Trip Type</i> |                 | <i>% guided</i> |
|---------------------------------|------------------|-----------------|-----------------|
|                                 | <i>Guided</i>    | <i>Unguided</i> |                 |
| <i>Port Edward Boat Launch</i>  | 17               | 575             | 2.9%            |
| <i>Prince Rupert Yacht Club</i> | 303              | 272             | 53%             |
| <i>Rushbrooke Boat Launch</i>   | 194              | 666             | 23%             |
| <b><i>TOTAL</i></b>             | <b>514</b>       | <b>1,513</b>    | <b>25%</b>      |

**Table 5. Numbers of non-lodge boat trips by month and subarea, 2016. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.**

| <i>Month</i>        | <i>Subarea</i> |            |           |           |            |            |            |           |            |           |            |           |           |            |                | <i>TOTAL</i> |
|---------------------|----------------|------------|-----------|-----------|------------|------------|------------|-----------|------------|-----------|------------|-----------|-----------|------------|----------------|--------------|
|                     | <i>3I</i>      | <i>3J</i>  | <i>3K</i> | <i>4A</i> | <i>4B</i>  | <i>4C</i>  | <i>4D</i>  | <i>4E</i> | <i>4F</i>  | <i>4G</i> | <i>4H</i>  | <i>4L</i> | <i>3?</i> | <i>4?</i>  | <i>Refusal</i> |              |
| <i>June</i>         | 27             | 62         | 4         | 41        | 98         | 97         | 53         | 10        | 129        | 21        | 63         | 6         | 13        | 67         | 15             | 703          |
| <i>July</i>         | 21             | 49         | 16        | 21        | 75         | 179        | 58         | 7         | 148        | 50        | 96         | 15        | 7         | 42         | 16             | 800          |
| <i>August</i>       | 9              | 21         | 10        | 31        | 84         | 125        | 33         | 4         | 91         | 15        | 52         | 9         | 4         | 21         | 16             | 524          |
| <b><i>TOTAL</i></b> | <b>57</b>      | <b>132</b> | <b>30</b> | <b>93</b> | <b>257</b> | <b>401</b> | <b>144</b> | <b>20</b> | <b>367</b> | <b>85</b> | <b>212</b> | <b>31</b> | <b>23</b> | <b>129</b> | <b>47</b>      | <b>2,027</b> |

**Table 6. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2016.**

|        | Subarea |    |    |    |    |    |    |    |    |    |    |    |    |    |       |
|--------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|
| Month  | 3I      | 3J | 3K | 4A | 4B | 4C | 4D | 4E | 4F | 4G | 4H | 4L | 3? | 4? | TOTAL |
| June   | 0       | 0  | 1  | 14 | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 18    |
| July   | 1       | 0  | 1  | 8  | 0  | 0  | 0  | 0  | 0  | 0  | 5  | 0  | 0  | 0  | 14    |
| August | 0       | 0  | 0  | 12 | 1  | 2  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 15    |
| TOTAL  | 1       | 0  | 2  | 34 | 2  | 2  | 0  | 0  | 1  | 0  | 5  | 0  | 0  | 1  | 47    |



**Figure 4.** Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and month, pooled over all subareas, 2016. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and month.

**Table 7. Sample sizes for CPE (harvest per effort) estimates by month, for guided and not guided non-lodge boat trips, 2016.**

| Subarea |           |    |    |    |    |     |     |    |    |     |    |     |    |
|---------|-----------|----|----|----|----|-----|-----|----|----|-----|----|-----|----|
| Month   | Trip Type | 3I | 3J | 3K | 4A | 4B  | 4C  | 4D | 4E | 4F  | 4G | 4H  | 4L |
| June    | Guided    | 12 | 35 | 0  | 12 | 4   | 7   | 3  | 0  | 20  | 4  | 31  | 7  |
|         | Unguided  | 17 | 35 | 4  | 44 | 108 | 110 | 61 | 11 | 134 | 23 | 40  | 3  |
|         | Combined  | 29 | 70 | 4  | 56 | 112 | 117 | 64 | 11 | 154 | 27 | 71  | 10 |
| July    | Guided    | 9  | 28 | 1  | 7  | 11  | 26  | 9  | 3  | 59  | 26 | 48  | 15 |
|         | Unguided  | 19 | 33 | 17 | 23 | 70  | 173 | 53 | 4  | 106 | 37 | 60  | 7  |
|         | Combined  | 28 | 61 | 18 | 30 | 81  | 199 | 62 | 7  | 165 | 63 | 108 | 22 |
| August  | Guided    | 8  | 12 | 2  | 14 | 10  | 27  | 1  | 1  | 44  | 7  | 28  | 13 |
|         | Unguided  | 7  | 13 | 10 | 29 | 81  | 116 | 35 | 4  | 62  | 12 | 35  | 2  |
|         | Combined  | 15 | 25 | 12 | 43 | 91  | 143 | 36 | 5  | 106 | 19 | 63  | 15 |

**Table 8. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each month and subarea, for guided and not guided non-lodge boat trips, 2016.**

| Subarea |           |    |    |          |    |        |    |        |        |    |        |    |        |
|---------|-----------|----|----|----------|----|--------|----|--------|--------|----|--------|----|--------|
| Month   | Trip Type | 3I | 3J | 3K       | 4A | 4B     | 4C | 4D     | 4E     | 4F | 4G     | 4H | 4L     |
| June    | Guided    | 3I | 3J | 3?,I,J,K | 4A | 4A,B,C | 4C | 4D,E,F | 4D,E,F | 4F | 4G,H,L | 4H | 4L     |
|         | Unguided  | 3I | 3J | 3?,I,J,K | 4A | 4B     | 4C | 4D     | 4E     | 4F | 4G     | 4H | 4G,H,L |
|         | Combined  | 3I | 3J | 3?,I,J,K | 4A | 4B     | 4C | 4D     | 4E     | 4F | 4G     | 4H | 4L     |
| July    | Guided    | 3I | 3J | 3?,I,J,K | 4A | 4B     | 4C | 4D     | 4D,E,F | 4F | 4G     | 4H | 4L     |
|         | Unguided  | 3I | 3J | 3K       | 4A | 4B     | 4C | 4D     | 4D,E,F | 4F | 4G     | 4H | 4L     |
|         | Combined  | 3I | 3J | 3K       | 4A | 4B     | 4C | 4D     | 4E     | 4F | 4G     | 4H | 4L     |
| August  | Guided    | 3I | 3J | 3?,I,J,K | 4A | 4B     | 4C | 4D,E,F | 4D,E,F | 4F | 4G     | 4H | 4L     |
|         | Unguided  | 3I | 3J | 3K       | 4A | 4B     | 4C | 4D     | 4D,E,F | 4F | 4G     | 4H | 4G,H,L |
|         | Combined  | 3I | 3J | 3K       | 4A | 4B     | 4C | 4D     | 4E     | 4F | 4G     | 4H | 4L     |



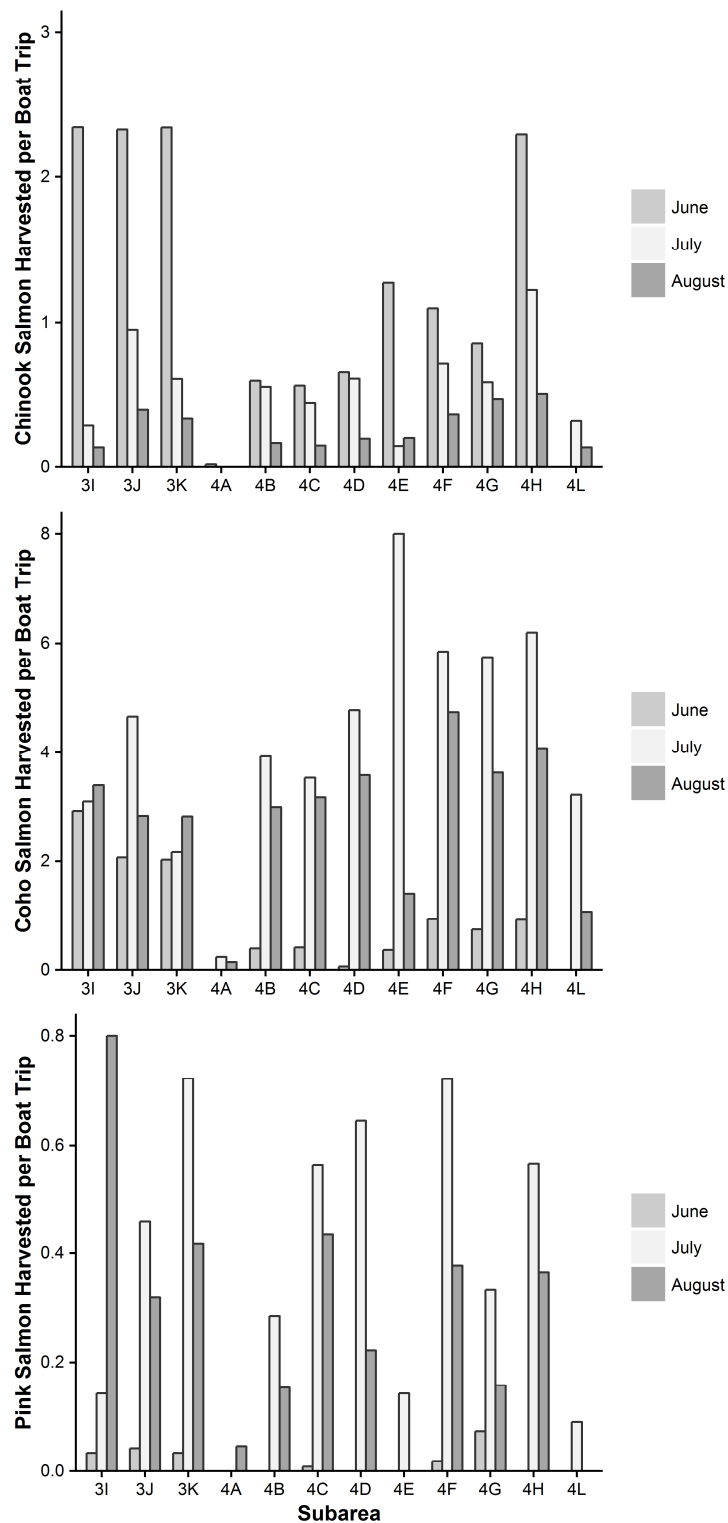


Figure 5. Average number of Chinook, Coho and Pink salmon harvested per non-lodge boat trip, by month and subarea, 2016. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

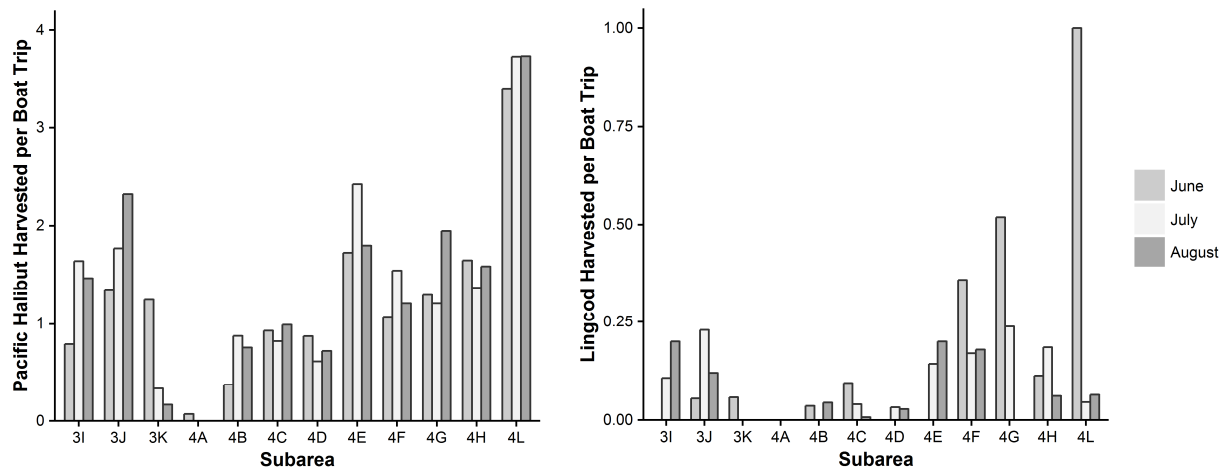
**Table 9. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among months, by species, 2016. Subareas treated as replicates.**

| Species             | Kruskal-Wallis Statistics |    |        | Median HPE |       |        |
|---------------------|---------------------------|----|--------|------------|-------|--------|
|                     | $\chi^2$                  | df | P      | June       | July  | August |
| Chinook Salmon      | 11.0                      | 2  | 0.004  | 1.0        | 0.6   | 0.2    |
| Chum Salmon         | 2.4                       | 2  | 0.30   | 0.02       | 0.003 | 0      |
| Coho Salmon         | 17.5                      | 2  | 0.0002 | 0.6        | 4.3   | 3.1    |
| Pink Salmon         | 16.1                      | 2  | 0.0003 | 0.004      | 0.4   | 0.3    |
| Sockeye Salmon      | 1.0                       | 2  | 0.62   | 0          | 0     | 0      |
| Arrow Flounder      | 3.1                       | 2  | 0.22   | 0          | 0     | 0      |
| Big Skate           | 1.0                       | 2  | 0.60   | 0          | 0     | 0      |
| Greenling           | 2.0                       | 2  | 0.37   | 0          | 0     | 0      |
| Lingcod             | 0.2                       | 2  | 0.92   | 0.06       | 0.08  | 0.05   |
| Pacific Cod         | 1.4                       | 2  | 0.51   | 0          | 0     | 0      |
| Pacific Halibut     | 0.2                       | 2  | 0.91   | 1.2        | 1.3   | 1.3    |
| Ratfish             | 2.2                       | 2  | 0.33   | 0          | 0     | 0      |
| Sablefish           | 2.2                       | 2  | 0.34   | 0          | 0     | 0      |
| Black Rockfish      | 1.0                       | 2  | 0.61   | 0          | 0     | 0      |
| Canary Rockfish     | 0.5                       | 2  | 0.79   | 0          | 0     | 0      |
| China Rockfish      | 0.9                       | 2  | 0.65   | 0          | 0     | 0      |
| Copper Rockfish     | 2.9                       | 2  | 0.24   | 0          | 0     | 0      |
| Quillback Rockfish  | 3.0                       | 2  | 0.22   | 0.03       | 0.02  | 0      |
| Redstripe Rockfish  | 0.0                       | 2  | 0.99   | 0          | 0     | 0      |
| Tiger Rockfish      | 6.3                       | 2  | 0.043  | 0          | 0     | 0      |
| Yelloweye Rockfish  | 1.0                       | 2  | 0.61   | 0.007      | 0.03  | 0.03   |
| Yellowtail Rockfish | 2.2                       | 2  | 0.33   | 0          | 0     | 0      |

significantly lower than in July or August (Table 9; Figure 5). There were no significant differences among months for HPE of Chum (*O. keta*) or Sockeye salmon (*O. nerka*), or for any species of groundfish (including Pacific Halibut and Lingcod; Figure 6). For the rockfish species, there were no significant differences among months in the number of fish harvested per boat trip. For Tiger rockfish (*S. nigrocinctus*), the omnibus test for an effect of month was significant ( $\chi^2 = 6.3$ ,  $df = 2$ ,  $P = 0.043$ ), but none of the *post-hoc* pairwise comparisons were statistically significant after controlling the false discovery rate (i.e., using the Benjamini–Hochberg procedure).

Guided trips had significantly higher HPE than non-guided trips (Table 10) for Chinook ( $\chi^2 = 5.2$ ,  $df = 1$ ,  $P = 0.022$ ) and Coho salmon ( $\chi^2 = 8.5$ ,  $df = 1$ ,  $P = 0.004$ ), and Pacific Halibut ( $\chi^2 = 20.7$ ,  $df = 1$ ,  $P < 0.0001$ ). Guided trips caught 167–181% more salmon and >200% more Pacific Halibut than unguided trips. Guided trips caught 266% more Lingcod than unguided trips, but the difference was not statistically significant.

For lodge boat trips, CPE estimates are shown in Table 11. Average numbers of Chinook Salmon retained (1.0 retained per trip) was 128% and 214% higher than the median guided and non-guided non-lodge boat trips, respectively. Similarly, Lingcod were caught more readily (0.3 fish per trip) by lodge boats than by non-lodge boats (< 0.1 fish per trip). For Coho Salmon, lodge boats did worse than guided non-lodge boats (54%, respectively), but were about equal (98%) to non-guided non-lodge boats. Lodge boats caught fewer Pacific Halibut (0.6 fish per boat trip) than either guided (2.1) or non-guided (0.9) non-lodge boats (Tables 10 and 11).



**Figure 6.** Average number of Pacific Halibut and Lingcod harvested per non-lodge boat trip, by month and subarea, 2016. Data are for guided and unguided trips combined. Note that Y axis scale varies between plots.

**Table 10.** Kruskal-Wallis tests for differences in HPE (fish caught per non-lodge boat trip) between guided and non-guided boats, by species, 2016. Months and subareas treated as replicates.

| Species             | Kruskal-Wallis Statistics |    |         | Median HPE |       |
|---------------------|---------------------------|----|---------|------------|-------|
|                     | $\chi^2$                  | df | P       | Guided     | Not   |
| Chinook Salmon      | 5.2                       | 1  | 0.022   | 0.8        | 0.5   |
| Chum Salmon         | 0.2                       | 1  | 0.66    | 0          | 0     |
| Coho Salmon         | 8.5                       | 1  | 0.004   | 4.3        | 2.3   |
| Pink Salmon         | 0.1                       | 1  | 0.72    | 0.09       | 0.2   |
| Sockeye Salmon      | 0.1                       | 1  | 0.74    | 0          | 0     |
| Arrow Flounder      | 0.6                       | 1  | 0.42    | 0          | 0     |
| Big Skate           | 2.0                       | 1  | 0.15    | 0          | 0     |
| Greenling           | 0.0                       | 1  | 0.95    | 0          | 0     |
| Lingcod             | 2.3                       | 1  | 0.13    | 0.09       | 0.03  |
| Pacific Cod         | 2.7                       | 1  | 0.10    | 0          | 0     |
| Pacific Halibut     | 20.7                      | 1  | <0.0001 | 2.1        | 0.9   |
| Ratfish             | 0.4                       | 1  | 0.54    | 0          | 0     |
| Sablefish           | 1.2                       | 1  | 0.28    | 0          | 0     |
| Black Rockfish      | 2.0                       | 1  | 0.16    | 0          | 0     |
| Canary Rockfish     | 0.1                       | 1  | 0.74    | 0          | 0     |
| China Rockfish      | 0.1                       | 1  | 0.77    | 0          | 0     |
| Copper Rockfish     | 0.3                       | 1  | 0.61    | 0          | 0     |
| Quillback Rockfish  | 3.6                       | 1  | 0.06    | 0          | 0.006 |
| Redstripe Rockfish  | 0.1                       | 1  | 0.79    | 0          | 0     |
| Tiger Rockfish      | 0.0                       | 1  | 0.90    | 0          | 0     |
| Yelloweye Rockfish  | 0.2                       | 1  | 0.64    | 0.02       | 0.02  |
| Yellowtail Rockfish | 0.2                       | 1  | 0.65    | 0          | 0     |

**Table 11. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, from census of Area 3 lodges, 2016.**

| <i>Species</i>            | <i>June</i> | <i>July</i> | <i>August</i> | <i>Total</i> |
|---------------------------|-------------|-------------|---------------|--------------|
| <i>Chinook Salmon:</i>    |             |             |               |              |
| <i>retained</i>           | 1.7         | 0.9         | 0.2           | 1.0          |
| <i>released</i>           | 0.2         | 0.07        | 0.02          | 0.1          |
| <i>Coho Salmon</i>        | 0.7         | 2.9         | 3.2           | 2.3          |
| <i>Pink Salmon</i>        | 0.002       | 0.01        | 0.05          | 0.02         |
| <i>Chum Salmon</i>        | 0.1         | 0.06        | 0.01          | 0.07         |
| <i>Sockeye Salmon</i>     | 0.007       | 0           | 0             | 0.002        |
| <i>Pacific Halibut</i>    | 0.5         | 0.6         | 0.6           | 0.6          |
| <i>Lingcod</i>            | 0.3         | 0.3         | 0.2           | 0.3          |
| <i>Yelloweye Rockfish</i> | 0.04        | 0.02        | 0             | 0.02         |
| <i>Other rockfish</i>     | 0.4         | 0.4         | 0.5           | 0.4          |

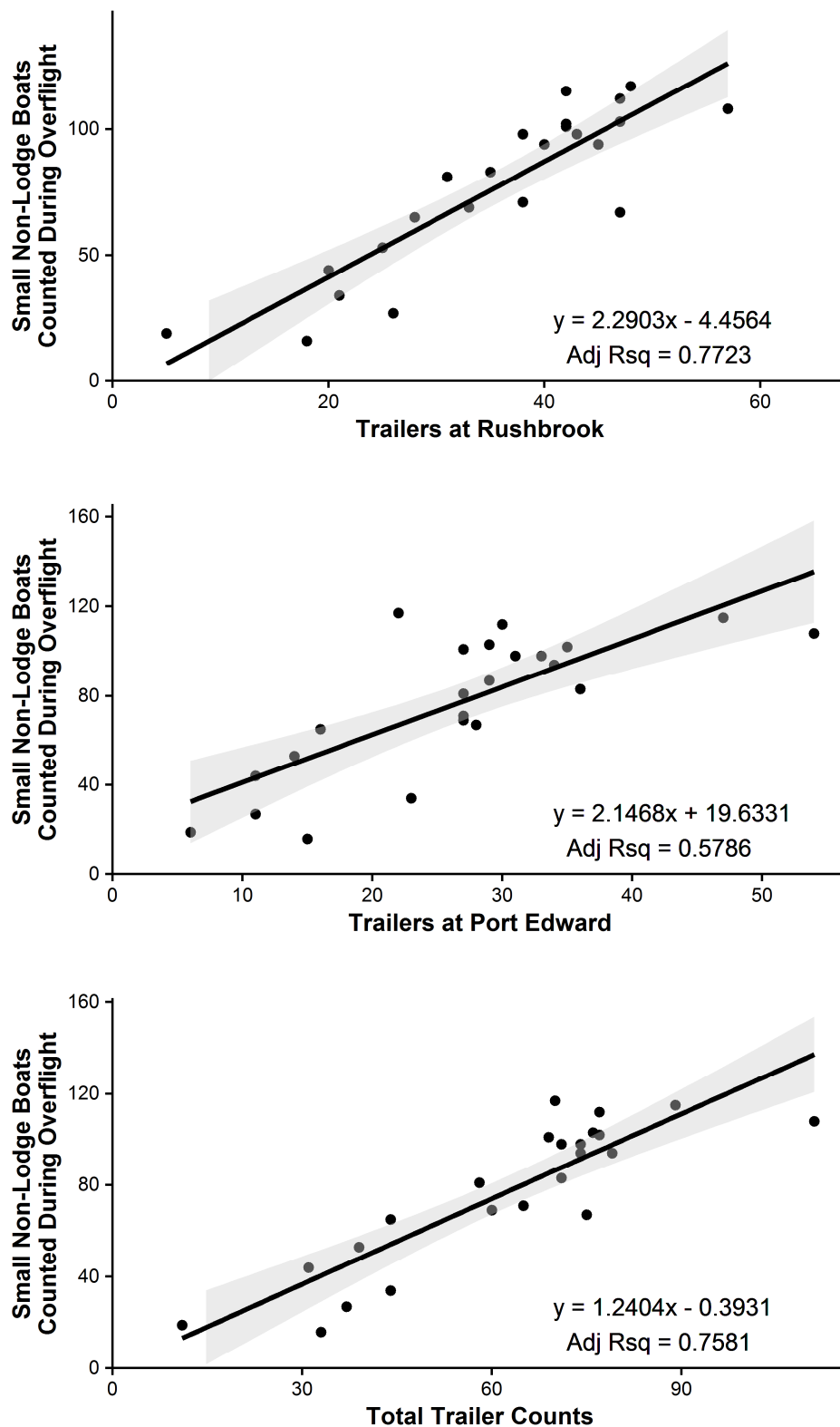
**Table 12. Total numbers of lodge boats that were visible to observers during each of the overflight surveys in 2016.**

| <i>Lodge Boats</i> |                          | <i>Lodge Boats</i> |                          | <i>Lodge Boats</i> |                          |
|--------------------|--------------------------|--------------------|--------------------------|--------------------|--------------------------|
| <i>Flight Date</i> | <i>Visible to Flight</i> | <i>Flight Date</i> | <i>Visible to Flight</i> | <i>Flight Date</i> | <i>Visible to Flight</i> |
| 1 Jun              | 6                        | 4 Jul              | 4                        | 2 Aug              | 12                       |
| 4 Jun              | 8                        | 7 Jul              | 12                       | 6 Aug              | 11                       |
| 7 Jun              | 11                       | 10 Jul             | 12                       | 8 Aug              | 4                        |
| 11 Jun             | 12                       | 12 Jul             | 12                       | 18 Aug             | 11                       |
| 19 Jun             | 12                       | 15 Jul             | 4                        | 21 Aug             | 12                       |
| 22 Jun             | 12                       | 19 Jul             | 11                       | 28 Aug             | 8                        |
| 24 Jun             | 4                        | 27 Jul             | 11                       | 29 Aug             | 4                        |
| 27 Jun             | 4                        | 30 Jul             | 13                       | 31 Aug             | 5                        |

### **Angler Effort Estimates**

Over the three month study period, a total of 24 effort surveys were conducted, including 16 weekday surveys and 8 conducted on weekends/holidays (Table 2, Appendix 3). The numbers of lodge boats that were visible during the overflights (Table 12) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts (“the non-lodge small boat counts”) were significantly correlated to trailer counts at both survey launch sites (Rushbrooke:  $F_{1,21} = 75.6$ ,  $P < 0.0001$ ; Port Edwards:  $F_{1,22} = 32.6$ ,  $P < 0.0001$ ), and when the two launch sites were combined ( $F_{1,21} = 69.5$ ,  $P < 0.0001$ ). The fit of the Port Edward regressions ( $R^2 = 0.58$ ) was lower than that at Rushbrooke ( $R^2 = 0.77$ ) or of the combined data ( $R^2 = 0.76$ ; Figure 7).

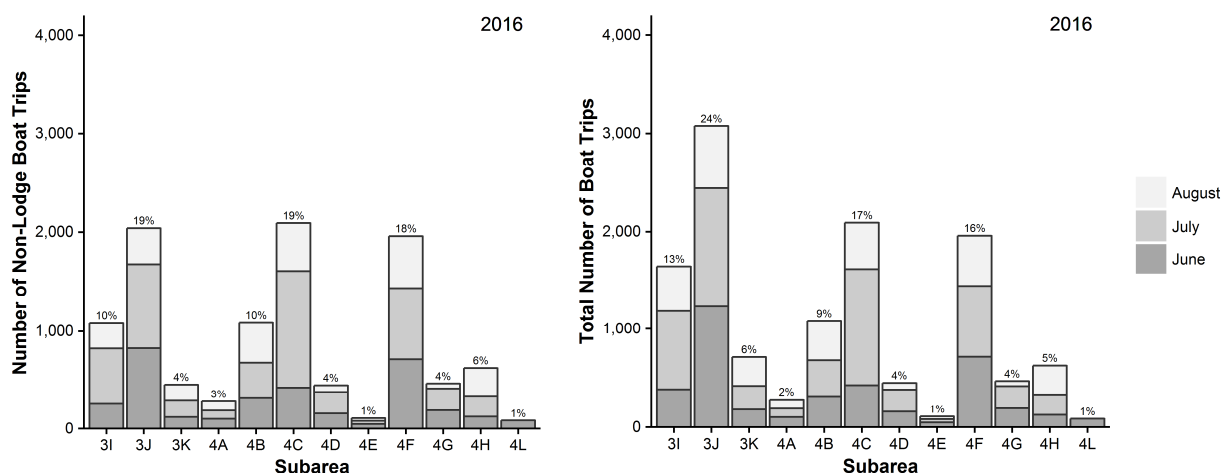
The non-lodge boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2016 (Table 13, Figure 8). Median daily non-lodge effort was 6.8 trips per day per subarea on weekdays, and was 8.1 on weekends/holidays, but the difference was not statistically significant ( $\chi^2 = 0.34$ ,  $df = 1$ ,  $P = 0.56$ ). There was also no significant effect of month (medians: Jun 5.7, July 7.9, August 6.9 trips per day per subarea;  $\chi^2 = 1.47$ ,  $df = 2$ ,  $P = 0.48$ ).



**Figure 7.** Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded area shows the 95% confidence interval of the predicted values. Note that X axis scale varies between plots.

**Table 13. Lodge, non-lodge and total effort estimates (boat trips per month), by month and subarea, 2016.**

| Month             |                | Subarea |        |       |       |       |        |       |      |       |       |       |      | TOTALS |        |         | SE of the Totals |        |         |
|-------------------|----------------|---------|--------|-------|-------|-------|--------|-------|------|-------|-------|-------|------|--------|--------|---------|------------------|--------|---------|
|                   |                | 3I      | 3J     | 3K    | 4A    | 4B    | 4C     | 4D    | 4E   | 4F    | 4G    | 4H    | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| June              | Creel Estimate | 257.4   | 827.4  | 120.4 | 101.9 | 315.3 | 425.9  | 157.6 | 47.9 | 716.2 | 189.6 | 124.9 | 83.8 | 1,205  | 2,163  | 3,368   | 8                | 9      | 12      |
|                   | Lodge Report   | 126.6   | 407.1  | 59.2  |       |       |        |       |      |       |       |       |      | 593    | 0      | 593     |                  |        |         |
|                   | TOTAL JUNE     | 384.0   | 1234.5 | 179.6 | 101.9 | 315.3 | 425.9  | 157.6 | 47.9 | 716.2 | 189.6 | 124.9 | 83.8 | 1,798  | 2,163  | 3,961   | 8                | 9      | 12      |
|                   | SE             | 2.4     | 7.9    | 1.3   | 0.8   | 2.8   | 2.3    | 1.6   | 0.5  | 7.2   | 2.3   | 1.2   | 1.1  |        |        |         |                  |        |         |
| July              | Creel Estimate | 567.2   | 849.8  | 169.4 | 85.4  | 364.8 | 1184.0 | 223.3 | 31.7 | 721.4 | 227.4 | 206.5 | 0.0  | 1,586  | 3,044  | 4,631   | 8                | 11     | 14      |
|                   | Lodge Report   | 237.4   | 355.7  | 70.9  |       |       |        |       |      |       |       |       |      | 664    | 0      | 664     |                  |        |         |
|                   | TOTAL JULY     | 804.6   | 1205.4 | 240.3 | 85.4  | 364.8 | 1184.0 | 223.3 | 31.7 | 721.4 | 227.4 | 206.5 | 0.0  | 2,250  | 3,044  | 5,295   | 8                | 11     | 14      |
|                   | SE             | 4.7     | 6.4    | 1.3   | 0.7   | 3.7   | 7.8    | 2.0   | 0.5  | 6.2   | 2.3   | 1.7   | 0.0  |        |        |         |                  |        |         |
| August            | Creel Estimate | 255.9   | 363.1  | 167.2 | 95.5  | 404.1 | 480.1  | 69.0  | 29.6 | 522.8 | 50.8  | 294.7 | 0.0  | 786    | 1,947  | 2,733   | 5                | 10     | 11      |
|                   | Lodge Report   | 192.7   | 273.4  | 125.9 |       |       |        |       |      |       |       |       |      | 592    | 0      | 592     |                  |        |         |
|                   | TOTAL AUGUST   | 448.6   | 636.5  | 293.1 | 95.5  | 404.1 | 480.1  | 69.0  | 29.6 | 522.8 | 50.8  | 294.7 | 0.0  | 1,378  | 1,947  | 3,325   | 5                | 10     | 11      |
|                   | SE             | 2.6     | 4.0    | 1.9   | 1.0   | 5.0   | 3.4    | 1.1   | 0.5  | 6.4   | 1.1   | 3.2   | 0.0  |        |        |         |                  |        |         |
| FULL SEASON TOTAL |                | 1,637   | 3,076  | 713   | 283   | 1,084 | 2,090  | 450   | 109  | 1,960 | 468   | 626   | 84   | 5,427  | 7,154  | 12,581  | 13               | 17     | 21      |
| FULL SEASON SE    |                | 6       | 11     | 3     | 1     | 7     | 9      | 3     | 1    | 11    | 3     | 4     | 1    |        |        |         |                  |        |         |



**Figure 8.** Total estimated numbers of boat trips, by month and subarea, 2016. Left panel: estimated monthly non-lodge boat trips. Right panel: total effort, calculated as the sum of lodge and non-lodge boat trips.

Total lodge effort in 2016 was 593, 664, and 592 boat trips in June, July and August, respectively (Table 13). To calculate total (lodge + non-lodge) fishing effort by subarea, we distributed the lodge boat trips among the three subareas in Management Area 3, according to the relative non-lodge effort in each (Table 13; Figure 8). In all, there were 3,961 (SE = 12) boat trips in June, 5,295 (SE = 14) boat trips in July, and 3,325 (SE = 11) boat trips in August. The full-season Area 3/4 effort estimate for 2016 was 12,581 boat trips (SE = 21).

### Catch Estimates

Monthly and full-season estimates were generated of the numbers of fish released and retained, by species (Tables 14-23; Appendix 4). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 10,043 (SE = 287) Chinook Salmon were retained by anglers in Areas 3 or 4 during 2016 the study period (Table 15; Figure 9). The largest proportional harvest came from subarea 3J (41%), followed by 4F (15%), and 3I (11%). Harvest declined over the study period: it was significantly higher in June (57% of harvest) and July (35% of harvest) than in August (9% of harvest;  $\chi^2 = 10.9$ ,  $df = 2$ ,  $P = 0.004$ ; Table 24). During the study, the number of released Chinook Salmon was estimated at 2,190 (SE = 178), or 18% of the total catch.

The species that was harvested in the greatest numbers during the 2016 study period was, by far, Coho Salmon (37,350 fish retained, SE = 842; Table 16; Figure 9). The largest proportional harvest came from subareas 3J (24%) and 4F (20%), followed by 4C (16%) and 3I (13%). In June, 12% of the total Coho Salmon harvest was recorded, significantly less than in July (58% of the total harvest) or August (30% of the total harvest;  $\chi^2 = 12.0$ ,  $df = 2$ ,  $P = 0.002$ ; Table 24). During the study period, an estimated 4,716 (SE = 540) Coho Salmon were released by anglers (11% of the total catch).

**Table 14. Total (lodge + non-lodge) estimated harvest, by month and subarea, for popular sportfish taxa, 2016.**

| Species            | Subarea |       |       |    |       |       |       |     |       |       |       |     | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|-------|----|-------|-------|-------|-----|-------|-------|-------|-----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K    | 4A | 4B    | 4C    | 4D    | 4E  | 4F    | 4G    | 4H    | 4L  | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| Chinook Salmon     | 1,136   | 4,143 | 637   | 2  | 458   | 840   | 254   | 71  | 1,494 | 319   | 689   | 0   | 5,916  | 4,127  | 10,043  | 249              | 143    | 287     |
| Coho Salmon        | 4,764   | 9,049 | 1,622 | 33 | 2,768 | 5,891 | 1,323 | 313 | 7,360 | 1,632 | 2,595 | 0   | 15,435 | 21,915 | 37,350  | 693              | 478    | 842     |
| Pink Salmon        | 311     | 556   | 203   | 4  | 166   | 878   | 159   | 5   | 732   | 98    | 224   | 0   | 1,070  | 2,266  | 3,336   | 203              | 125    | 238     |
| Chum Salmon        | 78      | 209   | 37    | 0  | 19    | 52    | 2     | 0   | 9     | 14    | 21    | 0   | 325    | 118    | 442     | 43               | 23     | 49      |
| Sockeye Salmon     | 12      | 0     | 1     | 0  | 4     | 6     | 9     | 0   | 14    | 0     | 9     | 0   | 14     | 42     | 56      | 6                | 10     | 12      |
| Pacific Halibut    | 1,807   | 4,172 | 284   | 7  | 744   | 1,849 | 325   | 213 | 2,509 | 619   | 956   | 285 | 6,263  | 7,508  | 13,771  | 295              | 190    | 350     |
| Lingcod            | 225     | 629   | 30    | 0  | 29    | 91    | 9     | 10  | 472   | 152   | 71    | 84  | 884    | 919    | 1,803   | 116              | 105    | 157     |
| other groundfish * | 137     | 41    | 17    | 2  | 35    | 128   | 2     | 0   | 172   | 14    | 12    | 8   | 195    | 373    | 568     | 82               | 54     | 98      |
| Rockfish †         | 358     | 1,127 | 385   | 18 | 343   | 344   | 127   | 32  | 1,066 | 276   | 172   | 42  | 1,869  | 2,422  | 4,291   | 221              | 185    | 289     |

\* includes Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and unknown groundfish

† includes Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

**Table 15. Total (lodge + non-lodge) estimated Chinook Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

| Month / Catch Type | Subarea |        |       |     |       |       |       |      |       |       |       |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|--------|-------|-----|-------|-------|-------|------|-------|-------|-------|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J     | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| June               |         |        |       |     |       |       |       |      |       |       |       |    |        |        |         |                  |        |         |
| Retained           | 826.0   | 2637.2 | 385.9 | 1.8 | 188.6 | 240.3 | 103.4 | 61.0 | 785.9 | 161.5 | 286.7 | 0  | 3,849  | 1,829  | 5,678   | 178              | 91     | 199     |
| Released           | 208.7   | 293.5  | 64.5  | 1.8 | 76.0  | 54.6  | 39.4  | 0    | 190.7 | 7.0   | 38.7  | 0  | 567    | 408    | 975     | 90               | 43     | 100     |
| July               |         |        |       |     |       |       |       |      |       |       |       |    |        |        |         |                  |        |         |
| Retained           | 256.4   | 1278.3 | 163.8 | 0   | 202.7 | 529.5 | 136.9 | 4.5  | 515.9 | 133.6 | 252.3 | 0  | 1,699  | 1,775  | 3,474   | 157              | 102    | 187     |
| Released           | 114.7   | 268.2  | 10.7  | 0   | 22.5  | 47.6  | 36.0  | 0    | 91.8  | 68.6  | 49.7  | 0  | 394    | 316    | 710     | 98               | 40     | 106     |
| August             |         |        |       |     |       |       |       |      |       |       |       |    |        |        |         |                  |        |         |
| Retained           | 53.4    | 227.4  | 87.3  | 0   | 66.6  | 70.5  | 13.4  | 5.9  | 192.4 | 24.1  | 149.7 | 0  | 368    | 523    | 891     | 76               | 42     | 87      |
| Released           | 17.8    | 227.1  | 0     | 0   | 4.4   | 63.8  | 3.8   | 0    | 69.1  | 21.4  | 98.2  | 0  | 245    | 261    | 506     | 90               | 50     | 103     |
| TOTAL              |         |        |       |     |       |       |       |      |       |       |       |    |        |        |         |                  |        |         |
| Retained           | 1,136   | 4,143  | 637   | 2   | 458   | 840   | 254   | 71   | 1,494 | 319   | 689   | 0  | 5,916  | 4,127  | 10,043  | 249              | 143    | 287     |
| Released           | 341     | 789    | 75    | 2   | 103   | 166   | 79    | 0    | 352   | 97    | 187   | 0  | 1,205  | 985    | 2,190   | 161              | 77     | 178     |
| Catch              | 1,477   | 4,932  | 712   | 4   | 561   | 1,006 | 333   | 71   | 1,846 | 416   | 875   | 0  | 7,121  | 5,112  | 12,233  | 297              | 162    | 338     |



**Table 16. Total (lodge + non-lodge) estimated Coho Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

| Month / Catch Type | Subarea |        |       |      |        |        |        |       |        |        |        |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|--------|-------|------|--------|--------|--------|-------|--------|--------|--------|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J     | 3K    | 4A   | 4B     | 4C     | 4D     | 4E    | 4F     | 4G     | 4H     | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <i>June</i>        |         |        |       |      |        |        |        |       |        |        |        |    |        |        |         |                  |        |         |
| Retained           | 874.2   | 1986.3 | 282.6 | 0    | 123.8  | 174.7  | 9.8    | 17.4  | 665.0  | 140.4  | 114.3  | 0  | 3,143  | 1,246  | 4,389   | 269              | 101    | 287     |
| Released           | 159.7   | 295.5  | 45.3  | 0    | 0      | 3.6    | 0      | 0     | 60.5   | 0      | 3.5    | 0  | 501    | 68     | 568     | 109              | 26     | 112     |
| <i>July</i>        |         |        |       |      |        |        |        |       |        |        |        |    |        |        |         |                  |        |         |
| Retained           | 2315.6  | 5197.9 | 482.3 | 19.9 | 1432.2 | 4188.6 | 1066.2 | 253.7 | 4218.9 | 1306.7 | 1278.9 | 0  | 7,996  | 13,765 | 21,761  | 558              | 372    | 671     |
| Released           | 992.7   | 808.0  | 0     | 8.5  | 175.6  | 339.1  | 115.3  | 4.5   | 441.6  | 198.5  | 84.1   | 0  | 1,801  | 1,367  | 3,168   | 452              | 139    | 473     |
| <i>August</i>      |         |        |       |      |        |        |        |       |        |        |        |    |        |        |         |                  |        |         |
| Retained           | 1573.8  | 1865.1 | 856.8 | 13.3 | 1212.3 | 1527.5 | 247.1  | 41.4  | 2476.0 | 184.6  | 1202.0 | 0  | 4,296  | 6,904  | 11,200  | 311              | 282    | 420     |
| Released           | 0       | 334.0  | 41.8  | 0    | 71.1   | 265.2  | 19.2   | 17.8  | 128.2  | 18.7   | 84.2   | 0  | 376    | 604    | 980     | 218              | 85     | 234     |
| <i>TOTAL</i>       |         |        |       |      |        |        |        |       |        |        |        |    |        |        |         |                  |        |         |
| Retained           | 4,764   | 9,049  | 1,622 | 33   | 2,768  | 5,891  | 1,323  | 313   | 7,360  | 1,632  | 2,595  | 0  | 15,435 | 21,915 | 37,350  | 693              | 478    | 842     |
| Released           | 1,152   | 1,437  | 87    | 9    | 247    | 608    | 134    | 22    | 630    | 217    | 172    | 0  | 2,677  | 2,039  | 4,716   | 514              | 165    | 540     |
| Catch              | 5,916   | 10,487 | 1,709 | 42   | 3,015  | 6,499  | 1,458  | 335   | 7,990  | 1,849  | 2,767  | 0  | 18,112 | 23,954 | 42,066  | 863              | 505    | 1,000   |

**Table 17. Total (lodge + non-lodge) estimated Pink Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

| Month / Catch Type | Subarea |        |       |     |       |        |       |       |        |       |       |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|--------|-------|-----|-------|--------|-------|-------|--------|-------|-------|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J     | 3K    | 4A  | 4B    | 4C     | 4D    | 4E    | 4F     | 4G    | 4H    | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <i>June</i>        |         |        |       |     |       |        |       |       |        |       |       |    |        |        |         |                  |        |         |
| Retained           | 9.1     | 36.2   | 4.2   | 0   | 0     | 3.6    | 0     | 0     | 14.0   | 14.0  | 0     | 0  | 49     | 32     | 81      | 17               | 11     | 20      |
| Released           | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 4.7    | 0     | 0     | 0  | 0      | 5      | 5       | 0                | 3      | 3       |
| <i>July</i>        |         |        |       |     |       |        |       |       |        |       |       |    |        |        |         |                  |        |         |
| Retained           | 82.1    | 395.3  | 124.0 | 0   | 103.6 | 666.4  | 144.1 | 4.5   | 520.3  | 75.8  | 116.6 | 0  | 601    | 1,631  | 2,233   | 115              | 109    | 159     |
| Released           | 1418.1  | 3914.4 | 263.5 | 8.5 | 490.9 | 1886.0 | 461.1 | 176.7 | 2242.8 | 483.7 | 487.5 | 0  | 5,596  | 6,237  | 11,833  | 891              | 381    | 969     |
| <i>August</i>      |         |        |       |     |       |        |       |       |        |       |       |    |        |        |         |                  |        |         |
| Retained           | 219.9   | 124.8  | 74.8  | 4.4 | 62.2  | 208.1  | 15.3  | 0     | 197.3  | 8.0   | 107.6 | 0  | 420    | 603    | 1,023   | 166              | 59     | 176     |
| Released           | 341.2   | 392.1  | 69.7  | 0   | 84.4  | 500.2  | 23.0  | 59.2  | 986.5  | 141.8 | 364.8 | 0  | 803    | 2,160  | 2,963   | 192              | 189    | 270     |
| <i>TOTAL</i>       |         |        |       |     |       |        |       |       |        |       |       |    |        |        |         |                  |        |         |
| Retained           | 311     | 556    | 203   | 4   | 166   | 878    | 159   | 5     | 732    | 98    | 224   | 0  | 1,070  | 2,266  | 3,336   | 203              | 125    | 238     |
| Released           | 1,759   | 4,307  | 333   | 9   | 575   | 2,386  | 484   | 236   | 3,234  | 625   | 852   | 0  | 6,399  | 8,402  | 14,801  | 911              | 426    | 1,006   |
| Catch              | 2,070   | 4,863  | 536   | 13  | 741   | 3,264  | 643   | 240   | 3,965  | 723   | 1,076 | 0  | 7,469  | 10,668 | 18,137  | 933              | 444    | 1,033   |

**Table 18. Total (lodge + non-lodge) estimated Chum Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

| Month / Catch Type | Subarea |       |      |    |     |      |     |    |     |      |     |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|------|----|-----|------|-----|----|-----|------|-----|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K   | 4A | 4B  | 4C   | 4D  | 4E | 4F  | 4G   | 4H  | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>June</b>        |         |       |      |    |     |      |     |    |     |      |     |    |        |        |         |                  |        |         |
| Retained           | 49.9    | 110.7 | 15.4 | 0  | 5.6 | 7.3  | 0   | 0  | 4.7 | 14.0 | 7.0 | 0  | 176    | 39     | 215     | 28               | 11     | 30      |
| Released           | 0       | 11.8  | 1.0  | 0  | 0   | 0    | 0   | 0  | 0   | 0    | 0   | 0  | 13     | 0      | 13      | 9                | 0      | 9       |
| <b>July</b>        |         |       |      |    |     |      |     |    |     |      |     |    |        |        |         |                  |        |         |
| Retained           | 28.6    | 98.3  | 0    | 0  | 4.5 | 41.6 | 0   | 0  | 4.4 | 0    | 9.6 | 0  | 127    | 60     | 187     | 31               | 19     | 36      |
| Released           | 0       | 13.9  | 0    | 0  | 9.0 | 0    | 0   | 0  | 4.4 | 0    | 0   | 0  | 14     | 13     | 27      | 11               | 5      | 12      |
| <b>August</b>      |         |       |      |    |     |      |     |    |     |      |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0     | 21.9 | 0  | 8.9 | 3.4  | 1.9 | 0  | 0   | 0    | 4.7 | 0  | 22     | 19     | 41      | 12               | 6      | 13      |
| Released           | 0       | 0     | 0    | 0  | 0   | 0    | 0   | 0  | 0   | 8.0  | 0   | 0  | 0      | 8      | 8       | 0                | 6      | 6       |
| <b>TOTAL</b>       |         |       |      |    |     |      |     |    |     |      |     |    |        |        |         |                  |        |         |
| Retained           | 78      | 209   | 37   | 0  | 19  | 52   | 2   | 0  | 9   | 14   | 21  | 0  | 325    | 118    | 442     | 43               | 23     | 49      |
| Released           | 0       | 26    | 1    | 0  | 9   | 0    | 0   | 0  | 4   | 8    | 0   | 0  | 27     | 21     | 48      | 14               | 8      | 16      |
| Catch              | 78      | 235   | 38   | 0  | 28  | 52   | 2   | 0  | 13  | 22   | 21  | 0  | 352    | 139    | 491     | 46               | 24     | 52      |

**Table 19. Total (lodge + non-lodge) estimated Sockeye Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

| Month / Catch Type | Subarea |    |     |    |     |     |     |    |     |    |     |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|----|-----|----|-----|-----|-----|----|-----|----|-----|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J | 3K  | 4A | 4B  | 4C  | 4D  | 4E | 4F  | 4G | 4H  | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>June</b>        |         |    |     |    |     |     |     |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 12.5    | 0  | 1.4 | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 14     | 0      | 14      | 6                | 0      | 6       |
| Released           | 0       | 0  | 0   | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| <b>July</b>        |         |    |     |    |     |     |     |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0   | 0  | 0   | 5.9 | 7.2 | 0  | 8.7 | 0  | 3.8 | 0  | 0      | 26     | 26      | 0                | 8      | 8       |
| Released           | 0       | 0  | 0   | 0  | 9.0 | 0   | 0   | 0  | 0   | 0  | 5.7 | 0  | 0      | 15     | 15      | 0                | 6      | 6       |
| <b>August</b>      |         |    |     |    |     |     |     |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0   | 0  | 4.4 | 0   | 1.9 | 0  | 4.9 | 0  | 4.7 | 0  | 0      | 16     | 16      | 0                | 6      | 6       |
| Released           | 0       | 0  | 0   | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| <b>TOTAL</b>       |         |    |     |    |     |     |     |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 12      | 0  | 1   | 0  | 4   | 6   | 9   | 0  | 14  | 0  | 9   | 0  | 14     | 42     | 56      | 6                | 10     | 12      |
| Released           | 0       | 0  | 0   | 0  | 9   | 0   | 0   | 0  | 0   | 0  | 6   | 0  | 0      | 15     | 15      | 0                | 6      | 6       |
| Catch              | 12      | 0  | 1   | 0  | 13  | 6   | 9   | 0  | 14  | 0  | 14  | 0  | 14     | 56     | 70      | 6                | 12     | 13      |

**Table 20. Total (lodge + non-lodge) estimated Pacific Halibut catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

| Month / Catch Type | Subarea |        |       |     |       |       |       |      |        |       |       |       | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|--------|-------|-----|-------|-------|-------|------|--------|-------|-------|-------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J     | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F     | 4G    | 4H    | 4L    | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <i>June</i>        |         |        |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 249.2   | 1356.8 | 183.5 | 7.3 | 118.2 | 396.8 | 137.9 | 82.7 | 762.7  | 245.8 | 205.8 | 284.9 | 1,789  | 2,242  | 4,032   | 139              | 100    | 172     |
| Released           | 62.1    | 945.6  | 95.7  | 0   | 31.0  | 127.4 | 91.1  | 13.1 | 432.5  | 217.7 | 89.7  | 301.7 | 1,103  | 1,304  | 2,408   | 254              | 173    | 307     |
| <i>July</i>        |         |        |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 1079.5  | 1742.9 | 65.4  | 0   | 319.8 | 975.7 | 136.9 | 77.0 | 1114.8 | 274.3 | 282.9 | 0     | 2,888  | 3,182  | 6,069   | 229              | 124    | 260     |
| Released           | 263.4   | 1824.9 | 0     | 0   | 31.5  | 220.1 | 43.2  | 9.1  | 682.0  | 191.3 | 156.8 | 0     | 2,088  | 1,334  | 3,422   | 344              | 151    | 375     |
| <i>August</i>      |         |        |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 477.8   | 1072.2 | 35.5  | 0   | 306.4 | 476.7 | 49.8  | 53.3 | 631.3  | 99.0  | 467.7 | 0     | 1,585  | 2,084  | 3,670   | 122              | 103    | 160     |
| Released           | 119.4   | 755.2  | 0     | 0   | 75.5  | 248.4 | 9.6   | 17.8 | 276.2  | 21.4  | 121.6 | 0     | 875    | 770    | 1,645   | 245              | 116    | 271     |
| <i>TOTAL</i>       |         |        |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 1,807   | 4,172  | 284   | 7   | 744   | 1,849 | 325   | 213  | 2,509  | 619   | 956   | 285   | 6,263  | 7,508  | 13,771  | 295              | 190    | 350     |
| Released           | 445     | 3,526  | 96    | 0   | 138   | 596   | 144   | 40   | 1,391  | 430   | 368   | 302   | 4,066  | 3,409  | 7,475   | 492              | 257    | 556     |
| Catch              | 2,251   | 7,698  | 380   | 7   | 882   | 2,445 | 468   | 253  | 3,900  | 1,049 | 1,324 | 587   | 10,329 | 10,916 | 21,245  | 574              | 320    | 657     |

**Table 21. Total (lodge + non-lodge) estimated Lingcod catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016.**

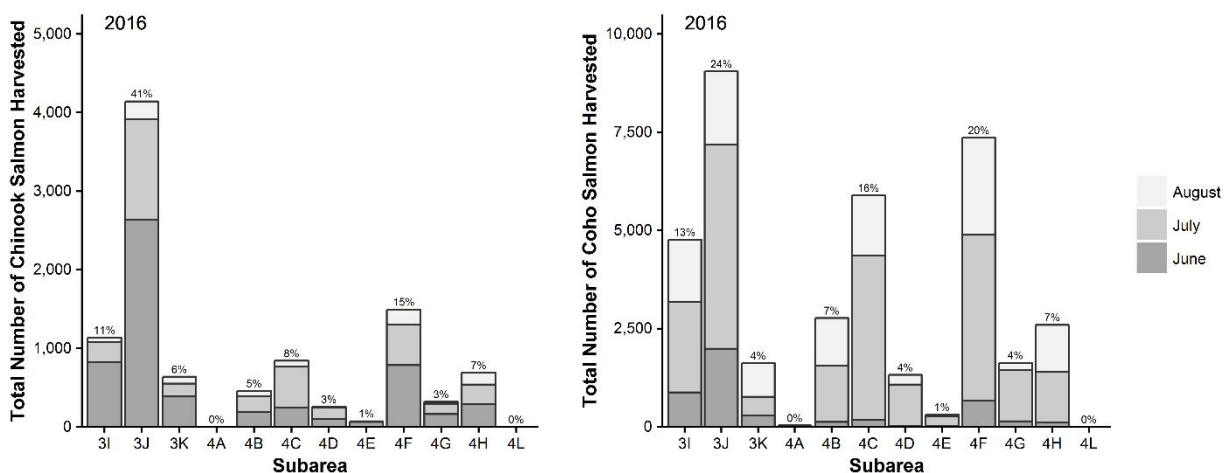
| Month / Catch Type | Subarea |       |      |    |      |      |     |     |       |      |      |      | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|------|----|------|------|-----|-----|-------|------|------|------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K   | 4A | 4B   | 4C   | 4D  | 4E  | 4F    | 4G   | 4H   | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <i>June</i>        |         |       |      |    |      |      |     |     |       |      |      |      |        |        |         |                  |        |         |
| Retained           | 0       | 198.3 | 30.2 | 0  | 11.3 | 40.0 | 0   | 0   | 255.8 | 98.3 | 14.1 | 83.8 | 228    | 503    | 732     | 23               | 93     | 95      |
| Released           | 0       | 0     | 0    | 0  | 0    | 3.6  | 2.5 | 0   | 37.2  | 28.1 | 0    | 0    | 0      | 71     | 71      | 0                | 26     | 26      |
| <i>July</i>        |         |       |      |    |      |      |     |     |       |      |      |      |        |        |         |                  |        |         |
| Retained           | 101.4   | 325.4 | 0    | 0  | 0    | 47.6 | 7.2 | 4.5 | 122.4 | 54.1 | 38.2 | 0    | 427    | 274    | 701     | 108              | 40     | 115     |
| Released           | 0       | 195.0 | 0    | 0  | 0    | 0    | 0   | 0   | 0     | 0    | 1.9  | 0    | 195    | 2      | 197     | 109              | 1      | 109     |
| <i>August</i>      |         |       |      |    |      |      |     |     |       |      |      |      |        |        |         |                  |        |         |
| Retained           | 123.6   | 105.2 | 0    | 0  | 17.8 | 3.4  | 1.9 | 5.9 | 93.7  | 0    | 18.7 | 0    | 229    | 141    | 370     | 36               | 31     | 48      |
| Released           | 0       | 0     | 0    | 0  | 4.4  | 0    | 5.7 | 0   | 0     | 2.7  | 0    | 0    | 0      | 13     | 13      | 0                | 6      | 6       |
| <i>TOTAL</i>       |         |       |      |    |      |      |     |     |       |      |      |      |        |        |         |                  |        |         |
| Retained           | 225     | 629   | 30   | 0  | 29   | 91   | 9   | 10  | 472   | 152  | 71   | 84   | 884    | 919    | 1,803   | 116              | 105    | 157     |
| Released           | 0       | 195   | 0    | 0  | 4    | 4    | 8   | 0   | 37    | 31   | 2    | 0    | 195    | 86     | 281     | 109              | 26     | 112     |
| Catch              | 225     | 824   | 30   | 0  | 33   | 95   | 17  | 10  | 509   | 183  | 73   | 84   | 1,079  | 1,005  | 2,084   | 159              | 109    | 193     |

**Table 22. Total (lodge + non-lodge) estimated catch (retained and released) of other groundfish, by month and subarea, shown separately for released and retained fish, 2016. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.**

| Month / Catch Type | Subarea |      |       |     |      |      |     |    |       |      |      |     | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|------|-------|-----|------|------|-----|----|-------|------|------|-----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J   | 3K    | 4A  | 4B   | 4C   | 4D  | 4E | 4F    | 4G   | 4H   | 4L  | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>June</b>        |         |      |       |     |      |      |     |    |       |      |      |     |        |        |         |                  |        |         |
| Retained           | 17.7    | 11.8 | 3.1   | 1.8 | 16.9 | 29.1 | 2.5 | 0  | 125.6 | 7.0  | 7.0  | 8.4 | 33     | 198    | 231     | 16               | 41     | 43      |
| Released           | 8.9     | 11.8 | 2.1   | 0   | 11.3 | 7.3  | 9.8 | 0  | 134.9 | 21.1 | 0    | 0   | 23     | 184    | 207     | 11               | 53     | 55      |
| <b>July</b>        |         |      |       |     |      |      |     |    |       |      |      |     |        |        |         |                  |        |         |
| Retained           | 0       | 0    | 0     | 0   | 0    | 89.2 | 0   | 0  | 26.2  | 7.2  | 0    | 0   | 0      | 123    | 123     | 0                | 32     | 32      |
| Released           | 20.3    | 27.9 | 0     | 0   | 49.5 | 53.5 | 0   | 0  | 100.6 | 50.5 | 0    | 0   | 48     | 254    | 302     | 26               | 70     | 75      |
| <b>August</b>      |         |      |       |     |      |      |     |    |       |      |      |     |        |        |         |                  |        |         |
| Retained           | 119.4   | 29.0 | 13.9  | 0   | 17.8 | 10.1 | 0   | 0  | 19.7  | 0    | 4.7  | 0   | 162    | 52     | 215     | 81               | 16     | 82      |
| Released           | 0       | 0    | 111.4 | 2.2 | 22.2 | 47.0 | 0   | 0  | 34.5  | 0    | 28.1 | 0   | 111    | 134    | 245     | 69               | 29     | 75      |
| <b>TOTAL</b>       |         |      |       |     |      |      |     |    |       |      |      |     |        |        |         |                  |        |         |
| Retained           | 137     | 41   | 17    | 2   | 35   | 128  | 2   | 0  | 172   | 14   | 12   | 8   | 195    | 373    | 568     | 82               | 54     | 98      |
| Released           | 29      | 40   | 114   | 2   | 83   | 108  | 10  | 0  | 270   | 72   | 28   | 0   | 182    | 573    | 755     | 75               | 93     | 119     |
| Catch              | 166     | 81   | 131   | 4   | 118  | 236  | 12  | 0  | 441   | 86   | 40   | 8   | 377    | 946    | 1,323   | 111              | 107    | 0       |

**Table 23. Total (lodge + non-lodge) estimated rockfish catch (retained and released), by month and subarea, shown separately for released and retained fish, 2016. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.**

| Month / Catch Type | Subarea |       |       |      |       |       |      |      |       |       |      |      | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|-------|------|-------|-------|------|------|-------|-------|------|------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K    | 4A   | 4B    | 4C    | 4D   | 4E   | 4F    | 4G    | 4H   | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>June</b>        |         |       |       |      |       |       |      |      |       |       |      |      |        |        |         |                  |        |         |
| Retained           | 0       | 269.5 | 33.9  | 7.3  | 123.8 | 138.3 | 36.9 | 21.8 | 558.1 | 147.5 | 3.5  | 41.9 | 303    | 1,079  | 1,383   | 25               | 113    | 115     |
| Released           | 0       | 35.5  | 3.1   | 0    | 101.3 | 94.6  | 32.0 | 0    | 237.2 | 49.2  | 3.5  | 8.4  | 39     | 526    | 565     | 21               | 95     | 98      |
| <b>July</b>        |         |       |       |      |       |       |      |      |       |       |      |      |        |        |         |                  |        |         |
| Retained           | 289.8   | 639.4 | 0     | 0    | 90.1  | 172.5 | 82.8 | 4.5  | 306.0 | 126.3 | 84.1 | 0    | 929    | 866    | 1,796   | 186              | 126    | 225     |
| Released           | 60.8    | 41.8  | 0     | 0    | 18.0  | 77.3  | 7.2  | 0    | 65.6  | 25.3  | 11.5 | 0    | 103    | 205    | 307     | 54               | 43     | 68      |
| <b>August</b>      |         |       |       |      |       |       |      |      |       |       |      |      |        |        |         |                  |        |         |
| Retained           | 68.2    | 217.8 | 350.7 | 11.1 | 128.8 | 33.6  | 7.7  | 5.9  | 202.2 | 2.7   | 84.2 | 0    | 637    | 476    | 1,113   | 117              | 75     | 139     |
| Released           | 0       | 0     | 0     | 4.4  | 31.1  | 47.0  | 11.5 | 0    | 44.4  | 0     | 9.4  | 0    | 0      | 148    | 148     | 0                | 31     | 31      |
| <b>TOTAL</b>       |         |       |       |      |       |       |      |      |       |       |      |      |        |        |         |                  |        |         |
| Retained           | 358     | 1,127 | 385   | 18   | 343   | 344   | 127  | 32   | 1,066 | 276   | 172  | 42   | 1,869  | 2,422  | 4,291   | 221              | 185    | 289     |
| Released           | 61      | 77    | 3     | 4    | 150   | 219   | 51   | 0    | 347   | 74    | 24   | 8    | 141    | 879    | 1,020   | 58               | 109    | 123     |
| Catch              | 419     | 1,204 | 388   | 23   | 493   | 563   | 178  | 32   | 1,413 | 351   | 196  | 50   | 2,010  | 3,301  | 5,311   | 229              | 215    | 314     |



**Figure 9. Total (lodge + non-lodge) estimated Chinook and Coho salmon harvest, by month and subarea, 2016. Note that Y axis scale varies between plots.**

Pink salmon was the least likely salmonid to be retained in 2016 (Table 17). We estimate that 18,137 (SE = 1,033) fish were caught, of which 18% were retained (3,336 fish, SE = 238), and 82% were released (14,801 fish, SE = 1,006). Harvest of Pink Salmon was significantly higher in July (67% of total) and August (31% of total) than it was in June (2% of total;  $\chi^2 = 20.6$ , df = 2,  $P < 0.0001$ ; Table 24). Very few Chum (442 fish, SE = 49) or Sockeye (56 fish, SE = 12) salmon were harvested in Area 3 and 4 during the study period (Tables 18 and 19).

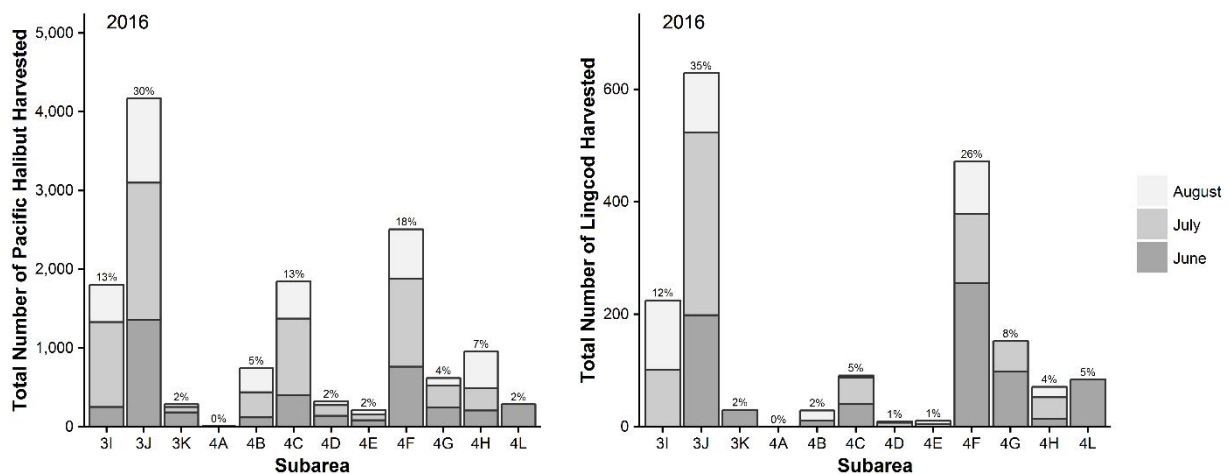
An estimated 13,771 (SE = 350) Pacific Halibut were harvested in Areas 3 or 4 during the 2016 study period (Table 20; Figure 10). Most of the harvest occurred in subarea 3J (30%), followed by 4F (18%), 3I (13%) and 4C (13%). Harvest varied among months, with 29% coming in June, peaking at 44% in July, and returning to 27% in August, though the differences were not statistically significant ( $\chi^2 = 0.8$ , df = 2,  $P = 0.68$ ; Table 24). We estimated that 7,475 (SE = 556) Pacific Halibut were released (35% of the total catch).

From June to August 2016, an estimated 1,803 (SE = 157) Lingcod were harvested, with 39-41% in June and July, and 21% in August (Table 21; Figure 10). The harvest occurred largely in subarea 3J (35%), 4F (26%), and 3I (12%). Of the captured Lingcod, 13% were released.

During the study period, an estimated total of 568 other groundfish (SE = 98) were retained (Table 22), including Pacific cod (*Gadus macrocephalus*), Sablefish (*Anoplopoma fimbria*), Ratfish (*Hydrolagus coliei*), Cabezon (*Scorpaenichthys marmoratus*), and various species of greenlings, dogfish, skates, flounders, soles, and others (755 fish, SE = 119, were released). Rockfish harvest was estimated at 4,291 fish (SE = 289; Table 23), including Black (*Sebastes melanops*), Canary (*S. pinniger*), China (*S. nebulosus*), Copper (*S. caurinus*), Quillback (*S. maliger*), Redstripe (*S. proriger*), Tiger, Yelloweye (*S. ruberrimus*), Yellowtail (*S. flavidus*) and other rockfish species. In all, 1,020 rockfish (SE = 123) were released.

**Table 24. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species, 2016. Subareas treated as replicates.**

| Species             | Kruskal-Wallis Statistics |    |         | Median Monthly Harvest |       |        |
|---------------------|---------------------------|----|---------|------------------------|-------|--------|
|                     | $\chi^2$                  | df | P       | June                   | July  | August |
| Chinook Salmon      | 10.9                      | 2  | 0.004   | 80.8                   | 85.8  | 17.7   |
| Chum Salmon         | 4.8                       | 2  | 0.09    | 2.1                    | 0.9   | 0      |
| Coho Salmon         | 12.0                      | 2  | 0.002   | 63.7                   | 707.1 | 332.5  |
| Pink Salmon         | 20.6                      | 2  | <0.0001 | 0.5                    | 46.8  | 31.1   |
| Sockeye Salmon      | 2.3                       | 2  | 0.32    | 0                      | 0     | 0      |
| Arrow Flounder      | 6.1                       | 2  | 0.047   | 0                      | 0     | 0      |
| Big Skate           | 2.1                       | 2  | 0.35    | 0                      | 0     | 0      |
| Greenling           | 4.1                       | 2  | 0.13    | 0                      | 0     | 0      |
| Lingcod             | 1.1                       | 2  | 0.58    | 6.5                    | 10.5  | 1.8    |
| Pacific Cod         | 2.5                       | 2  | 0.28    | 0                      | 0     | 0      |
| Pacific Halibut     | 0.8                       | 2  | 0.68    | 82.3                   | 133.4 | 91.5   |
| Ratfish             | 4.3                       | 2  | 0.12    | 0                      | 0     | 0      |
| Sablefish           | 4.4                       | 2  | 0.11    | 0                      | 0     | 0      |
| Black Rockfish      | 2.1                       | 2  | 0.35    | 0                      | 0     | 0      |
| Canary Rockfish     | 1.2                       | 2  | 0.56    | 0                      | 0     | 0      |
| China Rockfish      | 1.7                       | 2  | 0.42    | 0                      | 0     | 0      |
| Copper Rockfish     | 5.2                       | 2  | 0.07    | 0                      | 0     | 0      |
| Quillback Rockfish  | 5.9                       | 2  | 0.05    | 2.2                    | 1.5   | 0      |
| Redstripe Rockfish  | 0.0                       | 2  | 1.00    | 0                      | 0     | 0      |
| Tiger Rockfish      | 12.2                      | 2  | 0.002   | 0                      | 0     | 0      |
| Yelloweye Rockfish  | 1.3                       | 2  | 0.51    | 0.7                    | 4.2   | 1.7    |
| Yellowtail Rockfish | 4.1                       | 2  | 0.13    | 0                      | 0     | 0      |

**Figure 10. Total (lodge + non-lodge) estimated Pacific Halibut and Lingcod harvest, by month and subarea. Note that Y axis scale varies between plots.**

## **Biosampling**

During interviews, observers were told about 1,654 retained Chinook Salmon, and were permitted to biosample 630 fish. The observed sex ratio was 47:53 (male:female,  $n = 93$ ), and most fish were red-fleshed (89%; Table 25). The median fork length of biosampled Chinook Salmon ( $n = 548$ ) was 76 cm (95% of observations were between 53 and 101 cm; Figure 11). Of the biosampled Chinook Salmon, 11.4% were adipose-clipped (Appendix 5) and 9% had their heads collected and labelled for submission to the Salmon Head Recovery Program. Scale samples were obtained from 78% of the biosampled Chinook Salmon (Table 25).

For Coho Salmon, observers were told about 6,782 retained fish, and were permitted to biosample 1,718 fish. The observed sex ratio was 55:45 (male:female;  $n = 140$ ; Table 25). The median Coho Salmon fork length ( $n = 1450$ ) was 66 cm (95% of observations were between 54 and 76 cm; Figure 11). Of the biosampled Coho Salmon, 1.9% were adipose-clipped (Appendix 5) of which 79% had their heads were collected and labelled for submission to the Salmon Head Recovery Program).

Of the 2,668 Pacific Halibut that anglers reported as retained, 849 were biosampled (Table 25). None of the fish were sexed, and 848 were measured. The median size was 83 cm (95% of observations were between 65 and 128 cm; Figure 11). Average net and round weights, calculated from average lengths using regression formulas, were 7.8 kg (17.2 lb) and 10.3 kg (22.7 lb) in Area 3, and 6.7 kg (14.7 lb) and 8.8 kg (19.5 lb) in Area 4, respectively (Table 26).

## **Harvested Halibut Biomass**

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 98.8 tonnes (217,918 lb), including 48.7 t (107,464 lb) in Area 3 and 50.1 t (110,454 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 130.9 t (288,604 lb) including 64.6 t (142,322 lb) in Area 3 and 66.4 t (146,283 lb) in Area 4 (Table 26)<sup>1</sup>.

## **DISCUSSION**

### **Creel Estimates**

Recreational fishing effort, salmon CPE, and salmon catch in Areas 3 & 4 have been increasing over the last 20 years (Table 27). Using published creel data collected since 1995, we found eight years for which effort data were collected in June, July and August (see Van Tongeren and Winther 2010, Robichaud et al. 2016), and limited analyses to those three months (1995, 1996,

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<sup>1</sup> There are 2204.62 pounds in a metric tonne (t), thus, the total harvested Pacific Halibut net biomass was 217,918 lb, (107,464 lb in Area 3 and 110,454 lb in Area 4); or 288,605 lb round (142,322 lb in Area 3 and 146,283 lb in Area 4).

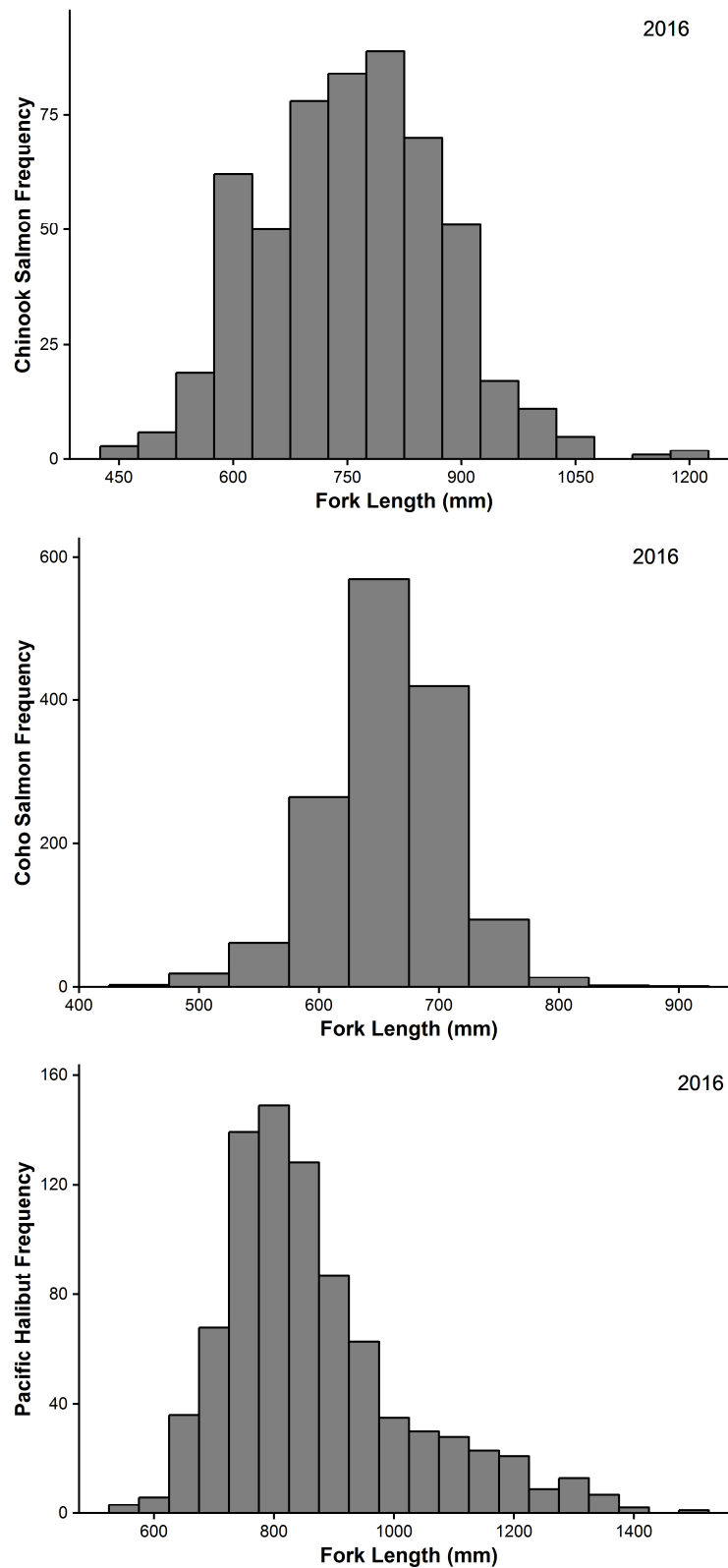
**Table 25. Biosampling results, 2016.**

|                      | Chinook Salmon |         | Coho Salmon |         | Pacific Halibut |         |
|----------------------|----------------|---------|-------------|---------|-----------------|---------|
|                      | Number         | Percent | Number      | Percent | Number          | Percent |
| Number Examined      | 630            |         | 1718        |         | 849             |         |
| <b>Subarea</b>       |                |         |             |         |                 |         |
| 3I                   | 32             | 6%      | 82          | 5%      | 34              | 4%      |
| 3J                   | 85             | 15%     | 96          | 6%      | 73              | 9%      |
| 3K                   | 4              | 1%      | 20          | 1%      |                 |         |
| 4A                   |                |         | 9           | 1%      | 1               | 0%      |
| 4B                   | 56             | 10%     | 236         | 14%     | 82              | 11%     |
| 4C                   | 74             | 13%     | 308         | 18%     | 168             | 22%     |
| 4D                   | 56             | 10%     | 238         | 14%     | 36              | 5%      |
| 4E                   | 5              | 1%      | 25          | 1%      | 17              | 2%      |
| 4F                   | 127            | 23%     | 483         | 29%     | 194             | 25%     |
| 4G                   | 23             | 4%      | 95          | 6%      | 32              | 4%      |
| 4H                   | 99             | 18%     | 71          | 4%      | 78              | 10%     |
| 4L                   | 1              | 0%      | 6           | 0%      | 61              | 8%      |
| <b>Sex</b>           |                |         |             |         |                 |         |
| Male                 | 44             | 47%     | 77          | 55%     |                 |         |
| Female               | 49             | 53%     | 63          | 45%     |                 |         |
| Undeterminable       | 415            |         | 1049        |         | 802             |         |
| Unknown              | 122            |         | 529         |         | 47              |         |
| <b>Trip Type</b>     |                |         |             |         |                 |         |
| Guided               | 267            | 42%     | 628         | 37%     | 340             | 40%     |
| Unguided             | 363            | 58%     | 1090        | 63%     | 509             | 60%     |
| <b>Adipose Clips</b> |                |         |             |         |                 |         |
| Clipped              | 70             | 11%     | 33          | 2%      |                 |         |
| Unmarked             | 545            | 89%     | 1682        | 98%     |                 |         |
| Unknown              | 10             |         | 3           |         |                 |         |
| <b>Head Tagged</b>   |                |         |             |         |                 |         |
| Yes                  | 57             | 9%      | 26          | 2%      |                 |         |
| No                   | 573            | 91%     | 1692        | 98%     |                 |         |
| <b>Scale Samples</b> |                |         |             |         |                 |         |
| Yes                  | 491            | 78%     |             |         |                 |         |
| No                   | 139            | 22%     |             |         |                 |         |
| <b>Flesh Colour</b>  |                |         |             |         |                 |         |
| Red                  | 179            | 89%     |             |         |                 |         |
| White                | 18             | 9%      |             |         |                 |         |
| Marble               | 5              | 2%      |             |         |                 |         |
| Unknown              |                |         |             |         |                 |         |

2001, 2002, 2008, 2009, 2015, 2016<sup>2</sup>). Meta-analysis revealed (Figure 12) that fishing effort has increased steadily (by 256%) from 4,911 trips in 1995 to 12,581 boat trips in 2016 (an average increase of 321 trips per year). Over the same period, Chinook Salmon harvest has increased six-fold from 1,657 in 1995 to > 10,000 in 2015 and 2016 (an average increase of 402 Chinook Salmon per year; Table 28, Figure 12). Given that harvest has increased more rapidly than effort, it follows that CPE must have also been increasing over the past 20 years. CPE is

<sup>2</sup> Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

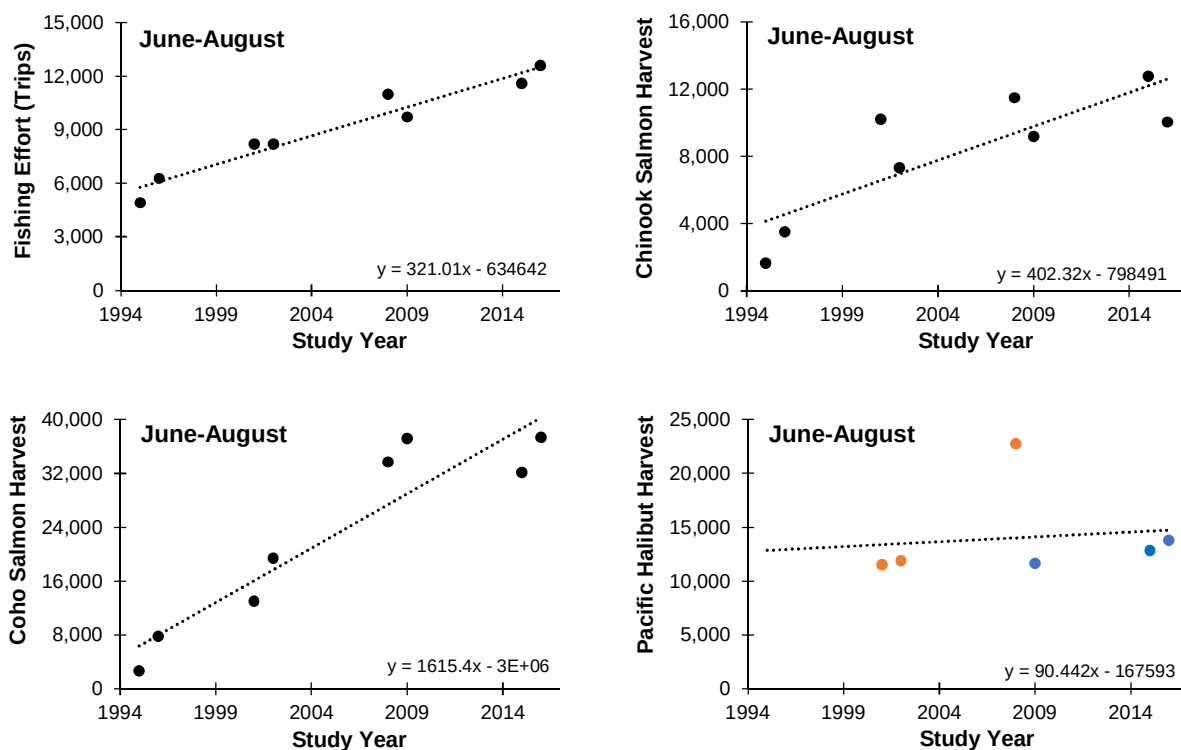




**Figure 11. Length-frequency distributions of biosampled Chinook (n = 548) and Coho (n = 1450) Salmon, and Pacific Halibut (n = 848), 2016. Note that X and Y axis scales vary between plots.**

**Table 26. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2016.**

|                                                     | Stat Area 3 |         |              |         | Stat Area 4 |         |              |         |
|-----------------------------------------------------|-------------|---------|--------------|---------|-------------|---------|--------------|---------|
|                                                     | Net Weight  |         | Round Weight |         | Net Weight  |         | Round Weight |         |
|                                                     | kg          | lb      | kg           | lb      | kg          | lb      | kg           | lb      |
| <i>Creel Interview (Non-Lodge) Average Weights</i>  |             |         |              |         |             |         |              |         |
| Guided                                              | 7.5         | 16.5    | 9.9          | 21.9    | 8.2         | 18.2    | 10.9         | 24.0    |
| Unguided                                            | 8.6         | 18.9    | 11.3         | 25.0    | 5.8         | 12.9    | 7.7          | 17.0    |
| Overall                                             | 7.8         | 17.2    | 10.3         | 22.7    | 6.7         | 14.7    | 8.8          | 19.5    |
| <i>Estimated Non-lodge Harvest (number of fish)</i> |             |         |              |         |             |         |              |         |
| Overall                                             |             | 5,204   |              |         |             | 7,508   |              |         |
| <i>Estimated Lodge Harvest (number of fish)</i>     |             |         |              |         |             |         |              |         |
| Overall                                             |             | 1,059   |              |         |             | 0       |              |         |
| <i>Biomass Harvested</i>                            |             |         |              |         |             |         |              |         |
| Non-lodge                                           | 40,502      | 89,292  | 53,640       | 118,256 | 50,101      | 110,454 | 66,353       | 146,283 |
| Lodge                                               | 8,243       | 18,172  | 10,916       | 24,066  | 0           | 0       | 0            | 0       |
| Overall                                             | 48,745      | 107,464 | 64,556       | 142,322 | 50,101      | 110,454 | 66,353       | 146,283 |



**Figure 12. Fishing effort, and harvest of Chinook and Coho salmon, and Pacific Halibut from 1995 to 2016. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Linear regression equations are shown in the lower right of each panel. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).**

**Table 27. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2016, as estimated using creel survey methods.**

| Month | Year  |       |      |       |      |       |       |       |      |       |      |        |        |      |      |      |      |      |        |        |
|-------|-------|-------|------|-------|------|-------|-------|-------|------|-------|------|--------|--------|------|------|------|------|------|--------|--------|
|       | 1995  | 1996  | 1997 | 1998  | 1999 | 2000  | 2001  | 2002  | 2003 | 2006  | 2007 | 2008   | 2009   | 2010 | 2011 | 2012 | 2013 | 2014 | 2015   | 2016   |
| May   | 965   | 391   | --   | --    | --   | --    | 449   | 960   | --   | --    | --   | 983    | --     | --   | --   | --   | --   | --   | --     | --     |
| June  | 1,590 | 2,490 | --   | --    | --   | --    | 3,801 | 3,463 | --   | --    | --   | 4,696  | 3,043  | --   | --   | --   | --   | --   | 3,424  | 3,961  |
| July  | 1,394 | 2,385 | --   | 1,307 | --   | 1,842 | 2,588 | 2,892 | --   | --    | --   | 3,750  | 3,804  | --   | --   | --   | --   | --   | 4,279  | 5,295  |
| Aug   | 1,927 | 1,400 | --   | 1,351 | --   | 1,327 | 1,808 | 1,843 | --   | 2,593 | --   | 2,529  | 2,865  | --   | --   | --   | --   | --   | 3,881  | 3,325  |
| Sept  | 864   | 488   | --   | 504   | --   | 954   | 540   | 676   | --   | 758   | --   | 394    | 817    | --   | --   | --   | --   | --   | --     | --     |
| TOTAL | 6,740 | 7,154 | --   | 3,162 | --   | 4,123 | 9,186 | 9,834 | --   | 3,351 | --   | 12,352 | 10,529 | --   | --   | --   | --   | --   | 11,585 | 12,581 |

**Table 28. Monthly Chinook Salmon, Coho Salmon and Pacific Halibut harvest in the Area 3 and 4 recreational fishery from 1995 to 2016, as estimated using creel survey methods.**

| Month                   | Year  |       |      |       |      |       |        |        |      |        |      |        |        |      |      |      |      |      |        |        |
|-------------------------|-------|-------|------|-------|------|-------|--------|--------|------|--------|------|--------|--------|------|------|------|------|------|--------|--------|
|                         | 1995  | 1996  | 1997 | 1998  | 1999 | 2000  | 2001   | 2002   | 2003 | 2006   | 2007 | 2008   | 2009   | 2010 | 2011 | 2012 | 2013 | 2014 | 2015   | 2016   |
| Chinook Salmon Harvest  |       |       |      |       |      |       |        |        |      |        |      |        |        |      |      |      |      |      |        |        |
| May                     | 141   | 45    | --   | --    | --   | --    | 509    | 522    | --   | --     | --   | 474    | --     | --   | --   | --   | --   | --   | --     | --     |
| June                    | 907   | 2,165 | --   | --    | --   | --    | 6,630  | 4,028  | --   | --     | --   | 7,708  | 4,797  | --   | --   | --   | --   | --   | 7,071  | 5,678  |
| July                    | 505   | 1,097 | --   | 254   | --   | 1,611 | 3,260  | 2,822  | --   | --     | --   | 3,080  | 3,873  | --   | --   | --   | --   | --   | 4,016  | 3,474  |
| Aug                     | 245   | 257   | --   | 35    | --   | 108   | 299    | 474    | --   | 1,207  | --   | 688    | 503    | --   | --   | --   | --   | --   | 1,669  | 891    |
| Sept                    | 15    | 62    | --   | 6     | --   | 93    | 61     | 59     | --   | 203    | --   | 20     | 4      | --   | --   | --   | --   | --   | --     | --     |
| TOTAL                   | 1,813 | 3,626 | --   | 295   | --   | 1,812 | 10,759 | 7,905  | --   | 1,410  | --   | 11,970 | 9,177  | --   | --   | --   | --   | --   | 12,756 | 10,043 |
| Coho Salmon Harvest     |       |       |      |       |      |       |        |        |      |        |      |        |        |      |      |      |      |      |        |        |
| May                     | 0     | 0     | --   | --    | --   | --    | 0      | 0      | --   | --     | --   | 0      | --     | --   | --   | --   | --   | --   | --     | --     |
| June                    | 335   | 474   | --   | --    | --   | --    | 1,771  | 1,332  | --   | --     | --   | 3,913  | 3,686  | --   | --   | --   | --   | --   | 4,436  | 4,389  |
| July                    | 402   | 3,565 | --   | 1,937 | --   | 568   | 4,736  | 9,798  | --   | --     | --   | 15,982 | 17,043 | --   | --   | --   | --   | --   | 15,101 | 21,761 |
| Aug                     | 1,907 | 3,771 | --   | 1,500 | --   | 2,125 | 6,488  | 8,292  | --   | 15,647 | --   | 13,791 | 16,459 | --   | --   | --   | --   | --   | 12,612 | 11,200 |
| Sept                    | 1,115 | 1,417 | --   | 79    | --   | 790   | 1,006  | 1,010  | --   | 3,215  | --   | 733    | 3,136  | --   | --   | --   | --   | --   | --     | --     |
| TOTAL                   | 3,759 | 9,227 | --   | 3,516 | --   | 3,483 | 14,001 | 20,432 | --   | 18,862 | --   | 34,419 | 40,324 | --   | --   | --   | --   | --   | 32,148 | 37,350 |
| Pacific Halibut Harvest |       |       |      |       |      |       |        |        |      |        |      |        |        |      |      |      |      |      |        |        |
| May                     | --    | --    | --   | --    | --   | --    | 281    | 467    | --   | --     | --   | 1,435  | --     | --   | --   | --   | --   | --   | --     | --     |
| June                    | --    | --    | --   | --    | --   | --    | 4,872  | 3,912  | --   | --     | --   | 8,367  | 3,029  | --   | --   | --   | --   | --   | 3,551  | 4,032  |
| July                    | --    | --    | --   | --    | --   | 3,963 | 3,535  | 4,716  | --   | --     | --   | 8,507  | 4,466  | --   | --   | --   | --   | --   | 4,096  | 6,069  |
| Aug                     | --    | --    | --   | 1,801 | --   | 2,530 | 3,085  | 3,247  | --   | 8,178  | --   | 5,865  | 4,142  | --   | --   | --   | --   | --   | 5,201  | 3,670  |
| Sept                    | --    | --    | --   | 328   | --   | 1,224 | 222    | 615    | --   | 1,812  | --   | 774    | 1,205  | --   | --   | --   | --   | --   | --     | --     |
| TOTAL                   | --    | --    | --   | 2,129 | --   | 7,717 | 11,995 | 12,957 | --   | 9,990  | --   | 24,948 | 12,842 | --   | --   | --   | --   | --   | 12,848 | 13,771 |

measured in ‘catch per boat trip’, thus CPE can increase if there are more anglers per boat, if more hours are spent fishing per boat trips, if more efficient equipment is used, or if fish abundances increase such that they are caught more readily. It is outside the scope of this report to determine whether the current salmon harvest levels are appropriate for our present-day stock abundances.

For Coho Salmon, Table 28 and Figure 12 show a steep (1413%) increase from 2,644 fish in 1995 to over 30,000 fish, yet the dramatic increase can be partly explained by the severe management restrictions in the mid-1990s. For example, if we had data showing higher harvests from before the management actions were not in effect (early 1990s), the slope would be considerably shallower. Indeed, harvest appears to have levelled-off since 2009, but it is difficult to draw firm conclusions without examining the 2010-2014 results.

Pacific Halibut harvest did not show a strong among-year trend (Figure 12). While fishing effort increased, the total harvest of Pacific Halibut in recent years (2009, 2015, 2016) is very similar to that from 2001-2002; Table 28). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 12).

### **Precision of the Results**

The precision of our Harvest estimates depended on that of its component parts; specifically, it depended on the precision of the Effort and CPE estimates (see Equation 21). In turn, the precision of either the Effort or CPE estimate depended on two factors: 1) real-world variability in the data; and 2) sampling error. When sample sizes are large, variability in the data is the main factor affecting precision. For example, catch rates (CPE) tend to follow a negative binomial distribution, where most catches are of zero fish; and the larger the catch, the rarer the event. If the fish were uniformly distributed and anglers had equal experience and ability, there would be considerably less variability in the CPE estimates. In addition to angler expertise, day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions typically result in a wide range of outcomes for any given fishing trip. This degree of variability translates into the size of the confidence limits around the CPE estimates for each species. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset, and hence larger confidence limits around the Effort estimates.

Sampling error is the other main source of estimation imprecision. As with any scientific study that involves sampling, the confidence you have in your final estimate is greater when a larger proportion of the population has been sampled. With catches expected to be widely variable, it follows that the precision of estimates drawn from a small number of samples would be low. One solution is to pool data among categories, but this is not ideal since we know a priori that catch rates differ substantially among areas and months. The other solution is to increase interviewing and survey effort. In this study, we sampled 19% of the non-lodge fishing effort, which although adequate, was down from 2015 when 26% of the non-lodge effort was surveyed. The added emphasis on biosampling in 2016 resulted in surveyors spending more time on

individual interviews and thus completed fewer interviews per day. The number of interviews per shift was 16.3 in 2015 (Robichaud et al. 2016) and 12.7 in 2016, whereas the number of biosampled fish rose by 182% from 1,753 to 3,197 during the same period. If current biosampling levels are considered good, then additional observer shifts would be required to increase the percent of effort we can survey. In terms of estimating angling effort, we performed eight aerial surveys per month and did not observe large variability in our boat count estimates (Table 13, Appendix 3). Effort surveys can be relatively expensive, given the cost of the overflights, and while there may be more cost-effective survey methods, it is doubtful that that would be more precise.

Overall, our level of survey effort, both for CPE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2016 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon and Pacific Halibut), for which harvest estimates were precise to within 4-6%. Pink Salmon estimates were precise to within 14%. Rockfish estimates were precise to within 13%. One major catch component, Lingcod, had catch estimates that were precise to within 17%, failing to meet the 'main taxa' precision target, likely due to highly variable HPE rates, especially in June. There were three other groups for which precision was less than 15%: 'other groundfish' was precise to within 34%; and those for Chum and Sockeye salmon were precise to within 22% and 41%, respectively. We did not have pre-defined precision targets for these three groups (the survey was not designed to meet any level of precision for these species).

### **Accuracy of the Results**

The accuracy the results depends on the quality of data provided by the anglers to the interviewers. Each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 957 of the 1,654 Chinook Salmon (58%) and 3,323 of the 6,782 Coho Salmon (49%) reported as kept during angler interviews were recorded as observed by the interviewer. While not every fish can be observed<sup>3</sup>, each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals. Similarly, adipose clip incidence rates are potentially inaccurate when derived from angler accounts or recollections. In 2016, the fin-clip data that we collected were those generated by the interviewers when observing or handling the fish.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the

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<sup>3</sup> Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, or boat launch traffic congestion causing tension, among other reasons.

surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled. In future years, weather data should be collected so trailer counts could be compared on good vs. bad weather days to determine if average effort should be adjusted downwards.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. The method for estimating the number of non-lodge boats fishing on aerial survey days is: 1) boats are counted during overflights, without knowledge of whether or not they are associated with a lodge; 2) the lodge reports how many boats were fishing on the date of the overflight; 3) DFO communicates with each of the lodges and uses average rates to estimate the approximate proportion of lodge vessels that were available to be counted during the overflight; and 4) the available lodge boats are subtracted from the overflight counts in order to obtain the number of non-lodge boats that were fishing. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g. vessels outside of the survey area). There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

As an alternative to overflights, fishing effort may be estimated using trailer counts as the main indicator of effort. This requires a known mathematical relationship between trailer counts and boat effort, and linear regression equations have been published to that end (Van Tongeren and Winther 2010, Robichaud et al. 2016; Figure 7). The numbers of trailers parked at boat launches can be used to estimate how many boats would have been seen by an overflight, had one occurred on that day. When relationships are tight, there would be little loss of accuracy from the trailer method as compared to overflights (the prediction would be close to correct); whereas weak relationships could produce wildly inaccurate effort estimates. A major disadvantage of the trailer count method is that it does not directly permit the allocation of effort among areas or subareas (though interview data can be used to do that indirectly if the interviewees are representative of anglers from all subareas in an unbiased way). For the 2016 dataset, we used the relationship in Figure 7 (bottom panel) to calculate June, July and August non-lodge effort estimates of 2,794 (SE = 31), 3,950 (SE = 31), and 3,101 (SE = 33) trips, respectively. Total effort was 93% of that estimated using aerial counts, and the standard error was 1.6 times larger. Effort estimates in Area 3 and subarea 4A were notably underestimated (50-70% of the aerial-based values), and those for subareas 4D and 4H were overestimated (160-193%). These discrepancies indicate an interview-bias that does not affect the standard (aerial-based) methods, but which could impact the accuracy of a method that uses interview data to partition effort among subzones. Trailer count data have the advantage of being collected every day: they are therefore not affected by 'good weather bias' (overflights do not occur on bad weather days when fewer boats are expected to be out); and they eliminate the need to calculate day-type and month-specific averages, which would be applied to all days on which no overflights occurred. Another advantage of collecting effort data every day is statistical: confidence bounds are tightened when based on thirty (daily) counts as compared to eight flights per month. Note that the trailer count method does not alleviate the 'visibility bias' (the proportion of angling boats that are missed during overflights), since the regression equations, which relate trailers to the number of observed boats, are built from overflight data which suffer from that bias. When the trailer-count-derived effort estimates were used, total harvest estimates were 8-23% lower than

those presented in Table 14, and the coefficients of variation were not markedly larger than those derived from the aerial survey method. As long as area-specific or subarea-specific estimates are not required, trailer counts can be used to provide reasonable effort and harvest estimates at a fraction of the cost of overflights.

## RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 27 and 28 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.
- Collect weather information along with trailer count data to determine the average effort reduction during foul weather, providing a potential adjustment factor for the overflight results.

## ACKNOWLEDGEMENTS

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## APPENDICES

### APPENDIX 1: DATA FORMS.

#### 2016 AREA 3/4 CREEL SURVEY AERIAL SURVEY VESSEL ENUMERATION FORM

DATE: \_\_\_\_\_

FLIGHT#: \_\_\_\_\_

SURVEYOR: \_\_\_\_\_

*Note: only survey shaded subareas when conditions permit*

| Sub Area | RIGHT FRONT COUNT     |                      |                       |                      | TOTAL COUNT |        |        |        | FLIGHT TIME |        | SEA STATE (1-4) |
|----------|-----------------------|----------------------|-----------------------|----------------------|-------------|--------|--------|--------|-------------|--------|-----------------|
|          | Small SPORT (Fishing) | Small SPORT (Moving) | Large SPORT (Fishing) | Large SPORT (Moving) | SS (F)      | SS (M) | LS (F) | LS (M) | START       | FINISH |                 |
| A        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| C        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| B        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| C        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| D        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| E        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| C        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| F        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| E        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| F        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| G        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| L        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| G        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| J        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| I        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| K        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| I        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| H        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| C        |                       |                      |                       |                      |             |        |        |        |             |        |                 |
| A        |                       |                      |                       |                      |             |        |        |        |             |        |                 |

WEATHER / SEA STATE / COMMENTS:

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SIGNATURE: \_\_\_\_\_

**SHIFT SUMMARY SHEET**

Landing Site: \_\_\_\_\_ Interviewer Name: \_\_\_\_\_

Interviewer Code: \_\_\_\_\_ Date: \_\_\_\_\_

Shift Code: \_\_\_\_\_ Shift Times: \_\_\_\_\_

| Time Blocks   |    |
|---------------|----|
| From - To     | TB |
| Before 7:00   | 1  |
| 7:00 - 7:59   | 2  |
| 8:00 - 8:59   | 3  |
| 9:00 - 9:59   | 4  |
| 10:00 - 10:59 | 5  |
| 11:00 - 11:59 | 6  |
| 12:00 - 12:59 | 7  |
| 13:00 - 13:59 | 8  |
| 14:00 - 14:59 | 9  |
| 15:00 - 15:59 | 10 |
| 16:00 - 16:59 | 11 |
| 17:00 - 17:59 | 12 |
| 18:00 - 18:59 | 13 |
| 19:00 - 19:59 | 14 |
| 20:00 - 20:59 | 15 |
| 21:00 - 21:59 | 16 |

\* Fill in using numbers only.

| Time Block | Fishing Conditions | Boats Missed | Not Fishing | Fishing |
|------------|--------------------|--------------|-------------|---------|
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |

Totals:

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

Comments: \_\_\_\_\_

|  |
|--|
|  |
|  |

Trailer Count: \_\_\_\_\_ Time: \_\_\_\_\_

Database Tally # :

(office use only)

[illegible]

## APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER BOAT TRIP) BY MONTH, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon caught per boat trip in June 2016, by trip type and subarea.

| Species / Trip Type   | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|-----------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
|                       | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| <i>Chinook Salmon</i> |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 3.25    | 2.89 | 3.09 |      | 1.57 | 2.57 | 1.04 | 1.04 | 1.10 | 2.86 | 3.74 |      |
| Unguided              | 1.71    | 1.77 | 1.68 | 0.02 | 0.45 | 0.44 | 0.66 | 1.27 | 1.10 | 0.83 | 1.18 | 1.00 |
| Combined              | 2.34    | 2.33 | 2.34 | 0.02 | 0.60 | 0.56 | 0.66 | 1.27 | 1.10 | 0.85 | 2.30 |      |
| <i>Chum Salmon</i>    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 0.17    | 0.14 | 0.13 |      | 0.13 | 0.14 |      |      |      | 0.02 | 0.03 |      |
| Unguided              | 0.06    |      | 0.02 |      |      | 0.01 |      |      | 0.01 | 0.09 | 0.08 | 0.08 |
| Combined              | 0.10    | 0.07 | 0.07 |      | 0.02 | 0.02 |      |      | 0.01 | 0.07 | 0.06 |      |
| <i>Coho Salmon</i>    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 6.00    | 2.94 | 3.20 |      | 1.13 | 1.43 | 0.65 | 0.65 | 0.75 | 1.33 | 1.52 |      |
| Unguided              | 0.76    | 1.20 | 0.98 |      | 0.26 | 0.35 | 0.07 | 0.36 | 0.96 | 0.48 | 0.45 | 0.44 |
| Combined              | 2.93    | 2.07 | 2.03 |      | 0.39 | 0.41 | 0.06 | 0.36 | 0.93 | 0.74 | 0.92 |      |
| <i>Pink Salmon</i>    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 0.08    | 0.06 | 0.05 |      |      |      | 0.09 | 0.09 | 0.10 |      |      |      |
| Unguided              |         | 0.03 | 0.02 |      |      | 0.01 |      |      | 0.01 | 0.09 |      | 0.03 |
| Combined              | 0.03    | 0.04 | 0.03 |      |      | 0.01 |      |      | 0.02 | 0.07 |      |      |
| <i>Sockeye Salmon</i> |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 0.08    |      | 0.02 |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              | 0.03    |      | 0.01 |      |      |      |      |      |      |      |      |      |

Average number of salmon caught per boat trip in July 2016, by trip type and subarea.

| Species / Trip Type   | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|-----------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
|                       | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| <i>Chinook Salmon</i> |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 0.33    | 1.04 | 0.86 |      | 1.73 | 0.62 | 1.56 | 1.15 | 1.14 | 0.85 | 1.52 | 0.40 |
| Unguided              | 0.26    | 0.88 | 0.59 |      | 0.37 | 0.42 | 0.45 | 0.46 | 0.48 | 0.41 | 0.98 | 0.14 |
| Combined              | 0.29    | 0.95 | 0.61 |      | 0.56 | 0.45 | 0.61 | 0.14 | 0.72 | 0.59 | 1.22 | 0.32 |
| <i>Chum Salmon</i>    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         | 0.07 | 0.16 |      |      | 0.04 |      |      |      |      | 0.04 |      |
| Unguided              | 0.05    | 0.09 |      |      | 0.01 | 0.03 |      | 0.01 | 0.01 |      | 0.05 |      |
| Combined              | 0.04    | 0.08 |      |      | 0.01 | 0.04 |      |      | 0.01 |      | 0.05 |      |
| <i>Coho Salmon</i>    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 2.67    | 5.21 | 5.28 |      | 8.55 | 6.50 | 9.89 | 8.62 | 8.20 | 7.85 | 8.40 | 3.87 |
| Unguided              | 3.32    | 4.18 | 1.76 | 0.30 | 3.20 | 3.09 | 3.91 | 4.33 | 4.54 | 4.27 | 4.43 | 1.86 |
| Combined              | 3.11    | 4.66 | 2.17 | 0.23 | 3.93 | 3.54 | 4.77 | 8.00 | 5.85 | 5.75 | 6.19 | 3.23 |
| <i>Pink Salmon</i>    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         | 0.54 | 0.49 |      |      | 0.81 | 1.00 | 0.41 | 0.34 | 0.31 | 0.56 |      |
| Unguided              | 0.21    | 0.39 | 0.41 |      | 0.33 | 0.53 | 0.58 | 0.80 | 0.93 | 0.35 | 0.57 | 0.29 |
| Combined              | 0.14    | 0.46 | 0.72 |      | 0.28 | 0.56 | 0.65 | 0.14 | 0.72 | 0.33 | 0.56 | 0.09 |
| <i>Sockeye Salmon</i> |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      | 0.11 | 0.04 | 0.03 |      |      |      |
| Unguided              |         |      |      |      |      | 0.01 | 0.02 | 0.01 |      |      | 0.03 |      |
| Combined              |         |      |      |      |      | 0.01 | 0.03 |      | 0.01 |      | 0.02 |      |

**Average number of salmon caught per boat trip in August 2016, by trip type and subarea.**

|                     | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Chinook Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 0.25    | 0.67 | 0.58 |      | 0.30 | 0.37 | 0.54 | 0.54 | 0.57 | 0.43 | 0.54 | 0.15 |
| Unguided            |         | 0.15 | 0.30 |      | 0.15 | 0.09 | 0.20 | 0.22 | 0.23 | 0.50 | 0.49 | 0.47 |
| Combined            | 0.13    | 0.40 | 0.33 |      | 0.16 | 0.15 | 0.19 | 0.20 | 0.37 | 0.47 | 0.51 | 0.13 |
| Chum Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      | 0.04 |      | 0.10 | 0.04 |      |      |      |      | 0.04 |      |
| Unguided            |         |      |      |      | 0.01 |      | 0.03 | 0.01 |      |      |      |      |
| Combined            |         |      | 0.08 |      | 0.02 | 0.01 | 0.03 |      |      |      | 0.02 |      |
| Coho Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 4.38    | 3.42 | 4.42 |      | 5.40 | 6.59 | 7.50 | 7.50 | 7.64 | 4.14 | 5.93 | 1.23 |
| Unguided            | 2.29    | 2.31 | 2.80 | 0.21 | 2.70 | 2.39 | 3.43 | 2.90 | 2.68 | 3.33 | 2.60 | 2.67 |
| Combined            | 3.40    | 2.84 | 2.83 | 0.14 | 3.00 | 3.18 | 3.58 | 1.40 | 4.74 | 3.63 | 4.08 | 1.07 |
| Pink Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 1.50    | 0.08 | 0.65 |      | 0.30 | 0.74 | 0.46 | 0.46 | 0.48 |      | 0.14 |      |
| Unguided            |         | 0.54 | 0.10 | 0.07 | 0.14 | 0.36 | 0.23 | 0.27 | 0.31 | 0.25 | 0.54 | 0.45 |
| Combined            | 0.80    | 0.32 | 0.42 | 0.05 | 0.15 | 0.43 | 0.22 |      | 0.38 | 0.16 | 0.37 |      |
| Sockeye Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      | 0.10 |      |      |      |      |      | 0.04 |      |
| Unguided            |         |      |      |      |      |      | 0.03 | 0.02 | 0.02 |      |      |      |
| Combined            |         |      |      |      | 0.01 |      | 0.03 |      | 0.01 |      | 0.02 |      |

**Average number of groundfish caught per boat trip in June 2016, by trip type and subarea.**

| Species / Trip Type | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
|                     | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow Flounder      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      | 0.02 | 0.03 |      |
| Unguided            |         | 0.03 | 0.02 |      |      |      |      |      |      |      |      |      |
| Combined            |         | 0.01 | 0.01 |      |      |      |      |      |      |      | 0.01 |      |
| Big Skate           |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |
| Cabezon             |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |
| Dogfish             |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      | 0.02 | 0.03 |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      | 0.01 |      |
| Greenling           |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      | 0.03 |      |      |      | 0.03 |      |      |      |
| Combined            |         |      |      |      | 0.03 |      |      |      | 0.03 |      |      |      |
| Lingcod             |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         | 0.11 | 0.09 |      |      |      | 0.65 | 0.65 | 0.75 | 0.29 | 0.10 | 1.29 |
| Unguided            |         |      | 0.03 |      | 0.04 | 0.10 |      |      | 0.30 | 0.61 | 0.13 | 0.30 |
| Combined            |         | 0.06 | 0.06 |      | 0.04 | 0.09 |      |      | 0.36 | 0.52 | 0.11 | 1.00 |
| Longnose Skate      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      | 0.01 |      |      |      |      |      |      |
| Combined            |         |      |      |      |      | 0.01 |      |      |      |      |      |      |
| Pacific Cod         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      | 0.05 | 0.06 |      |
| Unguided            |         |      |      |      | 0.03 | 0.03 |      |      | 0.03 |      |      |      |
| Combined            |         |      |      |      | 0.03 | 0.03 |      |      | 0.03 |      | 0.03 |      |
| Pacific Halibut     |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 0.83    | 2.29 | 1.95 |      | 0.65 | 1.57 | 2.26 | 2.26 | 2.40 | 2.45 | 2.26 | 4.00 |
| Unguided            | 0.76    | 0.40 | 0.63 | 0.09 | 0.35 | 0.89 | 0.85 | 1.73 | 0.87 | 1.30 | 1.18 | 1.26 |
| Combined            | 0.79    | 1.34 | 1.25 | 0.07 | 0.38 | 0.93 | 0.88 | 1.73 | 1.06 | 1.30 | 1.65 | 3.40 |
| Ratfish             |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 0.17    |      | 0.04 |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            | 0.07    |      | 0.02 |      |      |      |      |      |      |      |      |      |
| Rock Sole           |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      | 0.02 |      |      |      |      |      |      |      |      |
| Combined            |         |      |      | 0.02 |      |      |      |      |      |      |      |      |
| Sablefish           |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |
| Starry Flounder     |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      | 0.04 | 0.14 |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      | 0.01 |      |      |      |      |      |      |

**Average number of groundfish caught per boat trip in July 2016, by trip type and subarea.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow Flounder      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Big Skate           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Cabezon             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Dogfish             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Greenling           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Lingcod             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.22    | 0.46 | 0.35 |    |      | 0.08 |      | 0.17 | 0.20 | 0.27 | 0.15 | 0.07 |
| Unguided            | 0.05    | 0.03 |      |    |      | 0.03 | 0.04 | 0.12 | 0.15 | 0.22 | 0.22 |      |
| Combined            | 0.11    | 0.23 |      |    |      | 0.04 | 0.03 | 0.14 | 0.17 | 0.24 | 0.19 | 0.05 |
| Longnose Skate      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Pacific Cod         |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      | 0.08 |      |      |
| Unguided            |         |      |      |    |      | 0.03 |      | 0.02 | 0.03 |      |      |      |
| Combined            |         |      |      |    |      | 0.03 |      |      | 0.02 | 0.03 |      |      |
| Pacific Halibut     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 2.33    | 2.93 | 2.67 |    | 2.64 | 1.08 | 0.33 | 1.85 | 2.08 | 1.42 | 2.02 | 4.20 |
| Unguided            | 1.32    | 0.79 | 0.35 |    | 0.60 | 0.79 | 0.66 | 1.10 | 1.25 | 1.05 | 0.85 | 2.71 |
| Combined            | 1.64    | 1.77 | 0.33 |    | 0.88 | 0.82 | 0.61 | 2.43 | 1.55 | 1.21 | 1.37 | 3.73 |
| Ratfish             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Rock Sole           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Sablefish           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      | 0.02 | 0.03 |      |      |      |
| Combined            |         |      |      |    |      | 0.01 |      |      | 0.02 |      |      |      |
| Starry Flounder     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |

**Average number of groundfish caught per boat trip in August 2016, by trip type and subarea.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow Flounder      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    | 0.02 | 0.02 |      |      |      |      |      |      |
| Combined            |         |      |      |    | 0.02 | 0.01 |      |      |      |      |      |      |
| Big Skate           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            | 0.14    |      |      |    |      |      |      |      |      |      |      |      |
| Combined            | 0.07    |      |      |    |      |      |      |      |      |      |      |      |
| Cabezon             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Dogfish             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Greenling           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.17 | 0.08 |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         | 0.08 |      |    |      |      |      |      |      |      |      |      |
| Lingcod             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.25 | 0.12 |    | 0.20 |      | 0.09 | 0.09 | 0.07 |      | 0.04 | 0.08 |
| Unguided            | 0.43    |      |      |    | 0.02 | 0.01 | 0.03 | 0.17 | 0.26 |      | 0.09 | 0.06 |
| Combined            | 0.20    | 0.12 |      |    | 0.04 | 0.01 | 0.03 | 0.20 | 0.18 |      | 0.06 | 0.07 |
| Longnose Skate      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Pacific Cod         |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      |      |      | 0.03 | 0.02 |
| Combined            |         |      |      |    |      | 0.01 |      |      |      |      | 0.02 |      |
| Pacific Halibut     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 1.25    | 2.83 | 2.35 |    | 1.50 | 1.11 | 1.83 | 1.83 | 1.77 | 3.29 | 2.36 | 3.69 |
| Unguided            | 1.71    | 1.85 |      |    | 0.67 | 0.97 | 0.63 | 0.78 | 0.81 | 1.17 | 0.97 | 1.14 |
| Combined            | 1.47    | 2.32 | 0.17 |    | 0.76 | 0.99 | 0.72 | 1.80 | 1.21 | 1.95 | 1.59 | 3.73 |
| Ratfish             |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Rock Sole           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Sablefish           |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.75    |      | 0.23 |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      | 0.01 | 0.02 |      |      |      |
| Combined            | 0.40    |      |      |    |      |      |      |      | 0.01 |      |      |      |
| Starry Flounder     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    | 0.02 |      |      |      |      |      |      |      |
| Combined            |         |      |      |    | 0.02 |      |      |      |      |      |      |      |



**Average number of rockfish caught per boat trip in June 2016, by trip type and subarea.**

| Species / Trip Type | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
|                     | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      | 0.09 | 0.09 | 0.10 |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      | 0.10 | 0.04 |      | 0.02 |
| Combined            |         |      |      |      |      |      |      |      | 0.10 | 0.04 |      |      |
| Canary Rockfish     |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      | 0.09 | 0.09 | 0.10 |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      | 0.01 |      |      |      |
| China Rockfish      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      | 0.04 | 0.04 | 0.05 |      |      |      |
| Unguided            |         | 0.06 | 0.03 |      |      | 0.02 |      |      | 0.01 |      |      |      |
| Combined            |         | 0.03 | 0.02 |      |      | 0.02 |      |      | 0.02 |      |      |      |
| Copper Rockfish     |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      | 0.13 | 0.13 | 0.15 |      |      |      |
| Unguided            |         |      |      |      | 0.05 | 0.01 |      | 0.09 | 0.04 |      |      |      |
| Combined            |         |      |      |      | 0.04 | 0.01 |      | 0.09 | 0.05 |      |      |      |
| Dusky Rockfish      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      | 0.02 |      | 0.14 |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      | 0.10 |
| Quillback Rockfish  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      | 0.02 |      |      |      |      |      |      |      |      |      |
| Unguided            |         | 0.03 | 0.02 | 0.09 | 0.11 | 0.04 |      | 0.09 | 0.12 | 0.30 |      | 0.11 |
| Combined            |         | 0.01 | 0.02 | 0.07 | 0.11 | 0.03 |      | 0.09 | 0.10 | 0.26 |      |      |
| Redstripe Rockfish  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      | 0.04 | 0.04 | 0.05 |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      | 0.03 |      | 0.05 | 0.03 |
| Combined            |         |      |      |      |      |      |      |      | 0.03 |      | 0.03 |      |
| Tiger Rockfish      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      | 0.04 | 0.14 |      |      |      | 0.07 |      | 0.43 |
| Unguided            |         |      |      |      |      |      |      | 0.09 | 0.01 |      |      |      |
| Combined            |         |      |      |      |      | 0.01 |      | 0.09 | 0.01 |      |      | 0.30 |
| Widow Rockfish      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      | 0.01 |      |      |      |      |      |      |      |
| Combined            |         |      |      |      | 0.01 |      |      |      |      |      |      |      |
| Yelloweye Rockfish  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         | 0.03 | 0.02 |      |      |      | 0.04 | 0.04 | 0.05 | 0.05 |      |      |
| Unguided            |         |      | 0.06 |      |      | 0.05 |      | 0.18 | 0.14 | 0.04 |      | 0.02 |
| Combined            |         | 0.01 | 0.04 |      |      | 0.05 |      | 0.18 | 0.13 | 0.11 |      |      |
| Yellowtail Rockfish |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      | 0.02 |      | 0.14 |
| Unguided            |         |      |      |      | 0.04 |      |      |      |      |      |      |      |
| Combined            |         |      |      |      | 0.04 |      |      |      |      |      |      | 0.10 |

**Average number of rockfish caught per boat trip in July 2016, by trip type and subarea.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.28 | 0.34 |      |      |      |
| Unguided            |         | 0.18 |      |    |      | 0.05 |      |      |      | 0.03 |      |      |
| Combined            |         | 0.10 |      |    |      | 0.04 |      |      | 0.12 | 0.02 |      |      |
| Canary Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.08 |      | 0.01 | 0.02 |      |      |      |
| Combined            |         |      |      |    |      | 0.07 |      |      | 0.01 |      |      |      |
| China Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    | 0.36 |      |      |      |      |      |      |      |
| Unguided            | 0.05    |      |      |    | 0.03 |      |      |      |      |      |      |      |
| Combined            | 0.04    |      |      |    | 0.07 |      |      |      |      |      |      |      |
| Copper Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      | 0.04 |      |      |
| Unguided            |         |      |      |    | 0.04 |      |      |      |      |      |      |      |
| Combined            |         |      |      |    | 0.04 |      |      |      |      | 0.02 |      |      |
| Dusky Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Quillback Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      | 0.04 |      | 0.11 | 0.14 | 0.08 | 0.40 | 0.07 |
| Unguided            |         |      |      |    | 0.04 | 0.01 | 0.15 | 0.11 | 0.09 |      | 0.08 | 0.86 |
| Combined            |         |      |      |    | 0.04 | 0.02 | 0.13 |      | 0.11 | 0.03 | 0.22 | 0.32 |
| Redstripe Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.03 | 0.03 |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         |      |      |    |      | 0.01 |      |      | 0.01 |      |      |      |
| Tiger Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      | 0.01 | 0.01 |      |      |      |
| Combined            |         |      |      |    |      |      |      |      | 0.01 |      |      |      |
| Widow Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Yelloweye Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.07 | 0.05 |    | 0.09 | 0.04 |      | 0.06 | 0.07 |      | 0.04 | 0.07 |
| Unguided            | 0.11    | 0.21 |      |    | 0.01 | 0.01 |      | 0.09 | 0.12 | 0.05 | 0.03 | 1.14 |
| Combined            | 0.07    | 0.15 |      |    | 0.02 | 0.01 |      | 0.14 | 0.10 | 0.03 | 0.04 | 0.41 |
| Yellowtail Rockfish |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      | 0.02 | 0.04 |      |      |      |
| Combined            |         |      |      |    |      |      |      |      | 0.02 |      |      |      |

**Average number of rockfish caught per boat trip in August 2016, by trip type and subarea.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      | 0.11 | 0.18 |      | 0.09 | 0.06 |
| Combined            |         |      |      |    |      | 0.01 |      |      | 0.10 |      | 0.05 |      |
| Canary Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            | 0.29    |      |      |    |      |      |      |      |      |      |      |      |
| Combined            | 0.13    |      |      |    |      |      |      |      |      |      |      |      |
| China Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.67 | 0.31 |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      | 0.01 | 0.02 |      |      |      |
| Combined            |         | 0.32 |      |    |      |      |      |      | 0.01 |      |      |      |
| Copper Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    | 0.01 |      |      |      |      |      |      |      |
| Combined            |         |      |      |    | 0.01 |      |      |      |      |      |      |      |
| Dusky Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Quillback Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      | 0.04 |      |      |      | 0.14 | 0.04 |      |
| Unguided            | 0.29    |      |      |    |      | 0.02 |      |      |      |      |      |      |
| Combined            | 0.13    |      |      |    |      | 0.02 |      |      |      | 0.05 | 0.02 |      |
| Redstripe Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         | 0.23 |      |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         | 0.12 |      |    |      | 0.01 |      |      |      |      |      |      |
| Tiger Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Widow Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Yelloweye Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.33 | 0.15 |    | 0.10 |      | 0.02 | 0.02 | 0.02 |      |      | 0.08 |
| Unguided            |         |      |      |    |      | 0.03 | 0.06 | 0.07 | 0.06 |      | 0.09 | 0.06 |
| Combined            |         | 0.16 |      |    | 0.01 | 0.02 | 0.06 | 0.20 | 0.05 |      | 0.05 | 0.07 |
| Yellowtail Rockfish |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |

### APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2016. Weekend and holiday flights are shaded.

|        |             | Subarea |    |    |    |    |    |    |    |    |    |    |    |
|--------|-------------|---------|----|----|----|----|----|----|----|----|----|----|----|
| Month  | Date        | 3I      | 3J | 3K | 4A | 4B | 4C | 4D | 4E | 4F | 4G | 4H | 4I |
| June   | 1 Jun 2016  | 1       | 11 | 3  | 3  | 3  | 3  | 0  | 0  | 3  | 0  | 0  |    |
|        | 4 Jun 2016  | 10      | 8  | 9  | 2  | 10 | 16 | 6  | 1  | 26 | 3  | 1  | 5  |
|        | 7 Jun 2016  | 8       | 10 | 2  | 1  | 10 | 10 | 3  | 0  | 15 | 0  | 1  | 0  |
|        | 11 Jun 2016 | 14      | 21 | 1  | 0  | 6  | 8  | 4  | 1  | 26 | 3  | 1  | 0  |
|        | 19 Jun 2016 | 15      | 43 | 2  | 3  | 9  | 17 | 7  | 1  | 24 | 1  | 5  | 1  |
|        | 22 Jun 2016 | 8       | 44 | 8  | 2  | 8  | 10 | 6  | 1  | 8  | 6  | 7  | 4  |
|        | 24 Jun 2016 | 13      | 38 | 7  | 3  | 8  | 13 | 4  | 2  | 14 | 5  | 4  |    |
|        | 27 Jun 2016 | 3       | 28 | 0  | 5  | 7  | 9  | 2  | 3  | 29 | 16 | 5  | 2  |
| July   | 4 Jul 2016  | 21      | 50 | 7  | 1  | 2  | 18 | 5  | 0  | 28 | 5  | 6  |    |
|        | 7 Jul 2016  | 29      | 32 | 3  | 2  | 12 | 10 | 6  | 0  | 11 | 5  | 3  |    |
|        | 10 Jul 2016 | 25      | 23 | 6  | 3  | 5  | 43 | 7  | 0  | 17 | 3  | 5  | 0  |
|        | 12 Jul 2016 | 8       | 33 | 3  | 2  | 8  | 31 | 12 | 1  | 10 | 13 | 4  | 0  |
|        | 15 Jul 2016 | 7       | 15 | 11 | 2  | 8  | 28 | 1  | 0  | 4  | 1  | 8  | 0  |
|        | 19 Jul 2016 | 9       | 18 | 8  | 3  | 9  | 22 | 4  | 6  | 18 | 0  | 3  |    |
|        | 27 Jul 2016 | 16      | 17 | 6  | 2  | 4  | 20 | 7  | 0  | 24 | 5  | 4  | 0  |
|        | 30 Jul 2016 | 18      | 23 | 0  | 1  | 16 | 35 | 1  | 0  | 19 | 9  | 6  | 0  |
| August | 2 Aug 2016  | 6       | 17 | 14 | 1  | 9  | 17 | 6  | 0  | 25 | 4  | 10 | 0  |
|        | 6 Aug 2016  | 14      | 27 | 5  | 2  | 14 | 18 | 1  | 2  | 20 | 5  | 15 | 0  |
|        | 8 Aug 2016  | 8       | 12 | 8  | 4  | 5  | 13 | 0  | 0  | 14 | 0  | 7  |    |
|        | 18 Aug 2016 | 15      | 15 | 3  | 2  | 14 | 7  | 1  | 1  | 5  | 0  | 5  | 0  |
|        | 21 Aug 2016 | 9       | 13 | 0  | 7  | 18 | 13 | 2  | 2  | 8  | 0  | 11 |    |
|        | 28 Aug 2016 | 8       | 6  | 5  | 1  | 6  | 9  | 0  | 1  | 6  | 0  | 4  |    |
|        | 29 Aug 2016 | 1       | 1  | 1  | 1  | 1  | 5  | 0  | 0  | 8  | 0  | 2  |    |
|        | 31 Aug 2016 | 7       | 8  | 5  | 0  | 4  | 4  | 1  | 0  | 1  | 0  | 2  |    |

## APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA AND SPECIES.

**Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |        |       |     |       |       |       |      |       |       |       |    |        |
|--------------------|---------|--------|-------|-----|-------|-------|-------|------|-------|-------|-------|----|--------|
| Month / Catch Type | 3I      | 3J     | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L | TOTAL  |
| June               |         |        |       |     |       |       |       |      |       |       |       |    |        |
| Creel Estimate     | 603.4   | 1926.7 | 282.0 | 1.8 | 188.6 | 240.3 | 103.4 | 61.0 | 785.9 | 161.5 | 286.7 | 0  | 4,641  |
| Lodge Report       | 222.5   | 710.5  | 104.0 | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0  | 1,037  |
| JUNE RETAINED      | 826.0   | 2637.2 | 385.9 | 1.8 | 188.6 | 240.3 | 103.4 | 61.0 | 785.9 | 161.5 | 286.7 | 0  | 5,678  |
| SE                 | 63.9    | 165.8  | 3.6   | 1.5 | 28.8  | 32.1  | 13.8  | 15.4 | 59.6  | 38.9  | 29.2  | 0  | 199    |
| July               |         |        |       |     |       |       |       |      |       |       |       |    |        |
| Creel Estimate     | 162.1   | 808.0  | 103.5 | 0   | 202.7 | 529.5 | 136.9 | 4.5  | 515.9 | 133.6 | 252.3 | 0  | 2,849  |
| Lodge Report       | 94.4    | 470.4  | 60.3  | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0  | 625    |
| JULY RETAINED      | 256.4   | 1278.3 | 163.8 | 0   | 202.7 | 529.5 | 136.9 | 4.5  | 515.9 | 133.6 | 252.3 | 0  | 3,474  |
| SE                 | 40.8    | 144.3  | 47.3  | 0   | 31.8  | 61.7  | 23.4  | 4.5  | 60.6  | 19.3  | 31.3  | 0  | 187    |
| August             |         |        |       |     |       |       |       |      |       |       |       |    |        |
| Creel Estimate     | 34.1    | 145.2  | 55.7  | 0   | 66.6  | 70.5  | 13.4  | 5.9  | 192.4 | 24.1  | 149.7 | 0  | 758    |
| Lodge Report       | 19.3    | 82.2   | 31.5  | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0  | 133    |
| AUGUST RETAINED    | 53.4    | 227.4  | 87.3  | 0   | 66.6  | 70.5  | 13.4  | 5.9  | 192.4 | 24.1  | 149.7 | 0  | 891    |
| SE                 | 17.2    | 69.1   | 26.8  | 0   | 13.7  | 11.0  | 4.4   | 4.7  | 26.8  | 5.8   | 25.9  | 0  | 87     |
| TOTAL              |         |        |       |     |       |       |       |      |       |       |       |    |        |
| Creel Estimate     | 800     | 2,880  | 441   | 2   | 458   | 840   | 254   | 71   | 1,494 | 319   | 689   | 0  | 8,248  |
| Lodge Report       | 336     | 1,263  | 196   | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0  | 1,795  |
| TOTAL RETAINED     | 1,136   | 4,143  | 637   | 2   | 458   | 840   | 254   | 71   | 1,494 | 319   | 689   | 0  | 10,043 |
| SE                 | 78      | 230    | 54    | 2   | 45    | 70    | 28    | 17   | 89    | 44    | 50    | 0  | 287    |

**Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |       |      |     |      |      |      |    |       |      |      |    |       |
|--------------------|---------|-------|------|-----|------|------|------|----|-------|------|------|----|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A  | 4B   | 4C   | 4D   | 4E | 4F    | 4G   | 4H   | 4L | TOTAL |
| June               |         |       |      |     |      |      |      |    |       |      |      |    |       |
| Creel Estimate     | 159.7   | 224.6 | 49.4 | 1.8 | 76.0 | 54.6 | 39.4 | 0  | 190.7 | 7.0  | 38.7 | 0  | 842   |
| Lodge Report       | 49.0    | 68.9  | 15.1 | 0   | 0    | 0    | 0    | 0  | 0     | 0    | 0    | 0  | 133   |
| JUNE RELEASED      | 208.7   | 293.5 | 64.5 | 1.8 | 76.0 | 54.6 | 39.4 | 0  | 190.7 | 7.0  | 38.7 | 0  | 975   |
| SE                 | 65.2    | 62.7  | 0.8  | 1.5 | 17.9 | 11.7 | 10.0 | 0  | 32.7  | 6.1  | 13.1 | 0  | 100   |
| July               |         |       |      |     |      |      |      |    |       |      |      |    |       |
| Creel Estimate     | 101.3   | 236.8 | 9.4  | 0   | 22.5 | 47.6 | 36.0 | 0  | 91.8  | 68.6 | 49.7 | 0  | 664   |
| Lodge Report       | 13.4    | 31.3  | 1.2  | 0   | 0    | 0    | 0    | 0  | 0     | 0    | 0    | 0  | 46    |
| JULY RELEASED      | 114.7   | 268.2 | 10.7 | 0   | 22.5 | 47.6 | 36.0 | 0  | 91.8  | 68.6 | 49.7 | 0  | 710   |
| SE                 | 59.0    | 78.2  | 7.6  | 0   | 9.5  | 15.6 | 11.3 | 0  | 24.1  | 19.7 | 13.5 | 0  | 106   |
| August             |         |       |      |     |      |      |      |    |       |      |      |    |       |
| Creel Estimate     | 17.1    | 217.8 | 0    | 0   | 4.4  | 63.8 | 3.8  | 0  | 69.1  | 21.4 | 98.2 | 0  | 496   |
| Lodge Report       | 0.7     | 9.3   | 0    | 0   | 0    | 0    | 0    | 0  | 0     | 0    | 0    | 0  | 10    |
| AUGUST RELEASED    | 17.8    | 227.1 | 0    | 0   | 4.4  | 63.8 | 3.8  | 0  | 69.1  | 21.4 | 98.2 | 0  | 506   |
| SE                 | 12.6    | 88.7  | 0    | 0   | 3.2  | 22.0 | 2.2  | 0  | 32.4  | 11.8 | 28.9 | 0  | 103   |
| TOTAL              |         |       |      |     |      |      |      |    |       |      |      |    |       |
| Creel Estimate     | 278     | 679   | 59   | 2   | 103  | 166  | 79   | 0  | 352   | 97   | 187  | 0  | 2,001 |
| Lodge Report       | 63      | 109   | 16   | 0   | 0    | 0    | 0    | 0  | 0     | 0    | 0    | 0  | 189   |
| TOTAL RELEASED     | 341     | 789   | 75   | 2   | 103  | 166  | 79   | 0  | 352   | 97   | 187  | 0  | 2,190 |
| SE                 | 89      | 134   | 8    | 2   | 20   | 29   | 15   | 0  | 52    | 24   | 34   | 0  | 178   |

**Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |        |       |      |        |        |        |       |        |        |        |    |        |
|--------------------|---------|--------|-------|------|--------|--------|--------|-------|--------|--------|--------|----|--------|
| Month / Catch Type | 3I      | 3J     | 3K    | 4A   | 4B     | 4C     | 4D     | 4E    | 4F     | 4G     | 4H     | 4L | TOTAL  |
| June               |         |        |       |      |        |        |        |       |        |        |        |    |        |
| Creel Estimate     | 754.3   | 1713.9 | 243.9 | 0    | 123.8  | 174.7  | 9.8    | 17.4  | 665.0  | 140.4  | 114.3  | 0  | 3,958  |
| Lodge Report       | 119.9   | 272.4  | 38.8  | 0    | 0      | 0      | 0      | 0     | 0      | 0      | 0      | 0  | 431    |
| JUNE RETAINED      | 874.2   | 1986.3 | 282.6 | 0    | 123.8  | 174.7  | 9.8    | 17.4  | 665.0  | 140.4  | 114.3  | 0  | 4,389  |
| SE                 | 142.9   | 228.0  | 2.9   | 0    | 24.9   | 31.8   | 4.2    | 7.7   | 76.8   | 44.3   | 24.4   | 0  | 287    |
| July               |         |        |       |      |        |        |        |       |        |        |        |    |        |
| Creel Estimate     | 1762.5  | 3956.2 | 367.1 | 19.9 | 1432.2 | 4188.6 | 1066.2 | 253.7 | 4218.9 | 1306.7 | 1278.9 | 0  | 19,851 |
| Lodge Report       | 553.1   | 1241.6 | 115.2 | 0    | 0      | 0      | 0      | 0     | 0      | 0      | 0      | 0  | 1,910  |
| JULY RETAINED      | 2315.6  | 5197.9 | 482.3 | 19.9 | 1432.2 | 4188.6 | 1066.2 | 253.7 | 4218.9 | 1306.7 | 1278.9 | 0  | 21,761 |
| SE                 | 324.9   | 444.1  | 93.8  | 14.5 | 123.7  | 234.5  | 95.7   | 70.3  | 181.6  | 122.8  | 76.7   | 0  | 671    |
| August             |         |        |       |      |        |        |        |       |        |        |        |    |        |
| Creel Estimate     | 870.0   | 1031.1 | 473.7 | 13.3 | 1212.3 | 1527.5 | 247.1  | 41.4  | 2476.0 | 184.6  | 1202.0 | 0  | 9,279  |
| Lodge Report       | 703.8   | 834.1  | 383.1 | 0    | 0      | 0      | 0      | 0     | 0      | 0      | 0      | 0  | 1,921  |
| AUGUST RETAINED    | 1573.8  | 1865.1 | 856.8 | 13.3 | 1212.3 | 1527.5 | 247.1  | 41.4  | 2476.0 | 184.6  | 1202.0 | 0  | 11,200 |
| SE                 | 202.4   | 216.1  | 94.1  | 7.0  | 116.9  | 112.4  | 36.5   | 33.2  | 179.1  | 37.4   | 131.3  | 0  | 420    |
| TOTAL              |         |        |       |      |        |        |        |       |        |        |        |    |        |
| Creel Estimate     | 3,387   | 6,701  | 1,085 | 33   | 2,768  | 5,891  | 1,323  | 313   | 7,360  | 1,632  | 2,595  | 0  | 33,088 |
| Lodge Report       | 1,377   | 2,348  | 537   | 0    | 0      | 0      | 0      | 0     | 0      | 0      | 0      | 0  | 4,262  |
| TOTAL RETAINED     | 4,764   | 9,049  | 1,622 | 33   | 2,768  | 5,891  | 1,323  | 313   | 7,360  | 1,632  | 2,595  | 0  | 37,350 |
| SE                 | 409     | 544    | 133   | 16   | 172    | 262    | 103    | 78    | 266    | 136    | 154    | 0  | 842    |

**Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |       |      |     |       |       |       |      |       |       |      |    |       |
|--------------------|---------|-------|------|-----|-------|-------|-------|------|-------|-------|------|----|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H   | 4L | TOTAL |
| June               |         |       |      |     |       |       |       |      |       |       |      |    |       |
| Creel Estimate     | 159.7   | 295.5 | 45.3 | 0   | 0     | 3.6   | 0     | 0    | 60.5  | 0     | 3.5  | 0  | 568   |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0    | 0  | 0     |
| JUNE RELEASED      | 159.7   | 295.5 | 45.3 | 0   | 0     | 3.6   | 0     | 0    | 60.5  | 0     | 3.5  | 0  | 568   |
| SE                 | 66.5    | 85.8  | 0.8  | 0   | 0     | 2.7   | 0     | 0    | 25.4  | 0     | 2.0  | 0  | 112   |
| July               |         |       |      |     |       |       |       |      |       |       |      |    |       |
| Creel Estimate     | 992.7   | 808.0 | 0    | 8.5 | 175.6 | 339.1 | 115.3 | 4.5  | 441.6 | 198.5 | 84.1 | 0  | 3,168 |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0    | 0  | 0     |
| JULY RELEASED      | 992.7   | 808.0 | 0    | 8.5 | 175.6 | 339.1 | 115.3 | 4.5  | 441.6 | 198.5 | 84.1 | 0  | 3,168 |
| SE                 | 369.8   | 260.7 | 0    | 6.2 | 67.6  | 51.8  | 28.3  | 4.5  | 92.9  | 46.7  | 17.1 | 0  | 473   |
| August             |         |       |      |     |       |       |       |      |       |       |      |    |       |
| Creel Estimate     | 0       | 334.0 | 41.8 | 0   | 71.1  | 265.2 | 19.2  | 17.8 | 128.2 | 18.7  | 84.2 | 0  | 980   |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0    | 0  | 0     |
| AUGUST RELEASED    | 0       | 334.0 | 41.8 | 0   | 71.1  | 265.2 | 19.2  | 17.8 | 128.2 | 18.7  | 84.2 | 0  | 980   |
| SE                 | 0       | 215.1 | 35.6 | 0   | 19.4  | 64.2  | 8.3   | 14.2 | 43.5  | 9.6   | 22.0 | 0  | 234   |
| TOTAL              |         |       |      |     |       |       |       |      |       |       |      |    |       |
| Creel Estimate     | 1,152   | 1,437 | 87   | 9   | 247   | 608   | 134   | 22   | 630   | 217   | 172  | 0  | 4,716 |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0    | 0  | 0     |
| TOTAL RELEASED     | 1,152   | 1,437 | 87   | 9   | 247   | 608   | 134   | 22   | 630   | 217   | 172  | 0  | 4,716 |
| SE                 | 376     | 349   | 36   | 6   | 70    | 83    | 30    | 15   | 106   | 48    | 28   | 0  | 540   |

**Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |       |       |     |       |       |       |     |       |      |       |    |       |
|--------------------|---------|-------|-------|-----|-------|-------|-------|-----|-------|------|-------|----|-------|
| Month / Catch Type | 3I      | 3J    | 3K    | 4A  | 4B    | 4C    | 4D    | 4E  | 4F    | 4G   | 4H    | 4L | TOTAL |
| June               |         |       |       |     |       |       |       |     |       |      |       |    |       |
| Creel Estimate     | 8.9     | 35.5  | 4.1   | 0   | 0     | 3.6   | 0     | 0   | 14.0  | 14.0 | 0     | 0  | 80    |
| Lodge Report       | 0.2     | 0.7   | 0.1   | 0   | 0     | 0     | 0     | 0   | 0     | 0    | 0     | 0  | 1     |
| JUNE RETAINED      | 9.1     | 36.2  | 4.2   | 0   | 0     | 3.6   | 0     | 0   | 14.0  | 14.0 | 0     | 0  | 81    |
| SE                 | 6.4     | 16.1  | 0.1   | 0   | 0     | 2.7   | 0     | 0   | 5.7   | 8.4  | 0     | 0  | 20    |
| July               |         |       |       |     |       |       |       |     |       |      |       |    |       |
| Creel Estimate     | 81.0    | 390.1 | 122.4 | 0   | 103.6 | 666.4 | 144.1 | 4.5 | 520.3 | 75.8 | 116.6 | 0  | 2,225 |
| Lodge Report       | 1.1     | 5.3   | 1.6   | 0   | 0     | 0     | 0     | 0   | 0     | 0    | 0     | 0  | 8     |
| JULY RETAINED      | 82.1    | 395.3 | 124.0 | 0   | 103.6 | 666.4 | 144.1 | 4.5 | 520.3 | 75.8 | 116.6 | 0  | 2,233 |
| SE                 | 34.2    | 94.4  | 56.4  | 0   | 21.9  | 72.3  | 31.0  | 4.5 | 68.6  | 15.2 | 17.0  | 0  | 159   |
| August             |         |       |       |     |       |       |       |     |       |      |       |    |       |
| Creel Estimate     | 204.7   | 116.2 | 69.7  | 4.4 | 62.2  | 208.1 | 15.3  | 0   | 197.3 | 8.0  | 107.6 | 0  | 994   |
| Lodge Report       | 15.2    | 8.6   | 5.2   | 0   | 0     | 0     | 0     | 0   | 0     | 0    | 0     | 0  | 29    |
| AUGUST RETAINED    | 219.9   | 124.8 | 74.8  | 4.4 | 62.2  | 208.1 | 15.3  | 0   | 197.3 | 8.0  | 107.6 | 0  | 1,023 |
| SE                 | 151.6   | 48.2  | 47.9  | 3.1 | 14.9  | 36.9  | 6.4   | 0   | 29.4  | 4.1  | 31.6  | 0  | 176   |
| TOTAL              |         |       |       |     |       |       |       |     |       |      |       |    |       |
| Creel Estimate     | 295     | 542   | 196   | 4   | 166   | 878   | 159   | 5   | 732   | 98   | 224   | 0  | 3,298 |
| Lodge Report       | 16      | 15    | 7     | 0   | 0     | 0     | 0     | 0   | 0     | 0    | 0     | 0  | 38    |
| TOTAL RETAINED     | 311     | 556   | 203   | 4   | 166   | 878   | 159   | 5   | 732   | 98   | 224   | 0  | 3,336 |
| SE                 | 156     | 107   | 74    | 3   | 26    | 81    | 32    | 5   | 75    | 18   | 36    | 0  | 238   |

**Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |        |       |     |       |        |       |       |        |       |       |    |        |
|--------------------|---------|--------|-------|-----|-------|--------|-------|-------|--------|-------|-------|----|--------|
| Month / Catch Type | 3I      | 3J     | 3K    | 4A  | 4B    | 4C     | 4D    | 4E    | 4F     | 4G    | 4H    | 4L | TOTAL  |
| June               |         |        |       |     |       |        |       |       |        |       |       |    |        |
| Creel Estimate     | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 4.7    | 0     | 0     | 0  | 5      |
| Lodge Report       | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 0      | 0     | 0     | 0  | 0      |
| JUNE RELEASED      | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 4.7    | 0     | 0     | 0  | 5      |
| SE                 | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 3.3    | 0     | 0     | 0  | 3      |
| July               |         |        |       |     |       |        |       |       |        |       |       |    |        |
| Creel Estimate     | 1418.1  | 3914.4 | 263.5 | 8.5 | 490.9 | 1886.0 | 461.1 | 176.7 | 2242.8 | 483.7 | 487.5 | 0  | 11,833 |
| Lodge Report       | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 0      | 0     | 0     | 0  | 0      |
| JULY RELEASED      | 1418.1  | 3914.4 | 263.5 | 8.5 | 490.9 | 1886.0 | 461.1 | 176.7 | 2242.8 | 483.7 | 487.5 | 0  | 11,833 |
| SE                 | 413.3   | 779.5  | 121.3 | 6.2 | 94.3  | 233.9  | 77.2  | 99.8  | 240.0  | 74.7  | 51.1  | 0  | 969    |
| August             |         |        |       |     |       |        |       |       |        |       |       |    |        |
| Creel Estimate     | 341.2   | 392.1  | 69.7  | 0   | 84.4  | 500.2  | 23.0  | 59.2  | 986.5  | 141.8 | 364.8 | 0  | 2,963  |
| Lodge Report       | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 0      | 0     | 0     | 0  | 0      |
| AUGUST RELEASED    | 341.2   | 392.1  | 69.7  | 0   | 84.4  | 500.2  | 23.0  | 59.2  | 986.5  | 141.8 | 364.8 | 0  | 2,963  |
| SE                 | 140.4   | 122.5  | 47.9  | 0   | 22.7  | 65.8   | 10.1  | 47.5  | 136.9  | 51.5  | 85.1  | 0  | 270    |
| TOTAL              |         |        |       |     |       |        |       |       |        |       |       |    |        |
| Creel Estimate     | 1,759   | 4,307  | 333   | 9   | 575   | 2,386  | 484   | 236   | 3,234  | 625   | 852   | 0  | 14,801 |
| Lodge Report       | 0       | 0      | 0     | 0   | 0     | 0      | 0     | 0     | 0      | 0     | 0     | 0  | 0      |
| TOTAL RELEASED     | 1,759   | 4,307  | 333   | 9   | 575   | 2,386  | 484   | 236   | 3,234  | 625   | 852   | 0  | 14,801 |
| SE                 | 436     | 789    | 130   | 6   | 97    | 243    | 78    | 111   | 276    | 91    | 99    | 0  | 1,000  |

**Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |       |      |    |     |      |     |    |     |      |     |    |       |
|--------------------|---------|-------|------|----|-----|------|-----|----|-----|------|-----|----|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A | 4B  | 4C   | 4D  | 4E | 4F  | 4G   | 4H  | 4L | TOTAL |
| June               |         |       |      |    |     |      |     |    |     |      |     |    |       |
| Creel Estimate     | 26.6    | 59.1  | 8.2  | 0  | 5.6 | 7.3  | 0   | 0  | 4.7 | 14.0 | 7.0 | 0  | 133   |
| Lodge Report       | 23.2    | 51.6  | 7.2  | 0  | 0   | 0    | 0   | 0  | 0   | 0    | 0   | 0  | 82    |
| JUNE RETAINED      | 49.9    | 110.7 | 15.4 | 0  | 5.6 | 7.3  | 0   | 0  | 4.7 | 14.0 | 7.0 | 0  | 215   |
| SE                 | 14.1    | 24.4  | 0.1  | 0  | 4.2 | 3.7  | 0   | 0  | 3.3 | 8.4  | 2.8 | 0  | 30    |
| July               |         |       |      |    |     |      |     |    |     |      |     |    |       |
| Creel Estimate     | 20.3    | 69.7  | 0    | 0  | 4.5 | 41.6 | 0   | 0  | 4.4 | 0    | 9.6 | 0  | 150   |
| Lodge Report       | 8.3     | 28.7  | 0    | 0  | 0   | 0    | 0   | 0  | 0   | 0    | 0   | 0  | 37    |
| JULY RETAINED      | 28.6    | 98.3  | 0    | 0  | 4.5 | 41.6 | 0   | 0  | 4.4 | 0    | 9.6 | 0  | 187   |
| SE                 | 14.4    | 27.2  | 0    | 0  | 3.2 | 18.3 | 0   | 0  | 3.1 | 0    | 3.6 | 0  | 36    |
| August             |         |       |      |    |     |      |     |    |     |      |     |    |       |
| Creel Estimate     | 0       | 0     | 13.9 | 0  | 8.9 | 3.4  | 1.9 | 0  | 0   | 0    | 4.7 | 0  | 33    |
| Lodge Report       | 0.0     | 0.0   | 8.0  | 0  | 0   | 0    | 0   | 0  | 0   | 0    | 0   | 0  | 8     |
| AUGUST RETAINED    | 0       | 0     | 21.9 | 0  | 8.9 | 3.4  | 1.9 | 0  | 0   | 0    | 4.7 | 0  | 41    |
| SE                 | 0       | 0     | 11.9 | 0  | 4.4 | 2.4  | 1.6 | 0  | 0   | 0    | 3.3 | 0  | 13    |
| TOTAL              |         |       |      |    |     |      |     |    |     |      |     |    |       |
| Creel Estimate     | 47      | 129   | 22   | 0  | 19  | 52   | 2   | 0  | 9   | 14   | 21  | 0  | 315   |
| Lodge Report       | 32      | 80    | 15   | 0  | 0   | 0    | 0   | 0  | 0   | 0    | 0   | 0  | 127   |
| TOTAL RETAINED     | 78      | 209   | 37   | 0  | 19  | 52   | 2   | 0  | 9   | 14   | 21  | 0  | 442   |
| SE                 | 20      | 37    | 12   | 0  | 7   | 19   | 2   | 0  | 5   | 8    | 6   | 0  | 49    |

**Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |      |     |    |     |    |    |    |     |     |    |    |       |
|--------------------|---------|------|-----|----|-----|----|----|----|-----|-----|----|----|-------|
| Month / Catch Type | 3I      | 3J   | 3K  | 4A | 4B  | 4C | 4D | 4E | 4F  | 4G  | 4H | 4L | TOTAL |
| June               |         |      |     |    |     |    |    |    |     |     |    |    |       |
| Creel Estimate     | 0       | 11.8 | 1.0 | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 13    |
| Lodge Report       | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 0     |
| JUNE RELEASED      | 0       | 11.8 | 1.0 | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 13    |
| SE                 | 0       | 9.4  | 0.0 | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 9     |
| July               |         |      |     |    |     |    |    |    |     |     |    |    |       |
| Creel Estimate     | 0       | 13.9 | 0   | 0  | 9.0 | 0  | 0  | 0  | 4.4 | 0   | 0  | 0  | 27    |
| Lodge Report       | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 0     |
| JULY RELEASED      | 0       | 13.9 | 0   | 0  | 9.0 | 0  | 0  | 0  | 4.4 | 0   | 0  | 0  | 27    |
| SE                 | 0       | 10.5 | 0   | 0  | 4.5 | 0  | 0  | 0  | 3.1 | 0   | 0  | 0  | 12    |
| August             |         |      |     |    |     |    |    |    |     |     |    |    |       |
| Creel Estimate     | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 8.0 | 0  | 0  | 8     |
| Lodge Report       | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 0     |
| AUGUST RELEASED    | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 8.0 | 0  | 0  | 8     |
| SE                 | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 5.7 | 0  | 0  | 6     |
| TOTAL              |         |      |     |    |     |    |    |    |     |     |    |    |       |
| Creel Estimate     | 0       | 26   | 1   | 0  | 9   | 0  | 0  | 0  | 4   | 8   | 0  | 0  | 48    |
| Lodge Report       | 0       | 0    | 0   | 0  | 0   | 0  | 0  | 0  | 0   | 0   | 0  | 0  | 0     |
| TOTAL RELEASED     | 0       | 26   | 1   | 0  | 9   | 0  | 0  | 0  | 4   | 8   | 0  | 0  | 48    |
| SE                 | 0       | 14   | 0   | 0  | 4   | 0  | 0  | 0  | 3   | 6   | 0  | 0  | 16    |



**Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |     |     |    |     |     |     |    |     |    |     |    |       |
|--------------------|---------|-----|-----|----|-----|-----|-----|----|-----|----|-----|----|-------|
| Month / Catch Type | 3I      | 3J  | 3K  | 4A | 4B  | 4C  | 4D  | 4E | 4F  | 4G | 4H  | 4L | TOTAL |
| June               |         |     |     |    |     |     |     |    |     |    |     |    |       |
| Creel Estimate     | 8.9     | 0   | 1.0 | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 10    |
| Lodge Report       | 3.6     | 0   | 0.4 | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 4     |
| JUNE RETAINED      | 12.5    | 0   | 1.4 | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 14    |
| SE                 | 6.4     | 0   | 0.0 | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 6     |
| July               |         |     |     |    |     |     |     |    |     |    |     |    |       |
| Creel Estimate     | 0       | 0   | 0   | 0  | 0   | 5.9 | 7.2 | 0  | 8.7 | 0  | 3.8 | 0  | 26    |
| Lodge Report       | 0       | 0   | 0   | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 0     |
| JULY RETAINED      | 0       | 0   | 0   | 0  | 0   | 5.9 | 7.2 | 0  | 8.7 | 0  | 3.8 | 0  | 26    |
| SE                 | 0       | 0   | 0   | 0  | 0   | 4.2 | 3.9 | 0  | 4.4 | 0  | 2.7 | 0  | 8     |
| August             |         |     |     |    |     |     |     |    |     |    |     |    |       |
| Creel Estimate     | 0       | 0   | 0   | 0  | 4.4 | 0   | 1.9 | 0  | 4.9 | 0  | 4.7 | 0  | 16    |
| Lodge Report       | 0.0     | 0.0 | 0.0 | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 0     |
| AUGUST RETAINED    | 0       | 0   | 0   | 0  | 4.4 | 0   | 1.9 | 0  | 4.9 | 0  | 4.7 | 0  | 16    |
| SE                 | 0       | 0   | 0   | 0  | 3.2 | 0   | 1.6 | 0  | 3.7 | 0  | 3.3 | 0  | 6     |
| TOTAL              |         |     |     |    |     |     |     |    |     |    |     |    |       |
| Creel Estimate     | 9       | 0   | 1   | 0  | 4   | 6   | 9   | 0  | 14  | 0  | 9   | 0  | 52    |
| Lodge Report       | 4       | 0   | 0   | 0  | 0   | 0   | 0   | 0  | 0   | 0  | 0   | 0  | 4     |
| TOTAL RETAINED     | 12      | 0   | 1   | 0  | 4   | 6   | 9   | 0  | 14  | 0  | 9   | 0  | 56    |
| SE                 | 6       | 0   | 0   | 0  | 3   | 4   | 4   | 0  | 6   | 0  | 4   | 0  | 12    |

**Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |    |    |    |     |    |    |    |    |    |     |    |       |
|--------------------|---------|----|----|----|-----|----|----|----|----|----|-----|----|-------|
| Month / Catch Type | 3I      | 3J | 3K | 4A | 4B  | 4C | 4D | 4E | 4F | 4G | 4H  | 4L | TOTAL |
| June               |         |    |    |    |     |    |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| Lodge Report       | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| JUNE RELEASED      | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| SE                 | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| July               |         |    |    |    |     |    |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 9.0 | 0  | 0  | 0  | 0  | 0  | 5.7 | 0  | 15    |
| Lodge Report       | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| JULY RELEASED      | 0       | 0  | 0  | 0  | 9.0 | 0  | 0  | 0  | 0  | 0  | 5.7 | 0  | 15    |
| SE                 | 0       | 0  | 0  | 0  | 4.5 | 0  | 0  | 0  | 0  | 0  | 4.1 | 0  | 6     |
| August             |         |    |    |    |     |    |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| Lodge Report       | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| AUGUST RELEASED    | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| SE                 | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| TOTAL              |         |    |    |    |     |    |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 9   | 0  | 0  | 0  | 0  | 0  | 6   | 0  | 15    |
| Lodge Report       | 0       | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| TOTAL RELEASED     | 0       | 0  | 0  | 0  | 9   | 0  | 0  | 0  | 0  | 0  | 6   | 0  | 15    |
| SE                 | 0       | 0  | 0  | 0  | 4   | 0  | 0  | 0  | 0  | 0  | 4   | 0  | 6     |

**Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |        |       |     |       |       |       |      |        |       |       |       |        |
|--------------------|---------|--------|-------|-----|-------|-------|-------|------|--------|-------|-------|-------|--------|
| Month / Catch Type | 3I      | 3J     | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F     | 4G    | 4H    | 4L    | TOTAL  |
| June               |         |        |       |     |       |       |       |      |        |       |       |       |        |
| Creel Estimate     | 204.1   | 1111.1 | 150.2 | 7.3 | 118.2 | 396.8 | 137.9 | 82.7 | 762.7  | 245.8 | 205.8 | 284.9 | 3,708  |
| Lodge Report       | 45.1    | 245.7  | 33.2  | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 324    |
| JUNE RETAINED      | 249.2   | 1356.8 | 183.5 | 7.3 | 118.2 | 396.8 | 137.9 | 82.7 | 762.7  | 245.8 | 205.8 | 284.9 | 4,032  |
| SE                 | 42.5    | 132.6  | 2.2   | 4.8 | 23.2  | 38.6  | 21.1  | 20.5 | 58.1   | 49.4  | 22.1  | 29.2  | 172    |
| July               |         |        |       |     |       |       |       |      |        |       |       |       |        |
| Creel Estimate     | 931.9   | 1504.5 | 56.5  | 0   | 319.8 | 975.7 | 136.9 | 77.0 | 1114.8 | 274.3 | 282.9 | 0     | 5,674  |
| Lodge Report       | 147.7   | 238.4  | 8.9   | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 395    |
| JULY RETAINED      | 1079.5  | 1742.9 | 65.4  | 0   | 319.8 | 975.7 | 136.9 | 77.0 | 1114.8 | 274.3 | 282.9 | 0     | 6,069  |
| SE                 | 158.6   | 163.4  | 22.2  | 0   | 45.1  | 73.4  | 29.3  | 30.1 | 66.6   | 34.4  | 23.3  | 0     | 260    |
| August             |         |        |       |     |       |       |       |      |        |       |       |       |        |
| Creel Estimate     | 375.3   | 842.3  | 27.9  | 0   | 306.4 | 476.7 | 49.8  | 53.3 | 631.3  | 99.0  | 467.7 | 0     | 3,330  |
| Lodge Report       | 102.5   | 229.9  | 7.6   | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 340    |
| AUGUST RETAINED    | 477.8   | 1072.2 | 35.5  | 0   | 306.4 | 476.7 | 49.8  | 53.3 | 631.3  | 99.0  | 467.7 | 0     | 3,670  |
| SE                 | 78.2    | 91.1   | 23.7  | 0   | 39.9  | 40.8  | 11.9  | 13.9 | 65.3   | 22.5  | 47.4  | 0     | 160    |
| TOTAL              |         |        |       |     |       |       |       |      |        |       |       |       |        |
| Creel Estimate     | 1,511   | 3,458  | 235   | 7   | 744   | 1,849 | 325   | 213  | 2,509  | 619   | 956   | 285   | 12,712 |
| Lodge Report       | 295     | 714    | 50    | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 1,059  |
| TOTAL RETAINED     | 1,807   | 4,172  | 284   | 7   | 744   | 1,849 | 325   | 213  | 2,509  | 619   | 956   | 285   | 13,771 |
| SE                 | 182     | 229    | 33    | 5   | 65    | 92    | 38    | 39   | 110    | 64    | 57    | 29    | 350    |

**Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |        |      |    |      |       |      |      |       |       |       |       |       |
|--------------------|---------|--------|------|----|------|-------|------|------|-------|-------|-------|-------|-------|
| Month / Catch Type | 3I      | 3J     | 3K   | 4A | 4B   | 4C    | 4D   | 4E   | 4F    | 4G    | 4H    | 4L    | TOTAL |
| June               |         |        |      |    |      |       |      |      |       |       |       |       |       |
| Creel Estimate     | 62.1    | 945.6  | 95.7 | 0  | 31.0 | 127.4 | 91.1 | 13.1 | 432.5 | 217.7 | 89.7  | 301.7 | 2,408 |
| Lodge Report       | 0       | 0      | 0    | 0  | 0    | 0     | 0    | 0    | 0     | 0     | 0     | 0     | 0     |
| JUNE RELEASED      | 62.1    | 945.6  | 95.7 | 0  | 31.0 | 127.4 | 91.1 | 13.1 | 432.5 | 217.7 | 89.7  | 301.7 | 2,408 |
| SE                 | 28.5    | 252.2  | 1.4  | 0  | 9.5  | 29.5  | 30.2 | 7.4  | 85.2  | 90.1  | 32.3  | 108.0 | 307   |
| July               |         |        |      |    |      |       |      |      |       |       |       |       |       |
| Creel Estimate     | 263.4   | 1824.9 | 0    | 0  | 31.5 | 220.1 | 43.2 | 9.1  | 682.0 | 191.3 | 156.8 | 0     | 3,422 |
| Lodge Report       | 0       | 0      | 0    | 0  | 0    | 0     | 0    | 0    | 0     | 0     | 0     | 0     | 0     |
| JULY RELEASED      | 263.4   | 1824.9 | 0    | 0  | 31.5 | 220.1 | 43.2 | 9.1  | 682.0 | 191.3 | 156.8 | 0     | 3,422 |
| SE                 | 106.9   | 326.5  | 0    | 0  | 9.3  | 54.1  | 17.5 | 9.1  | 126.2 | 45.5  | 37.9  | 0     | 375   |
| August             |         |        |      |    |      |       |      |      |       |       |       |       |       |
| Creel Estimate     | 119.4   | 755.2  | 0    | 0  | 75.5 | 248.4 | 9.6  | 17.8 | 276.2 | 21.4  | 121.6 | 0     | 1,645 |
| Lodge Report       | 0       | 0      | 0    | 0  | 0    | 0     | 0    | 0    | 0     | 0     | 0     | 0     | 0     |
| AUGUST RELEASED    | 119.4   | 755.2  | 0    | 0  | 75.5 | 248.4 | 9.6  | 17.8 | 276.2 | 21.4  | 121.6 | 0     | 1,645 |
| SE                 | 40.8    | 241.7  | 0    | 0  | 20.6 | 78.0  | 4.6  | 9.5  | 77.9  | 9.7   | 25.8  | 0     | 271   |
| TOTAL              |         |        |      |    |      |       |      |      |       |       |       |       |       |
| Creel Estimate     | 445     | 3,526  | 96   | 0  | 138  | 596   | 144  | 40   | 1,391 | 430   | 368   | 302   | 7,475 |
| Lodge Report       | 0       | 0      | 0    | 0  | 0    | 0     | 0    | 0    | 0     | 0     | 0     | 0     | 0     |
| TOTAL RELEASED     | 445     | 3,526  | 96   | 0  | 138  | 596   | 144  | 40   | 1,391 | 430   | 368   | 302   | 7,475 |
| SE                 | 118     | 478    | 1    | 0  | 25   | 99    | 35   | 15   | 171   | 101   | 56    | 108   | 556   |

**Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |       |      |    |      |      |     |     |       |      |      |      |       |
|--------------------|---------|-------|------|----|------|------|-----|-----|-------|------|------|------|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A | 4B   | 4C   | 4D  | 4E  | 4F    | 4G   | 4H   | 4L   | TOTAL |
| June               |         |       |      |    |      |      |     |     |       |      |      |      |       |
| Creel Estimate     | 0       | 47.3  | 7.2  | 0  | 11.3 | 40.0 | 0   | 0   | 255.8 | 98.3 | 14.1 | 83.8 | 558   |
| Lodge Report       | 0       | 151.0 | 23.0 | 0  | 0    | 0    | 0   | 0   | 0     | 0    | 0    | 0    | 174   |
| JUNE RETAINED      | 0       | 198.3 | 30.2 | 0  | 11.3 | 40.0 | 0   | 0   | 255.8 | 98.3 | 14.1 | 83.8 | 732   |
| SE                 | 0       | 22.8  | 0.1  | 0  | 6.7  | 11.9 | 0   | 0   | 50.8  | 49.1 | 7.8  | 57.7 | 95    |
| July               |         |       |      |    |      |      |     |     |       |      |      |      |       |
| Creel Estimate     | 60.8    | 195.0 | 0    | 0  | 0    | 47.6 | 7.2 | 4.5 | 122.4 | 54.1 | 38.2 | 0    | 530   |
| Lodge Report       | 40.6    | 130.4 | 0    | 0  | 0    | 0    | 0   | 0   | 0     | 0    | 0    | 0    | 171   |
| JULY RETAINED      | 101.4   | 325.4 | 0    | 0  | 0    | 47.6 | 7.2 | 4.5 | 122.4 | 54.1 | 38.2 | 0    | 701   |
| SE                 | 31.8    | 103.3 | 0    | 0  | 0    | 13.1 | 3.9 | 4.5 | 29.5  | 18.6 | 12.0 | 0    | 115   |
| August             |         |       |      |    |      |      |     |     |       |      |      |      |       |
| Creel Estimate     | 51.2    | 43.6  | 0    | 0  | 17.8 | 3.4  | 1.9 | 5.9 | 93.7  | 0    | 18.7 | 0    | 236   |
| Lodge Report       | 72.4    | 61.6  | 0.0  | 0  | 0    | 0    | 0   | 0   | 0     | 0    | 0    | 0    | 134   |
| AUGUST RETAINED    | 123.6   | 105.2 | 0    | 0  | 17.8 | 3.4  | 1.9 | 5.9 | 93.7  | 0    | 18.7 | 0    | 370   |
| SE                 | 27.4    | 23.5  | 0    | 0  | 8.9  | 2.4  | 1.6 | 4.7 | 28.6  | 0    | 6.5  | 0    | 48    |
| TOTAL              |         |       |      |    |      |      |     |     |       |      |      |      |       |
| Creel Estimate     | 112     | 286   | 7    | 0  | 29   | 91   | 9   | 10  | 472   | 152  | 71   | 84   | 1,324 |
| Lodge Report       | 113     | 343   | 23   | 0  | 0    | 0    | 0   | 0   | 0     | 0    | 0    | 0    | 479   |
| TOTAL RETAINED     | 225     | 629   | 30   | 0  | 29   | 91   | 9   | 10  | 472   | 152  | 71   | 84   | 1,803 |
| SE                 | 42      | 108   | 0    | 0  | 11   | 18   | 4   | 7   | 65    | 52   | 16   | 58   | 157   |

**Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2016.**

|                    | Subarea |       |    |    |     |     |     |    |      |      |     |    |       |
|--------------------|---------|-------|----|----|-----|-----|-----|----|------|------|-----|----|-------|
| Month / Catch Type | 3I      | 3J    | 3K | 4A | 4B  | 4C  | 4D  | 4E | 4F   | 4G   | 4H  | 4L | TOTAL |
| June               |         |       |    |    |     |     |     |    |      |      |     |    |       |
| Creel Estimate     | 0       | 0     | 0  | 0  | 0   | 3.6 | 2.5 | 0  | 37.2 | 28.1 | 0   | 0  | 71    |
| Lodge Report       | 0       | 0     | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 0   | 0  | 0     |
| JUNE RELEASED      | 0       | 0     | 0  | 0  | 0   | 3.6 | 2.5 | 0  | 37.2 | 28.1 | 0   | 0  | 71    |
| SE                 | 0       | 0     | 0  | 0  | 0   | 2.7 | 1.8 | 0  | 16.9 | 19.0 | 0   | 0  | 26    |
| July               |         |       |    |    |     |     |     |    |      |      |     |    |       |
| Creel Estimate     | 0       | 195.0 | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 1.9 | 0  | 197   |
| Lodge Report       | 0       | 0     | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 0   | 0  | 0     |
| JULY RELEASED      | 0       | 195.0 | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 1.9 | 0  | 197   |
| SE                 | 0       | 108.6 | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 1.4 | 0  | 109   |
| August             |         |       |    |    |     |     |     |    |      |      |     |    |       |
| Creel Estimate     | 0       | 0     | 0  | 0  | 4.4 | 0   | 5.7 | 0  | 0    | 2.7  | 0   | 0  | 13    |
| Lodge Report       | 0       | 0     | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 0   | 0  | 0     |
| AUGUST RELEASED    | 0       | 0     | 0  | 0  | 4.4 | 0   | 5.7 | 0  | 0    | 2.7  | 0   | 0  | 13    |
| SE                 | 0       | 0     | 0  | 0  | 3.2 | 0   | 4.7 | 0  | 0    | 1.9  | 0   | 0  | 6     |
| TOTAL              |         |       |    |    |     |     |     |    |      |      |     |    |       |
| Creel Estimate     | 0       | 195   | 0  | 0  | 4   | 4   | 8   | 0  | 37   | 31   | 2   | 0  | 281   |
| Lodge Report       | 0       | 0     | 0  | 0  | 0   | 0   | 0   | 0  | 0    | 0    | 0   | 0  | 0     |
| TOTAL RELEASED     | 0       | 195   | 0  | 0  | 4   | 4   | 8   | 0  | 37   | 31   | 2   | 0  | 281   |
| SE                 | 0       | 109   | 0  | 0  | 3   | 3   | 5   | 0  | 17   | 19   | 1   | 0  | 112   |

**Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2016. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.**

|                    | Subarea |      |      |     |      |      |     |    |       |     |     |     |       |
|--------------------|---------|------|------|-----|------|------|-----|----|-------|-----|-----|-----|-------|
| Month / Catch Type | 3I      | 3J   | 3K   | 4A  | 4B   | 4C   | 4D  | 4E | 4F    | 4G  | 4H  | 4L  | TOTAL |
| June               |         |      |      |     |      |      |     |    |       |     |     |     |       |
| Creel Estimate     | 17.7    | 11.8 | 3.1  | 1.8 | 16.9 | 29.1 | 2.5 | 0  | 125.6 | 7.0 | 7.0 | 8.4 | 231   |
| Lodge Report       | 0       | 0    | 0    | 0   | 0    | 0    | 0   | 0  | 0     | 0   | 0   | 0   | 0     |
| JUNE RETAINED      | 17.7    | 11.8 | 3.1  | 1.8 | 16.9 | 29.1 | 2.5 | 0  | 125.6 | 7.0 | 7.0 | 8.4 | 231   |
| SE                 | 12.8    | 9.4  | 0.0  | 1.5 | 6.7  | 10.6 | 1.8 | 0  | 37.2  | 6.1 | 3.5 | 6.4 | 43    |
| July               |         |      |      |     |      |      |     |    |       |     |     |     |       |
| Creel Estimate     | 0       | 0    | 0    | 0   | 0    | 89.2 | 0   | 0  | 26.2  | 7.2 | 0   | 0   | 123   |
| Lodge Report       | 0       | 0    | 0    | 0   | 0    | 0    | 0   | 0  | 0     | 0   | 0   | 0   | 0     |
| JULY RETAINED      | 0       | 0    | 0    | 0   | 0    | 89.2 | 0   | 0  | 26.2  | 7.2 | 0   | 0   | 123   |
| SE                 | 0       | 0    | 0    | 0   | 0    | 28.9 | 0   | 0  | 13.4  | 5.2 | 0   | 0   | 32    |
| August             |         |      |      |     |      |      |     |    |       |     |     |     |       |
| Creel Estimate     | 119.4   | 29.0 | 13.9 | 0   | 17.8 | 10.1 | 0   | 0  | 19.7  | 0   | 4.7 | 0   | 215   |
| Lodge Report       | 0.0     | 0.0  | 0.0  | 0   | 0    | 0    | 0   | 0  | 0     | 0   | 0   | 0   | 0     |
| AUGUST RETAINED    | 119.4   | 29.0 | 13.9 | 0   | 17.8 | 10.1 | 0   | 0  | 19.7  | 0   | 4.7 | 0   | 215   |
| SE                 | 76.8    | 21.4 | 11.9 | 0   | 8.9  | 5.5  | 0   | 0  | 11.8  | 0   | 3.3 | 0   | 82    |
| TOTAL              |         |      |      |     |      |      |     |    |       |     |     |     |       |
| Creel Estimate     | 137     | 41   | 17   | 2   | 35   | 128  | 2   | 0  | 172   | 14  | 12  | 8   | 568   |
| Lodge Report       | 0       | 0    | 0    | 0   | 0    | 0    | 0   | 0  | 0     | 0   | 0   | 0   | 0     |
| TOTAL RETAINED     | 137     | 41   | 17   | 2   | 35   | 128  | 2   | 0  | 172   | 14  | 12  | 8   | 568   |
| SE                 | 78      | 23   | 12   | 2   | 11   | 31   | 2   | 0  | 41    | 8   | 5   | 6   | 98    |

**Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2016. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.**

|                    | Subarea |      |       |     |      |      |     |    |       |      |      |    |       |
|--------------------|---------|------|-------|-----|------|------|-----|----|-------|------|------|----|-------|
| Month / Catch Type | 3I      | 3J   | 3K    | 4A  | 4B   | 4C   | 4D  | 4E | 4F    | 4G   | 4H   | 4L | TOTAL |
| June               |         |      |       |     |      |      |     |    |       |      |      |    |       |
| Creel Estimate     | 8.9     | 11.8 | 2.1   | 0   | 11.3 | 7.3  | 9.8 | 0  | 134.9 | 21.1 | 0    | 0  | 207   |
| Lodge Report       | 0       | 0    | 0     | 0   | 0    | 0    | 0   | 0  | 0     | 0    | 0    | 0  | 0     |
| JUNE RELEASED      | 8.9     | 11.8 | 2.1   | 0   | 11.3 | 7.3  | 9.8 | 0  | 134.9 | 21.1 | 0    | 0  | 207   |
| SE                 | 6.4     | 9.4  | 0.0   | 0   | 4.2  | 3.8  | 7.0 | 0  | 50.8  | 13.6 | 0    | 0  | 55    |
| July               |         |      |       |     |      |      |     |    |       |      |      |    |       |
| Creel Estimate     | 20.3    | 27.9 | 0     | 0   | 49.5 | 53.5 | 0   | 0  | 100.6 | 50.5 | 0    | 0  | 302   |
| Lodge Report       | 0       | 0    | 0     | 0   | 0    | 0    | 0   | 0  | 0     | 0    | 0    | 0  | 0     |
| JULY RELEASED      | 20.3    | 27.9 | 0     | 0   | 49.5 | 53.5 | 0   | 0  | 100.6 | 50.5 | 0    | 0  | 302   |
| SE                 | 14.4    | 21.1 | 0     | 0   | 29.4 | 15.2 | 0   | 0  | 53.4  | 31.2 | 0    | 0  | 75    |
| August             |         |      |       |     |      |      |     |    |       |      |      |    |       |
| Creel Estimate     | 0       | 0    | 111.4 | 2.2 | 22.2 | 47.0 | 0   | 0  | 34.5  | 0    | 28.1 | 0  | 245   |
| Lodge Report       | 0       | 0    | 0     | 0   | 0    | 0    | 0   | 0  | 0     | 0    | 0    | 0  | 0     |
| AUGUST RELEASED    | 0       | 0    | 111.4 | 2.2 | 22.2 | 47.0 | 0   | 0  | 34.5  | 0    | 28.1 | 0  | 245   |
| SE                 | 0       | 0    | 69.2  | 1.6 | 11.4 | 12.9 | 0   | 0  | 11.2  | 0    | 20.0 | 0  | 75    |
| TOTAL              |         |      |       |     |      |      |     |    |       |      |      |    |       |
| Creel Estimate     | 29      | 40   | 114   | 2   | 83   | 108  | 10  | 0  | 270   | 72   | 28   | 0  | 755   |
| Lodge Report       | 0       | 0    | 0     | 0   | 0    | 0    | 0   | 0  | 0     | 0    | 0    | 0  | 0     |
| TOTAL RELEASED     | 29      | 40   | 114   | 2   | 83   | 108  | 10  | 0  | 270   | 72   | 28   | 0  | 755   |
| SE                 | 16      | 23   | 69    | 2   | 32   | 20   | 7   | 0  | 75    | 34   | 20   | 0  | 119   |

**Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2016. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.**

|                    | Subarea |       |       |      |       |       |      |      |       |       |      |      |       |
|--------------------|---------|-------|-------|------|-------|-------|------|------|-------|-------|------|------|-------|
| Month / Catch Type | 3I      | 3J    | 3K    | 4A   | 4B    | 4C    | 4D   | 4E   | 4F    | 4G    | 4H   | 4L   | TOTAL |
| June               |         |       |       |      |       |       |      |      |       |       |      |      |       |
| Creel Estimate     | 0       | 59.1  | 10.3  | 7.3  | 123.8 | 138.3 | 36.9 | 21.8 | 558.1 | 147.5 | 3.5  | 41.9 | 1,149 |
| Lodge Report       | 0       | 210.3 | 23.7  | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 234   |
| JUNE RETAINED      | 0       | 269.5 | 33.9  | 7.3  | 123.8 | 138.3 | 36.9 | 21.8 | 558.1 | 147.5 | 3.5  | 41.9 | 1,383 |
| SE                 | 0       | 24.9  | 0.1   | 6.1  | 36.2  | 32.8  | 17.5 | 7.5  | 73.6  | 63.5  | 2.9  | 21.4 | 115   |
| July               |         |       |       |      |       |       |      |      |       |       |      |      |       |
| Creel Estimate     | 202.6   | 473.6 | 0     | 0    | 90.1  | 172.5 | 82.8 | 4.5  | 306.0 | 126.3 | 84.1 | 0    | 1,543 |
| Lodge Report       | 87.2    | 165.8 | 0     | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 253   |
| JULY RETAINED      | 289.8   | 639.4 | 0     | 0    | 90.1  | 172.5 | 82.8 | 4.5  | 306.0 | 126.3 | 84.1 | 0    | 1,796 |
| SE                 | 83.2    | 166.5 | 0     | 0    | 22.0  | 54.7  | 34.8 | 4.5  | 76.4  | 67.8  | 29.3 | 0    | 225   |
| August             |         |       |       |      |       |       |      |      |       |       |      |      |       |
| Creel Estimate     | 68.2    | 217.8 | 69.7  | 11.1 | 128.8 | 33.6  | 7.7  | 5.9  | 202.2 | 2.7   | 84.2 | 0    | 832   |
| Lodge Report       | 0.0     | 0.0   | 281.0 | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 281   |
| AUGUST RETAINED    | 68.2    | 217.8 | 350.7 | 11.1 | 128.8 | 33.6  | 7.7  | 5.9  | 202.2 | 2.7   | 84.2 | 0    | 1,113 |
| SE                 | 35.7    | 94.8  | 59.3  | 5.6  | 45.1  | 7.7   | 3.1  | 4.7  | 54.1  | 1.9   | 22.2 | 0    | 139   |
| TOTAL              |         |       |       |      |       |       |      |      |       |       |      |      |       |
| Creel Estimate     | 271     | 751   | 80    | 18   | 343   | 344   | 127  | 32   | 1,066 | 276   | 172  | 42   | 3,523 |
| Lodge Report       | 87      | 376   | 305   | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 768   |
| TOTAL RETAINED     | 358     | 1,127 | 385   | 18   | 343   | 344   | 127  | 32   | 1,066 | 276   | 172  | 42   | 4,291 |
| SE                 | 91      | 193   | 59    | 8    | 62    | 64    | 39   | 10   | 119   | 93    | 37   | 21   | 289   |

**Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2016. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.**

|                    | Subarea |      |     |     |       |      |      |    |       |      |      |     |       |
|--------------------|---------|------|-----|-----|-------|------|------|----|-------|------|------|-----|-------|
| Month / Catch Type | 3I      | 3J   | 3K  | 4A  | 4B    | 4C   | 4D   | 4E | 4F    | 4G   | 4H   | 4L  | TOTAL |
| June               |         |      |     |     |       |      |      |    |       |      |      |     |       |
| Creel Estimate     | 0       | 35.5 | 3.1 | 0   | 101.3 | 94.6 | 32.0 | 0  | 237.2 | 49.2 | 3.5  | 8.4 | 565   |
| Lodge Report       | 0       | 0    | 0   | 0   | 0     | 0    | 0    | 0  | 0     | 0    | 0    | 0   | 0     |
| JUNE RELEASED      | 0       | 35.5 | 3.1 | 0   | 101.3 | 94.6 | 32.0 | 0  | 237.2 | 49.2 | 3.5  | 8.4 | 565   |
| SE                 | 0       | 20.9 | 0.0 | 0   | 46.3  | 32.0 | 14.9 | 0  | 70.9  | 25.1 | 2.9  | 6.4 | 98    |
| July               |         |      |     |     |       |      |      |    |       |      |      |     |       |
| Creel Estimate     | 60.8    | 41.8 | 0   | 0   | 18.0  | 77.3 | 7.2  | 0  | 65.6  | 25.3 | 11.5 | 0   | 307   |
| Lodge Report       | 0       | 0    | 0   | 0   | 0     | 0    | 0    | 0  | 0     | 0    | 0    | 0   | 0     |
| JULY RELEASED      | 60.8    | 41.8 | 0   | 0   | 18.0  | 77.3 | 7.2  | 0  | 65.6  | 25.3 | 11.5 | 0   | 307   |
| SE                 | 43.3    | 31.6 | 0   | 0   | 12.8  | 26.2 | 5.5  | 0  | 25.0  | 15.6 | 8.2  | 0   | 68    |
| August             |         |      |     |     |       |      |      |    |       |      |      |     |       |
| Creel Estimate     | 0       | 0    | 0   | 4.4 | 31.1  | 47.0 | 11.5 | 0  | 44.4  | 0    | 9.4  | 0   | 148   |
| Lodge Report       | 0       | 0    | 0   | 0   | 0     | 0    | 0    | 0  | 0     | 0    | 0    | 0   | 0     |
| AUGUST RELEASED    | 0       | 0    | 0   | 4.4 | 31.1  | 47.0 | 11.5 | 0  | 44.4  | 0    | 9.4  | 0   | 148   |
| SE                 | 0       | 0    | 0   | 3.1 | 17.0  | 15.8 | 7.0  | 0  | 17.9  | 0    | 6.7  | 0   | 31    |
| TOTAL              |         |      |     |     |       |      |      |    |       |      |      |     |       |
| Creel Estimate     | 61      | 77   | 3   | 4   | 150   | 219  | 51   | 0  | 347   | 74   | 24   | 8   | 1,020 |
| Lodge Report       | 0       | 0    | 0   | 0   | 0     | 0    | 0    | 0  | 0     | 0    | 0    | 0   | 0     |
| TOTAL RELEASED     | 61      | 77   | 3   | 4   | 150   | 219  | 51   | 0  | 347   | 74   | 24   | 8   | 1,020 |
| SE                 | 43      | 38   | 0   | 3   | 51    | 44   | 17   | 0  | 77    | 30   | 11   | 6   | 123   |

## APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month and subarea, in 2016.

| Species        |                 | Subarea |     |    |    |    |     |     |     |    |     |     |     |    |     |       |
|----------------|-----------------|---------|-----|----|----|----|-----|-----|-----|----|-----|-----|-----|----|-----|-------|
| Month          | Adipose         | 3I      | 3J  | 3K | 3? | 4A | 4B  | 4C  | 4D  | 4E | 4F  | 4G  | 4H  | 4L | 4?  | TOTAL |
| Chinook Salmon |                 |         |     |    |    |    |     |     |     |    |     |     |     |    |     |       |
| June           | Clipped         | 4       | 5   | 0  | 1  | 0  | 6   | 3   | 1   | 0  | 8   | 1   | 12  | 0  | 6   | 47    |
|                | Not Clipped     | 24      | 71  | 0  | 24 | 0  | 35  | 35  | 21  | 3  | 55  | 4   | 63  | 0  | 31  | 366   |
|                | Percent Clipped | 14%     | 7%  | -  | 4% | -  | 15% | 8%  | 5%  | 0% | 13% | 20% | 16% | -  | 16% | 11.4% |
| July           | Clipped         | 0       | 1   | 0  | 0  | 0  | 1   | 2   | 3   | 0  | 6   | 5   | 1   | 0  | 0   | 19    |
|                | Not Clipped     | 2       | 7   | 2  | 2  | 0  | 9   | 29  | 26  | 1  | 39  | 12  | 13  | 1  | 4   | 147   |
|                | Percent Clipped | 0%      | 13% | 0% | 0% | -  | 10% | 6%  | 10% | 0% | 13% | 29% | 7%  | 0% | 0%  | 11.4% |
| August         | Clipped         | 0       | 0   | 0  | 0  | 0  | 0   | 2   | 0   | 0  | 1   | 0   | 1   | 0  | 0   | 4     |
|                | Not Clipped     | 1       | 0   | 2  | 0  | 0  | 5   | 3   | 3   | 1  | 11  | 1   | 5   | 0  | 0   | 32    |
|                | Percent Clipped | 0%      | -   | 0% | -  | -  | 0%  | 40% | 0%  | 0% | 8%  | 0%  | 17% | -  | -   | 11.1% |
| Total          | Clipped         | 4       | 6   | 0  | 1  | 0  | 7   | 7   | 4   | 0  | 15  | 6   | 14  | 0  | 6   | 70    |
|                | Not Clipped     | 27      | 78  | 4  | 26 | 0  | 49  | 67  | 50  | 5  | 105 | 17  | 81  | 1  | 35  | 545   |
|                | Percent Clipped | 13%     | 7%  | 0% | 4% | -  | 13% | 9%  | 7%  | 0% | 13% | 26% | 15% | 0% | 15% | 11.4% |
| Coho Salmon    |                 |         |     |    |    |    |     |     |     |    |     |     |     |    |     |       |
| June           | Clipped         | 1       | 3   | 0  | 0  | 0  | 1   | 2   | 0   | 0  | 0   | 1   | 0   | 0  | 0   | 8     |
|                | Not Clipped     | 55      | 55  | 1  | 1  | 0  | 18  | 24  | 4   | 3  | 62  | 6   | 8   | 0  | 9   | 246   |
|                | Percent Clipped | 2%      | 5%  | 0% | 0% | -  | 5%  | 8%  | 0%  | 0% | 0%  | 14% | 0%  | -  | 0%  | 3.1%  |
| July           | Clipped         | 0       | 0   | 0  | 0  | 0  | 3   | 1   | 1   | 0  | 8   | 4   | 0   | 0  | 0   | 17    |
|                | Not Clipped     | 17      | 32  | 7  | 0  | 7  | 117 | 175 | 179 | 15 | 264 | 76  | 36  | 2  | 17  | 944   |
|                | Percent Clipped | 0%      | 0%  | 0% | -  | 0% | 3%  | 1%  | 1%  | 0% | 3%  | 5%  | 0%  | 0% | 0%  | 1.8%  |
| August         | Clipped         | 0       | 0   | 0  | 0  | 0  | 3   | 0   | 2   | 0  | 3   | 0   | 0   | 0  | 0   | 8     |
|                | Not Clipped     | 9       | 5   | 12 | 8  | 2  | 94  | 106 | 52  | 7  | 146 | 8   | 25  | 4  | 14  | 492   |
|                | Percent Clipped | 0%      | 0%  | 0% | 0% | 0% | 3%  | 0%  | 4%  | 0% | 2%  | 0%  | 0%  | 0% | 0%  | 1.6%  |
| Total          | Clipped         | 1       | 3   | 0  | 0  | 0  | 7   | 3   | 3   | 0  | 11  | 5   | 0   | 0  | 0   | 33    |
|                | Not Clipped     | 81      | 92  | 20 | 9  | 9  | 229 | 305 | 235 | 25 | 472 | 90  | 69  | 6  | 40  | 1,682 |
|                | Percent Clipped | 1%      | 3%  | 0% | 0% | 0% | 3%  | 1%  | 1%  | 0% | 2%  | 5%  | 0%  | 0% | 0%  | 1.9%  |